CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY



UNIVERSITY GRADUATION THESIS
SOFTWARE ENGINEERING (HIGH-QUALITY PROGRAM)

Topic

A Hybrid Cascade Framework Integrating Protein Language Models and Multi-scale CNNs for Peptide Screening and Toxicity Prediction

Author: Ho Quoc Bao
ID: B2203539
Course: 48

Can Tho, 8/2026

CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY



UNIVERSITY GRADUATION THESIS
SOFTWARE ENGINEERING (HIGH-QUALITY PROGRAM)

Topic

A Hybrid Cascade Framework Integrating Protein Language Models and Multi-scale CNNs for Peptide Screening and Toxicity Prediction

Supervisor:
Ho Quang Thai, Ph.D.

Author:
Ho Quoc Bao
ID: B2203539

Can Tho, 8/2026

CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

SOCIALIST REPUBLIC OF VIETNAM
Independence – Freedom – Happiness

THESIS REVISION CONFIRMATION IN ACCORDANCE WITH THE COMMITTEE'S COMMENTS

Thesis Title: A Hybrid Cascade Framework Integrating Protein Language Models and Multi-scale CNNs for Peptide Screening and Toxicity Prediction

Student: Ho Quoc Bao Student ID: B2203539

Class Code: DI2296F1

The thesis was presented before the Thesis Examination Committee of the Software Engineering Program.

Members of the Examination Committee:

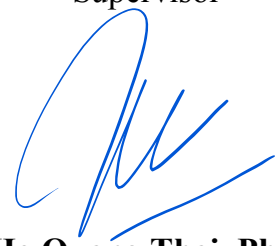
- Truong Minh Thai, Ph.D.
- Nguyen Thanh Khoa, Ph.D.
- Ho Quang Thai, Ph.D.

Reporting date: August 20, 2026

The thesis has been revised in accordance with the Committee's comments.

Can Tho, August 20, 2026

Supervisor



Ho Quang Thai, Ph.D.

COMMENTS FROM THE SUPERVISOR

COMMENTS FROM THE REVIEWERS

ACKNOWLEDGEMENTS

This graduation thesis was completed with the support and companionship of many individuals and organizations, to whom I would like to express my deepest gratitude.

First and foremost, I would like to express my most sincere and respectful thanks to my supervisor, Dr. Ho Quang Thai. His guidance, professional orientation, and feedback throughout this research have been instrumental in the completion of this thesis.

I would also like to extend my sincere thanks to the professors and lecturers of the High-Quality Program in Software Engineering at the College of Information and Communication Technology, Can Tho University. The academic environment and the foundational knowledge I received during my four years here have equipped me with the essential tools and mindset to undertake this research.

A special and profound thanks goes to the Kaggle platform and the open-source AI community. The free, high-performance computational resources provided by Kaggle enabled me to conduct large-scale experiments and train complex deep learning models without being constrained by my personal hardware limitations. Furthermore, I would like to acknowledge the assistance of advanced Large Language Models (such as Claude and Gemini), which served as invaluable technical sounding boards and coding assistants during the arduous debugging and writing phases.

Due to limitations in time and practical experience, this thesis inevitably contains certain shortcomings. I sincerely welcome and appreciate any comments, critiques, and feedback from the professors and the evaluation committee to further improve this work.

Sincerely,
Ho Quoc Bao

TABLE OF CONTENTS

THESIS REVISION CONFIRMATION I

COMMENTS FROM THE SUPERVISOR II

COMMENTS FROM THE REVIEWERS III

ACKNOWLEDGEMENTS IV

LIST OF FIGURES VIII

LIST OF TABLES X

GLOSSARY & ABBREVIATIONS XI

ABSTRACT XXI

CHAPTER 1: INTRODUCTION 1

1.1. Background and Motivation 1

1.2. Research Objectives 1

1.2.1 Overall Objective 1

1.2.2. Specific Objectives 2

Research Question: 2

1.3. Technical Challenges 2

1.4 . Research Subject and Scope 4

1.4.1. Subject of study 4

1.4.2. Scope of study 4

1.5. Thesis Structure 5

CHAPTER 2: LITERATURE REVIEW 7

2.1. Fundamentals of Peptides 7

2.1.1 Antimicrobial Peptides (AMPs) and Their Mechanisms 7

2.1.2. Hemolytic Peptides and Mechanisms of Hemolysis 7

2.2. Traditional Machine Learning in Peptide Prediction 8

2.2.1. Manual Feature Extraction and Compositional Metrics 8

2.2.2. Evolutionary Information and PSSMs 9

2.2.3. Limitations of the Traditional Pipeline 11

2.3. Representation Learning and Protein Language Models (PLMs) 11

2.3.1. Discrete Sequence Representations: One-Hot Encoding 11

2.3.2. Deep Contextual Embeddings and Protein Language Models (PLMs) 12

2.3.3. End-to-End Fine-Tuning vs. Selective Fine-Tuning 15

2.4. Advanced Convolutional Architectures for Motif Scanning 16

2.4.1. Standard 1D Convolutional Neural Networks 16

2.4.2. Multi-scale Convolutional Neural Networks (mCNNs) 16

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY 18

3.1. Overview of the Hybrid Cascade Architecture 18
3.2. Dataset Curation and Preprocessing 20
3.2.1. Full Dataset Collection and Homology Reduction 21
3.3. Scientific Justification for Proxy Task.....23
3.3.1. The "Proxy Subset" Methodology.....23
3.3.2. Scientific Justification (Multi-Fidelity Architecture Search).....23
3.4. Representation Learning and Motif Scanning Architectures 23
3.4.1. Static Feature Extraction (Baselines).....23
3.4.2. Contextual Representation Learning (ESM-2).....24
3.4.3. Universal Downstream Motif Scanners (1D CNN & mCNN) and Hyperparameter Rationales 24
3.5. Progressive Architecture Evolution and Ablation Strategy 27
3.5.1. Stage 1: Shallow Baselines and Static Descriptors 27
3.5.2. Stage 2: Spatial Feature Extraction via Deep Learning 27
3.5.3. Stage 3: Contextual Paradigms and Hybrid Architectures 27
3.6. Final Model Training and Dual-Task Deployment 27
3.6.1. AMP Classification Task.....28
3.6.2. Toxicity Prediction Task and Transfer Justification 28
3.7. Model Interpretability and Explainable AI (XAI).....28

CHAPTER 4: EXPERIMENTAL RESULTS AND DISCUSSION 30

4.1. Data Collection and Corpus Composition 30
4.1.1. Origin and Aggregation of the AMP Dataset 30
4.1.2. Filtering Protocol for the Toxicity Dataset 30
4.2. Dataset Statistics and Data Splitting.....31
4.2.1. Global Distribution and Sequence Length Constraints.....31
4.2.2. Post-Split Dataset Statistics 33
4.3. Experimental Setup and Evaluation Metrics.....33
4.3.1. Installation Environment and Hardware Configurations 33
4.3.2. Justification of Evaluation Metrics 34
4.4. Phase 1 Results: Architecture Screening and Ablation Study 35
4.4.1. Stage 1: Shallow Baselines and Static Descriptors 36
4.4.2. Stage 2: Spatial Feature Extraction via Deep Learning 37
4.4.3. Stage 3: Contextual Paradigms and Protein Language Models.....40
4.4.4. Resolving Geometric Instability: The Multi-Scale CNN (mCNN) Hybrid.....46
4.5. Return on Investment (ROI) Analysis and Final Architecture Selection 48
4.6. Data-Centric Intervention: Resolving Spurious Length Correlations 50
4.6.1. Discovery of Spurious Correlation at Scale 51

4.6.2. Engineering the Solution: Length-Matched Negative Sampling *52*
4.6.3. Validation of Ablation Integrity on Sub-Sampled Datasets *52*
4.7. Phase 2: Full-Scale System Training and Global Benchmarking 53
4.7.1. Deployment-Driven Multi-Tiered Strategy *53*
4.7.2. Architecture Flowchart and Algorithmic Configurations *54*
4.7.3. Global Independent Benchmarking *56*
4.7.4. The Universal "650M Paradox" and Architectural Limitations *59*
4.8. Interpretability and Sequence Analysis via XAI 59
4.9. Chapter Conclusion and key findings 61

CHAPTER 5: REAL-WORLD DEPLOYMENT AND CONCLUSION ... 62

5.1. System Architecture and Software Integration 62
5.2. System Deployment Strategies 63
5.2.1. Tier 1: Lightweight Portable Release (General Accessibility) *63*
5.2.2. Tier 2: Pro Academic Docker Release (Research & Server) *64*
5.3. Limitations and Future Work 65
5.4. Conclusion 67

REFERENCES 68

APPENDIX 70

APPENDIX A: OPEN-SOURCE REPOSITORY AND SOURCE CODE 70

APPENDIX B: EVALUATION OF POSITION-SPECIFIC SCORING MATRIX (PSSM) AND ALIGNMENT-BASED LIMITATIONS 70

B.1. The Alignment Blind Spot of Short Peptides 70
B.2. Sparse Matrices and Convolutional Model Collapse 71
B.3. Justification for Alignment-Free Architectures 71

APPENDIX C: SOFTWARE USER INTERFACE (STREAMLIT DEMO) 71

C.1. Single Sequence Analysis (Cascade Stages 2 & 3) 72
C.2. High-Throughput Batch Screening Module 73

LIST OF FIGURES

Figure 2.1: Workflow of the Dipeptide Composition (DPC) and Amino Acid Composition (AAC) feature extraction. 9

Figure 2.2: Workflow of the Position-Specific Scoring Matrix (PSSM) generation process, extracting evolutionary information and substitution probabilities while preserving the sequence's spatial order. 10

..... 11

Figure 2.3: The traditional machine learning pipeline for peptide classification, highlighting the reliance on manual feature extraction to generate fixed-length inputs for standard statistical classifiers. 11

Figure 2.4: One-hot encoding representation of a peptide sequence, illustrating the mapping of individual amino acids into a sparse binary matrix. 12

..... 13

Figure 2.5: Generation of deep contextual embeddings via a Transformer-based Protein Language Model, highlighting the capture of bi-directional sequence dependencies 14

Figure 2.6: Architectural comparison of parameter freezing strategies during the fine-tuning of large-scale Protein Language Models on resource-constrained datasets. 15

Figure 2.7: Architecture of a Multi-scale Convolutional Neural Network (mCNN) utilizing parallel variable-length windows and 1-max pooling for comprehensive motif extraction. 17

Figure 3.1: The Simplified Waterfall Cascade architecture of the proposed peptide screening pipeline, demonstrating the conditional execution flow from AMP Gatekeeping to Hemolytic Assessment and XAI extraction. 20

Figure 3.2: Data aggregation and splitting workflow for the AMP Gatekeeper (Stage 1). 21

Figure 3.3: Strict-mode data filtering for the Toxicity Assessor (Stage 2), explicitly discarding non-AMP imposters. 22

Figure 3.4: Modular pipeline architecture and combinatorial testing framework for peptide classification. 26

Figure 4.1: Global class distribution and sequence length density of the AMP and Toxicity datasets. 32

Figure 4.2: Bottleneck Test: F1-Score vs. Kernel Size 45

Figure 4.3: ROI Scatter Plots for Compact and Mid-tier PLMs (8M, 35M, 150M) coupled with mCNN configurations. 48

Figure 4.4: ROI Scatter Plots for the Massive 650M PLM coupled with mCNN configurations. 49

Figure 4.5: Initial AMP ROC Curve and Performance Metrics (Before length-matching intervention), showcasing the strong initial performance of the 650M model before the length-matching intervention. 51
Figure 4.6: The Generalized Hybrid Deep Learning Architecture for AMP and Toxicity Prediction 54
Figure 4.7: ROC Curve and Performance Metrics for the AMP Task 57
Figure 4.8: ROC Curve and Performance Metrics for the Toxicity Task 58
Figure 4.9: XAI heatmap of Stage 1 (AMP Gatekeeper) predicting a verified Antimicrobial Peptide, highlighting critical cationic and hydrophobic motifs. 60
Figure 4.10: XAI heatmap of Stage 2 (Toxicity Assessor) flagging a hemolytic sequence, emphasizing concentrated hydrophobic clusters associated with membrane interaction and hemolytic activity. 60
Figure 5.1: The decoupled Streamlit-based graphical user interface, demonstrating real-time peptide sequence entry, automated cascade screening, and interactive XAI heatmap visualization. 63
Figure C.1: Stage 2 (Hemolytic Assessment) interface. Following AMP detection, the sequence is evaluated for toxicity. The UI renders the XAI Hemolytic Attention heatmap and automatically extracts critical sub-sequences, specifically flagging Hemolytic Hydrophobic Clusters responsible for membrane penetration. 72
Figure C.2: Stage 3 (Wet-Lab Synthesis Recommendation). Based on the cascade results, the system outputs actionable synthesis advice. The UI also features a built-in Literature-Derived Motif Dictionary, allowing researchers to cross-reference AI findings with established wet-lab databases. 72
Figure C.3: The Batch Screening interface. Users can upload standard .fasta or .csv files. The system processes sequences in parallel utilizing the ESM2-35M core, outputting a comprehensive tabular results matrix tracking both AMP and Toxicity probabilities. 73
Figure C.4: Batch Analytics Dashboard. The system automatically filters the batch to isolate 'Ideal Candidates' (High AMP, Low Toxicity). Additionally, it generates a global Amino Acid Composition (AAC) profile of the successful candidates to aid in downstream synthesis planning. 73

LIST OF TABLES

Table 4.1: Sources and distribution of raw data aggregated for the AMP classification task. 30
Table 4.2: Filtering protocol and source tracking for the Toxicity assessment task. 31
Table 4.3: Global Dataset Statistics prior to data splitting. 33
Table 4.4: Detailed statistical breakdown of the Training, Validation, and Testing subsets. 33
Table 4.5: Performance evaluation of static descriptors paired with Classic ML algorithms. 36
Table 4.6: Performance collapse when applying Convolutional Networks to non-spatial AAC and DPC features. 38
Table 4.7: The performance ceiling of One-Hot Encoding across Shallow and Deep Learning architectures. 39
Table 4.8: Performance trajectory of Native End-to-End Fine-Tuning across scaling ESM-2 parameter sizes. 41
Table 4.9: Performance evaluation of Mean-Pooled ESM-2 embeddings paired with Classic Machine Learning algorithms. 43
Table 4.10: Single Kernel Performance Comparison across scaling ESM-2 models. 45
Table 4.11: Performance Evaluation of the Dynamic mCNN Hybrid Architecture across scaling ESM-2 models. 47
Table 4.12: Initial Benchmark Results (Before Data Intervention) 50
Table 4.13: Final Independent Benchmark Results on Hold-out Test Datasets. 56

GLOSSARY & ABBREVIATIONS

Ordinal numbers	Abbreviation / Term	Full Form / Description
1	**1-Max Pooling**	A down-sampling technique that extracts the absolute maximum value from a feature map. In peptide analysis, this ensures the most significant motif detected by a filter is kept, ignoring its exact sequence position.
2	**Ablation Study**	A systematic experimental procedure in machine learning where specific components of a model (e.g., layers, parameter sizes, or features) are removed or altered to determine their exact contribution to the overall performance of the system.
3	**Algorithmic Seed (random_state)**	A specific starting point (number) used to initialize a pseudorandom number generator. Setting a fixed seed ensures that the "random" sampling selects the exact same sequences every time the code is run, enabling fair comparisons between different models.
4	**Amino Acid Composition (AAC)**	A basic feature extraction method that calculates the percentage or frequency of each of the 20 standard amino acids within a given peptide sequence.
5	**Amino Acids (AAs)**	The fundamental organic molecules that combine to form peptides and proteins. Peptides are generally distinguished from proteins by their shorter chain length.
6	**Amphiphilic Alpha-helices**	A specific 3D structural conformation of a peptide chain where one side of the helix is hydrophilic (water-attracting/polar) and the other side is hydrophobic (water-repelling/non-polar), allowing the peptide to interact with and insert into lipid membranes.
7	**AMPs**	**Antimicrobial Peptides**: A large group of endogenous compounds exhibiting a broad spectrum of activities against pathogens, including antibacterial, anticancer, antiviral, and antifungal properties.
8	**Anti-gravity**	A colloquial reference to the highly efficient, cloud-based Google IDE utilized during the local deployment and testing phases, which significantly streamlined the system integration and web application development.

9	**Architecture Transfer Justification**	The scientific reasoning for reusing the network design (mCNN) optimized on the AMP dataset for the Toxicity dataset, based on the shared physical mechanisms of peptide-membrane interactions.
10	**Baseline Configurations**	The combination of traditional encoding methods (One-hot, AAC, DPC) with standard classifiers (RF, SVM, 1D CNN) used to set a fundamental benchmark.
11	**Cationic Peptides**	Peptides that carry a net positive electrical charge. This positive charge is crucial for their initial electrostatic attraction to the negatively charged components of bacterial cell walls.
12	**CD-HIT**	A widely used bioinformatics clustering program designed to reduce sequence redundancy in biological datasets by grouping and removing highly similar sequences.
13	**CNNs**	**Convolutional Neural Networks**: A class of deep learning architectures highly effective at automatically learning and extracting complex, local features directly from raw data (e.g., amino acid sequences).
14	**Computational ROI**	*Return on Investment.* The empirical evaluation of trade-offs between a model's predictive performance (F1-Score, AUC) and the computational cost (Training Time) required to achieve it.
15	**Contextual Embeddings**	Dynamic representations where an amino acid's vector changes based on its position and interactions with neighboring residues.
16	**Contextualized Embeddings**	Dense mathematical vectors where the representation of an amino acid changes dynamically depending on the other amino acids surrounding it in the sequence, allowing the model to understand local and global structural motifs.
17	**Dipeptide Composition (DPC)**	A feature extraction method that calculates the occurrence frequency of two adjacent amino acids (pairs), providing a very basic level of sequence-order information.
18	**Dual-Task Deployment**	The process of training independent models for two separate but related biological objectives (AMP classification and Toxicity prediction).

19	**Dynamic Early Stopping**	A strict regularization heuristic that automatically halts the training loop if the network fails to improve its classification boundaries (monitored via eval_auc with a patience of 3 epochs) to prevent global overfitting.
20	**Empirical Evaluation**	The process of assessing a model or hypothesis based on verifiable observation, experimentation, and statistical results rather than pure theory or logic.
21	**ESM-2 (Evolutionary Scale Modeling)**	A state-of-the-art family of Protein Language Models developed by Meta AI, trained on millions of protein sequences to generate high-dimensional embeddings that capture deep evolutionary and structural biochemical patterns.
22	**Evaluation Metrics**	Mathematical standardizations used to assess the performance of a classification model. In this thesis, metrics typically include Accuracy, Sensitivity (Recall), Specificity, Precision, and the Matthews Correlation Coefficient (MCC).
23	**False Negative (FN)**	A "missed threat" where the model incorrectly predicts a biologically active or deadly peptide as safe/inactive. This is the most critical and dangerous error in clinical screening.
24	**False Positive (FP)**	A "false alarm" where the model incorrectly flags a safe/inactive sequence as toxic/AMP. It poses no biological danger but wastes wet-lab synthesis resources.
25	**FASTA Format**	A text-based format for representing either nucleotide sequences or amino acid (protein) sequences, where base pairs or amino acids are represented using single-letter codes.
26	**Feature Engineering**	The manual process of using domain knowledge to extract specific variables (features) from raw data that make machine learning algorithms work.
27	**Feature Extraction**	The computational process of transforming raw data (such as a string of amino acid characters) into a structured numerical representation (vectors or matrices) that a neural network can process and learn from.
28	**Feature-based (Core Freezing)**	A training paradigm where the massive PLM (ESM-2) is kept static to generate embeddings, and only the subsequent CNN layers are updated during training.

29	**Fine-tuning End-to-End**	A training paradigm where all parameters of the PLM and its native classifier are updated simultaneously for the specific classification task.
30	**Focal Loss**	A specialized loss function dynamically scaled based on prediction confidence, designed to address extreme class imbalance by down-weighting the loss assigned to well-classified majority examples and focusing the model on hard-to-predict minority examples (e.g., the rare hemolytic class).
31	**Hemolysis**	The destruction or rupturing of red blood cells; considered a severe, intrinsic undesired toxic property in many therapeutic peptides.
32	**Hemolysis / Hemolytic Activity**	The rupture or destruction of red blood cells (erythrocytes). In pharmacology, high hemolytic activity indicates high toxicity to the host organism.
33	**Hemolytic Motifs**	Specific, recurring subsequences or patterns of amino acids within a peptide chain that are statistically associated with and predictive of hemolytic toxicity.
34	**High-Capacity Tier**	The gigabyte-scale operational tier (utilizing the ESM2-650M core). Initially evaluated as a theoretical apex architecture, but ultimately retained strictly for architectural comparison due to its susceptibility to the "650M Paradox" and spatial overfitting.
35	**High-Throughput Screening (HTS)**	A method typically used in drug discovery that utilizes automation to quickly assay the biological or biochemical activity of a large number of compounds (in this case, massive batches of peptide sequences).
36	**Homology**	The state of having an identical or highly similar sequence of amino acids, often indicating a shared evolutionary ancestry or functional similarity.
37	**Hybrid Architectures**	The final, most complex models created by fusing advanced contextual embeddings (ESM-2) with spatial deep learning networks (mCNN).
38	**Hybrid Cascade Framework**	A sequential, multi-stage machine learning architecture where the output or filtering decision of the first model (Stage 1) determines whether the data is passed to the subsequent model (Stage 2).

39	**Hyperparameter Rationales**	The logical, mathematical, and biological justifications for choosing specific configuration values (e.g., kernel sizes, number of branches) in the neural network to avoid arbitrary parameter selection.
40	**Ideal Candidates**	Within the context of this pipeline, these are peptide sequences that successfully pass both cascade stages: they possess high antimicrobial probability and display negligible/zero hemolytic toxicity.
41	**In silico**	Computational methods or experiments performed on a computer or via computer simulation, as opposed to physical wet-laboratory experiments.
42	**In vitro SPPS**	Solid-Phase Peptide Synthesis (SPPS) conducted in a controlled laboratory environment (*in vitro*). The final goal of the computational pipeline is to select the safest, most effective candidates for this costly physical synthesis.
43	**Inference**	The operational phase in deep learning where a fully trained model is deployed into a real-world system (e.g., the FastAPI backend) to make predictions on new, unseen peptide sequences.
44	**Innate Immune Response**	The nonspecific, first line of defense mechanism in organisms against invading pathogens. AMPs are a primary biochemical component of this system.
45	**Kernel / Filter**	A mathematical sliding window used in a CNN to extract features from the input data. Its size determines how much of the sequence the network "looks at" at any one time.
46	**Lipid Bilayer**	The universal structural component of all cell membranes, consisting of two layers of lipid molecules. AMPs target this structure to cause permeation and cell death.
47	**Masked Language Modeling**	A self-supervised training technique where certain tokens (amino acids) in a sequence are intentionally hidden (masked), and the neural network is tasked with predicting them based on the unmasked context.
48	**Masked Token Importance (XAI)**	An Explainable AI technique that quantifies the importance of a specific amino acid by temporarily hiding (masking) it from the model and measuring how much the model's confidence drops without it.

49	**Modular Pipeline**	An experimental design where models are built by plugging different feature extractors into different downstream classifiers, allowing for comprehensive combinatorial testing.
50	**Motif**	A specific, recurring sequence or pattern of amino acids that is statistically associated with a particular biological function (e.g., an antimicrobial signature).
51	**Multi-scale Convolutional Neural Network (mCNN)**	A deep learning architecture that utilizes multiple parallel convolutional filters of varying sizes (window lengths) simultaneously, allowing it to capture both short and long sequence motifs within biological data.
52	**Occlusion-based Sensitivity Analysis**	An Explainable AI (XAI) technique that evaluates feature importance by systematically masking input tokens and measuring the resultant degradation in predictive confidence.
53	**Offline Deployment**	The implementation of a software system or machine learning model on local hardware without requiring continuous connection to cloud servers or high-performance computing clusters.
54	**Out-Of-Memory (OOM)**	A fatal computational error that occurs when a system's GPU VRAM is completely exhausted, commonly encountered when training massive LLMs (like ESM-2) without sufficient hardware.
55	**Overfitting**	A modeling error that occurs when a machine learning algorithm learns the exact noise and details of the training data to the extent that it negatively impacts the model's performance on new, unseen data.
56	**Over-parameterization**	A mathematical condition where a model's parameter space (e.g., 650 million parameters) is excessively large relative to the training dataset size (e.g., ~2,500 toxicity sequences), leading to the immediate memorization of dataset noise and a loss of generalization.
57	**Parameter Bloat**	A detrimental condition where a model has an excessively large number of parameters (e.g., using too many CNN branches), leading to computational inefficiency and a high risk of overfitting.
58	**Performance Plateau**	An empirical phenomenon observed during ablation testing where increasing model complexity (e.g., scaling up from K5 to K11) no longer yields improvements and induces accuracy degradation.

59	**Performance-Driven Multi-Tiered Strategy**	A deployment-driven system design that bifurcates the screening pipeline into distinct operational tiers (Standard-Scale and High-Capacity) based on hardware constraints and empirical predictive generalization.
60	**Position-Specific Scoring Matrix (PSSM)**	A numerical matrix representing evolutionary protein profiles, used to identify conserved sequence motifs by calculating the substitution frequency of amino acids at specific sites.
61	**Progressive Architecture Evolution**	An experimental strategy where model complexity is systematically increased step-by-step (e.g., from SVM to CNN, then to ESM-2 hybrids) to find the most efficient design.
62	**Protein Language Models (PLMs)**	Advanced neural networks (like ESM-2) trained on massive protein databases to capture the contextual and evolutionary grammar of amino acid sequences.
63	**Proxy Subset**	A scaled-down, random sample of a larger dataset used in machine learning to quickly test and debug model architectures before executing resource-intensive full training.
64	**Random Forest (RF)**	A traditional ensemble machine learning algorithm that constructs multiple decision trees during training and outputs the majority vote of the classes for prediction.
65	**Receptive Field Upper Bound**	The maximum effective size of a convolutional filter, theoretically and empirically restricted by the inherently short length of antimicrobial peptide (AMP) sequences.
66	**Regex Pattern Matching**	A computational method used to search a biological sequence for specific strings of characters (motifs), accommodating biological variations (e.g., matching either Leucine or Isoleucine).
67	**Reproducibility**	The ability of an entire experiment or study to be duplicated, either by the same researcher or by someone else working independently, yielding the same results.
68	**Resource Optimization Mode**	A programmed halt in the system's execution pipeline. If a peptide is deemed functionally useless (e.g., non-antimicrobial), the system conserves GPU and memory resources by bypassing the toxicity evaluation.
69	**Resource-Aware Multi-Tiered Strategy**	A deployment-driven system design that bifurcates the screening pipeline into distinct operational tiers based on hardware constraints and predictive requirements.

70	**Return on Investment (ROI) Analysis**	An architectural evaluation metric used in this thesis to quantify the trade-off between the computational cost (memory, inference time, parameter count) of deploying a massive deep learning model versus the marginal gain in classification accuracy it provides over a smaller model.
71	**ROC-AUC**	*Receiver Operating Characteristic - Area Under the Curve*. A robust, threshold-independent metric that evaluates a model's absolute ability to distinguish between two classes (e.g., Toxic vs. Safe) regardless of class imbalance.
72	**Selected Optimal Pipeline**	The optimal architectural configuration of this research: the **ESM2-35M core coupled with the Standard mCNN (**$K \in \{3,5,7\}$**)**. It provides the absolute best Return on Investment (ROI), balancing deep evolutionary grammar with seamless software deployment across both AMP and Toxicity tasks.
73	**Selective Fine-Tuning**	A transfer learning optimization technique where specific layers or components of a massive pre-trained neural network are kept mathematically static (frozen) while others are allowed to update, stabilizing training on small datasets.
74	**Sensitivity / Recall**	The proportion of actual positive sequences correctly identified by the model. It is prioritized as the supreme safety standard to ensure no toxic peptides bypass the screening system.
75	**Sequence Homology**	The degree of similarity between two biological sequences. High homology in machine learning datasets can lead to data leakage and overfitting, necessitating clustering tools to ensure models learn generalized rules rather than memorizing redundant data.
76	**Shallow Baselines**	Traditional machine learning models (RF, SVM) combined with basic encoding techniques (One-hot, AAC) that lack deep feature extraction layers.
77	**Short-length AMPs**	Antimicrobial peptides consisting fewer than 30 amino acids, known for enhanced stability, easier synthesis, and lower toxicity to human cells.
78	**Short-Length Peptides**	In the context of this computational study, these are peptide sequences constrained to a maximum length of 30 amino acids, noted for their lower manufacturing costs and higher stability compared to longer peptide chains.

79	**Single-branch 1D CNN**	A deep learning architecture that utilizes only one specific convolutional kernel size (e.g., K3 or K9) to sequentially scan for local sequence motifs.
80	**Spatial Alignment**	Maintaining the exact positional relationship of amino acids within a sequence during the feature extraction process to accurately locate biological motifs.
81	**Standard Tier**	The lightweight, resource-aware operational tier (utilizing the ESM2-35M core) engineered for broad applicability. It enables rapid, high-throughput sequence screening on standard local hardware without risking OOM failures.
82	**Static Features**	Fixed numerical representations (e.g., AAC, DPC) that calculate amino acid properties globally without adapting to the specific surrounding sequence context.
83	**Support Vector Machine (SVM)**	A supervised machine learning algorithm that classifies data by finding an optimal boundary (hyperplane) that separates different classes in a high-dimensional feature space.
84	**Symmetric Zero-padding**	The practice of adding an equal number of zeros to both ends of an input sequence, which is only mathematically perfectly aligned when using odd-numbered convolutional kernels.
85	**Tensor Batching**	The computational process of grouping multiple independent data inputs (e.g., 128 peptide sequences) into a single multi-dimensional array (tensor) to be processed simultaneously by the GPU, drastically reducing overall inference time.
86	**The "650M Paradox"**	An unexpected empirical finding in this research where the massive, highly complex ESM2-650M model severely underperforms the lightweight ESM2-35M model across both AMP and Toxicity tasks post-data intervention, caused by spatial noise, hyper-dimensional over-parameterization, and overfitting on short-length peptide distributions.
87	**True Negative (TN)**	The model correctly identifies a background noise sequence (Non-AMP) or a Safe peptide.
88	**True Positive (TP)**	The model correctly identifies a biologically active Antimicrobial Peptide (AMP) or a highly Toxic sequence.
89	**Universal Downstream Scanners**	The CNN architectures (Single 1D CNN and mCNN) that are consistently attached to the tail-end of both static features and PLM embeddings to standardize the classification process.

90	**Waterfall Cascade**	A sequential architectural design where data must sequentially pass strict criteria at one stage before proceeding to the next, preventing unnecessary computations on invalid inputs.
91	**Wet-lab**	Traditional physical laboratories where biological matter, chemicals, and drugs are tested *in vitro* or *in vivo*.
92	**Word2Vec / Skip-Gram**	Early NLP algorithms that map words (or amino acids) into a continuous vector space by predicting surrounding context, useful for capturing basic similarity but lacking deep structural context.
93	**XAI (Explainable AI)**	Techniques and algorithms designed to make the internal mechanics, representations, and decisions of artificial intelligence systems transparent and interpretable to human experts.

ABSTRACT

Motivation: Antimicrobial peptides (AMPs) represent a highly promising alternative to conventional antibiotics in the global fight against multidrug-resistant pathogens. However, their clinical translation is severely hindered by hemolytic toxicity—the unintended destruction of human red blood cells. Traditional wet-lab screening for safety and efficacy is both cost-prohibitive and time-consuming, highlighting the urgent need for advanced computational prediction models.

Methods: In this study, we propose a novel Hybrid Cascade Framework integrating Protein Language Models (ESM-2) with multi-scale Convolutional Neural Networks (mCNNs). The architecture operates in two sequential stages: an *AMP Gatekeeper* to rapidly identify true antimicrobial sequences, and a *Toxicity Assessor* to evaluate their hemolytic potential, effectively addressing data imbalance and spurious correlations.

Results & Conclusion: Through ROI analysis and data-centric intervention, the study found that the ESM-2 35M configuration provided the best performance–efficiency trade-off among the evaluated architectures after length-matched negative sampling. The results also showed that the larger ESM-2 650M configuration did not consistently improve predictive performance on the evaluated short-peptide tasks.

CHAPTER 1: INTRODUCTION

1.1. Background and Motivation

Antimicrobial peptides (AMPs) represent a large group of endogenous compounds that exhibit a broad spectrum of activities, including antibacterial, anticancer, antiviral, and antifungal properties. Consequently, AMPs are increasingly recognized as promising alternatives to conventional antibiotics and are being utilized in the development of new therapeutic drugs to combat the growing crisis of multidrug-resistant pathogens. In particular, short-length AMPs (≤30 amino acids) have drawn significant attention because they demonstrate enhanced antimicrobial activities, higher stability, and lower toxicity to human cells. [1,2]

Despite these remarkable therapeutic advantages, the widespread clinical application of AMPs is severely hindered by their potential to cause hemolysis, which is the destruction of human red blood cells. Hemolytic activity is an intrinsic, undesired property associated with many therapeutic peptides. When peptides interact with and rupture the red blood cell membrane, the released products can lead to systemic inflammation, widespread tissue damage, anemia, renal failure, and even shock. Therefore, accurately predicting and mitigating the hemolytic activity of candidate peptides is a crucial prerequisite for their safe and effective use in clinical treatments. [5,6,7]

Traditionally, the identification and optimization of AMPs through wet-laboratory experiments have proven to be both cost-prohibitive and highly time-consuming. To address these financial and temporal constraints, *in silico* computational methods are now commonly utilized to accelerate the discovery and screening of new AMP candidates. Recently, deep learning approaches—specifically Convolutional Neural Networks (CNNs)—have emerged as highly effective tools for this task because of their capacity to automatically learn and extract complex, local features directly from raw amino acid sequences. Leveraging these advanced sequence-based deep learning models provides a rational, highly efficient basis for screening vast amounts of genomic data to discover novel, potent AMPs with minimized hemolytic toxicity. [1,2]

1.2. Research Objectives

1.2.1 Overall Objective

The primary objective of this thesis is to develop a highly robust, end-to-end computational framework capable of accurately predicting both the antimicrobial activity and the hemolytic toxicity of short-length peptides. By constructing a two-stage Hybrid Cascade Framework that integrates advanced representation learning from a pre-trained Protein Language Model with local motif extraction from a multi-scale Convolutional Neural Network (mCNN), this research aims to provide an efficient, privacy-preserving, and offline *in silico* tool to accelerate the discovery of novel, safe therapeutic peptides.

1.2.2. Specific Objectives

To realize the overall objective, this study focuses on the following specific goals:

- **To curate and preprocess high-quality datasets:** To aggregate and rigorously clean data from multiple verified AMP and hemolysis databases (e.g., HemoPI, RNN-Hem, and HLPpred-Fuse). This includes strict measures to eliminate sequence redundancy and mitigate data bias, ensuring an objective evaluation environment. [3,6]
- **To develop an AMP Gatekeeper (Stage 1):** To design an initial screening model utilizing ESM-2 embeddings and dynamic mCNNs to rapidly identify true antimicrobial sequences from raw inputs, effectively filtering out non-functional (imposter) peptides.
- **To develop a Toxicity Assessor (Stage 2):** To construct a secondary model specialized in classifying the hemolytic potential of the verified AMPs. This includes implementing a Focal Loss function tailored specifically to handle the severe class imbalance inherent in clinical toxicity datasets.
- **To perform a comprehensive Ablation Study and ROI Analysis:** To systematically evaluate the performance trade-offs among various ESM-2 configurations and mCNN window sizes.

Research Question:

To address the limitations of current in silico screening methods and fulfill the outlined objectives, this thesis seeks to answer the following central research question: *How can a two-stage Hybrid Cascade Framework—integrating evolutionary representations from a scaled Protein Language Model (PLM) with local motif scanning via a multi-scale Convolutional Neural Network (mCNN)—be optimized to accurately predict both the antimicrobial efficacy and hemolytic toxicity of short-length peptides, while maintaining a computational footprint suitable for practical, offline deployment?*

1.3. Technical Challenges

Developing an automated, high-accuracy *in silico* framework for peptide classification presents several multi-faceted technical challenges:

- **Sequence Representation and Feature Extraction Bottlenecks:** Historically, predictive models have relied on manual feature engineering to generate inputs that represent the compositional, physicochemical, and structural properties of a peptide sequence. This conventional process is highly time-consuming and requires extensive expert biological knowledge. Furthermore, transforming sequences into profiles like Position-Specific Scoring Matrices (PSSMs) often leads to the loss of crucial sequence order and positional information when these matrices are summed or compressed. [1,4]
- **Variable Peptide Lengths:** Peptides naturally vary in length, yet algorithms like Convolutional Neural Networks (CNNs) typically require fixed-dimension input layers. Padding or truncating sequences to fit these models can dilute critical biochemical signals. This is especially challenging for short-length Antimicrobial Peptides (AMPs) (≤30 amino acids) which have proven difficult for general AMP models to predict accurately despite their enhanced stability and lower toxicity. [3,4,6]
- **Data Scarcity and Severe Class Imbalance:** Constructing a robust model is hindered by the scarcity of experimentally validated negative samples (e.g., proven non-AMP or non-hemolytic peptides). Consequently, negative datasets are often artificially generated by randomly selecting unannotated sequences from databases like UniProt. This creates inherently imbalanced datasets, causing models to become biased toward predicting the majority class correctly while neglecting the minority class. [1,2]
- **Sequence Redundancy and Overfitting:** Biological datasets frequently contain high sequence homology. If highly similar sequences overlap between training and testing sets, the neural network's performance will be artificially overestimated and biased. Applying strict clustering software is mandatory to remove sequence redundancy and prevent the neural network from simply memorizing homologous sequences rather than learning true biological rules. [1,3,7]
- **Untangling Antimicrobial Efficacy from Hemolytic Toxicity:** There is a massive overlap between AMPs and hemolytic peptides, making discrimination incredibly difficult. A visual data mining study of metadata networks revealed that 9s4.46% of known hemolytic peptides also exhibit antimicrobial activity. Because cationic AMPs interact with microbial membranes, they can also inadvertently interact with and damage the structurally similar membranes of human red blood cells. [5,6]

- **Computational Cost of Advanced Architectures:** While Protein Language Models (PLMs) and deep learning networks automatically extract complex spatial features without domain expertise, they are highly resource-intensive. Models utilizing large-scale Transformers and multi-layered CNNs demand significant memory, processing power, and specialized hardware (like GPUs) to encode sequences and process massive datasets during virtual screening. [1,3]

1.4. Research Subject and Scope

1.4.1. Subject of study

This research primarily focuses on the intersection of deep learning and bioinformatics, specifically targeting the *in silico* sequence-based prediction of biological activities. The core biological subjects of investigation are short-length Antimicrobial Peptides (AMPs) and their associated hemolytic toxicity. From a computational perspective, the study centers on advanced representation learning paradigms, specifically exploring how pre-trained Protein Language Models (PLMs)—namely the ESM-2 architecture—can be coupled with Multi-scale Convolutional Neural Networks (mCNNs) to automatically extract, represent, and classify complex 3D spatial and evolutionary motifs from 1D amino acid sequences.

1.4.2. Scope of study

- **Data Scope:** The sequence data investigated in this study is strictly bounded to short-length peptides, defined computationally and biologically as sequences containing 30 or fewer amino acid residues. The models are trained, validated, and tested using an aggregated corpus of rigorously curated, publicly available benchmark datasets, specifically HemoPI-1, HemoPI-2, HemoPI-3, RNN-Hem, and Hlppredfuse, complemented by the open-source Viri dataset. The source benchmark datasets had undergone sequence redundancy and homology reduction as part of their original preprocessing. The merged corpus was additionally checked for exact duplicate sequences, while the Viri dataset was incorporated as an additional data source. [3,6]
- **Algorithmic and Functional Scope:** Functionally, the study is restricted to binary classification tasks executed within a custom two-stage pipeline, programmed utilizing Python deep learning ecosystems. Stage 1 (The AMP Gatekeeper) is limited to classifying sequences as either AMP or non-AMP. Stage 2 (The Toxicity Assessor) is strictly conditionally activated, evaluating the hemolytic versus non-hemolytic potential solely for sequences that successfully pass Stage 1. Algorithmically, the scope is confined to utilizing ESM-2 for feature embedding and parallel-window mCNNs for motif scanning, incorporating Focal Loss optimization specifically to resolve the severe class imbalance inherent in the toxicity datasets. [4]

- **Deployment and Evaluation Scope:** The evaluation of the proposed Hybrid Cascade Framework extends beyond standard predictive metrics (Accuracy, Precision, Recall, MCC) to include a computational Return on Investment (ROI) analysis. This specific scope bounds the architectural comparison to evaluating the structural trade-offs between a massive parameter model (ESM-2 650M, initially hypothesized as a high-capacity ceiling) and a compact parameter variant (the compact ESM-2 35M configuration). The deployment scope is specifically targeted at determining the feasibility of running these models in hardware-constrained, offline environments while maintaining optimal predictive reliability.

1.5. Thesis Structure

The remainder of this thesis is logically organized into four subsequent chapters, each detailing a specific phase of the research, from foundational theory to system implementation and empirical evaluation. The structure is as follows:

- **Chapter 1: Introduction** This chapter introduces the research background, motivation, objectives, research questions, technical challenges, and scope of the study. It establishes the practical and computational context of antimicrobial peptide screening and hemolytic toxicity prediction, and outlines the overall direction of the proposed Hybrid Cascade Framework.
- **Chapter 2: Literature Review** This chapter provides a comprehensive survey of the current bioinformatics research landscape. It critically examines the evolution of peptide prediction from traditional Machine Learning paradigms relying on hand-crafted physicochemical features to state-of-the-art representation learning utilizing Protein Language Models (PLMs). Furthermore, it reviews advanced convolutional architectures for biological motif scanning and establishes the current methodologies and limitations in hemolytic toxicity analysis.
- **Chapter 3: System Design and Methodology** This chapter details the core engineering and architectural design of the proposed hybrid framework. It outlines the rigorous dataset curation and preprocessing protocols, including sequence deduplication. The chapter provides an in-depth mathematical and structural breakdown of the generalized architecture (combining frozen ESM-2 embeddings with a Multi-branch Convolutional Neural Network - mCNN). It also introduces the theoretical foundation for the Resource-Aware Multi-Tiered Strategy and the algorithmic interventions (such as Focal Loss) required to handle highly imbalanced biological datasets.

- **Chapter 4: Experimental Results and Discussion** This chapter presents a rigorous empirical evaluation of the proposed models. It begins by defining the experimental setup and evaluation metrics. The chapter systematically advances through a multiphase ablation study: evaluating classic descriptors against PLMs, exposing the geometric instability of single-kernel CNNs, and determining the Return on Investment (ROI) sweet spot. Crucially, it presents the global independent benchmarking results, discusses the discovery of the "650M Paradox" (overfitting on constrained datasets), and evaluates the full-scale models and identifies the configuration providing the most favorable performance efficiency trade-off within the evaluated architectures.
- **Chapter 5: Real-World Deployment and Conclusion** The final chapter bridges the gap between theoretical deep learning performance and practical software application. It details the decoupled Client-Server system architecture (FastAPI and Streamlit) and proposes two viable deployment strategies: a Lightweight Portable Release for general accessibility and a Pro Academic Docker Release for strict scientific reproducibility. The chapter concludes by summarizing the primary technical contributions of the thesis, acknowledging the current limitations dictated by data availability, and outlining potential trajectories for future research and algorithmic refinement.

CHAPTER 2: LITERATURE REVIEW

2.1. Fundamentals of Peptides

Peptides are relatively small chains of amino acids that can be chemically synthesized or purified from living organisms. Within biological systems, organisms naturally produce peptides to carry out critical physiological functions, including healing, serving as chemical messengers, and providing defense against infections. Due to their high efficacy, target selectivity, and generally good tolerability, peptides are becoming highly relevant in medical applications and drug development. [5]

2.1.1 Antimicrobial Peptides (AMPs) and Their Mechanisms

Antimicrobial peptides (AMPs) are produced by most organisms as an innate immune response against microbes. These peptides represent a vast repertoire of antimicrobial agents and are highly valued for their broad-spectrum activity against bacteria, fungi, viruses, and transformed cancer cells. [3]

The primary mechanism of action for cationic AMPs relies on their structural conformation and electrostatic properties. These peptides often fold into amphiphilic alpha-helices, displaying distinct hydrophobic and hydrophilic surfaces. Through strong electrostatic forces, cationic AMPs bind to and accumulate at the negatively charged surfaces of Gram-negative or Gram-positive bacteria. This accumulation leads to the destabilization and permeation of the microbial surface. Upon breaching the outer defenses, AMPs interact with the cytoplasmic membrane by either carpeting the lipid bilayer or forming membrane-spanning channels, ultimately causing cell death. Additionally, some AMPs operate by targeting intracellular molecules, inhibiting crucial processes such as protein folding, enzymatic activities, and DNA transcription. [3]

2.1.2. Hemolytic Peptides and Mechanisms of Hemolysis

While AMPs offer significant therapeutic potential, their clinical application is severely hindered by peptide-associated hemolysis. Hemolysis is the destruction or lysis of red blood cells (RBCs), which releases intracellular products into the bloodstream, subsequently leading to systemic inflammation and widespread tissue damage. [5]

The mechanism of hemolysis is closely tied to the antimicrobial mechanism of action. Ideally, the positive charge of AMPs allows them to selectively disrupt the negatively charged membranes of bacteria while leaving the neutral membranes of mammalian cells intact. However, it has been demonstrated that many AMPs lack this perfect selectivity and can inadvertently disrupt mammalian cell membranes, causing RBC lysis. Consequently, there is a massive overlap between therapeutic and toxic properties; visual data mining reveals that 94.46% of peptides occupying the hemolytic chemical space also possess antimicrobial activity. [5]

Recent motif discovery efforts have identified specific sequence patterns—known as hemolytic motifs—that act as toxic signatures within a peptide chain. Sequence analysis demonstrates a positive association between the number of these hemolytic motifs present in a peptide and the statistical likelihood of that peptide exhibiting hemolytic toxicity. [5]

2.2. Traditional Machine Learning in Peptide Prediction

Historically, the *in silico* prediction of biological sequences relied heavily on traditional machine learning architectures. In this paradigm, algorithms such as Random Forest (RF) and Support Vector Machines (SVM) have been widely employed to classify peptides based on mathematically derived representations of their biological properties. The primary challenge in this approach is that traditional classifiers require fixed-length numerical inputs, whereas natural peptide sequences vary significantly in length. [1,2]

2.2.1. Manual Feature Extraction and Compositional Metrics

To transform variable-length amino acid sequences into fixed-length vectors suitable for traditional models, researchers developed extensive manual feature engineering techniques. Early feature sets primarily focused on calculating the frequencies of residues within a sequence. Two of the most common methods are Amino Acid Composition (AAC) and Dipeptide Composition (DPC). These methods generate arrays representing the global frequency of individual amino acids or adjacent amino acid pairs, respectively. [1]

While these compositional features are computationally inexpensive and perform adequately in baseline models, they have a fundamental limitation: they completely fail to capture the sequence order. The critical spatial arrangement of the peptide, which often dictates its biological function, is lost. To partially address this, researchers introduced advanced descriptors like pseudo-amino acid composition (PseAAC) and Composition-Transition-Distribution (CTD) to incorporate physicochemical properties and some degree of sequence correlation. [1]

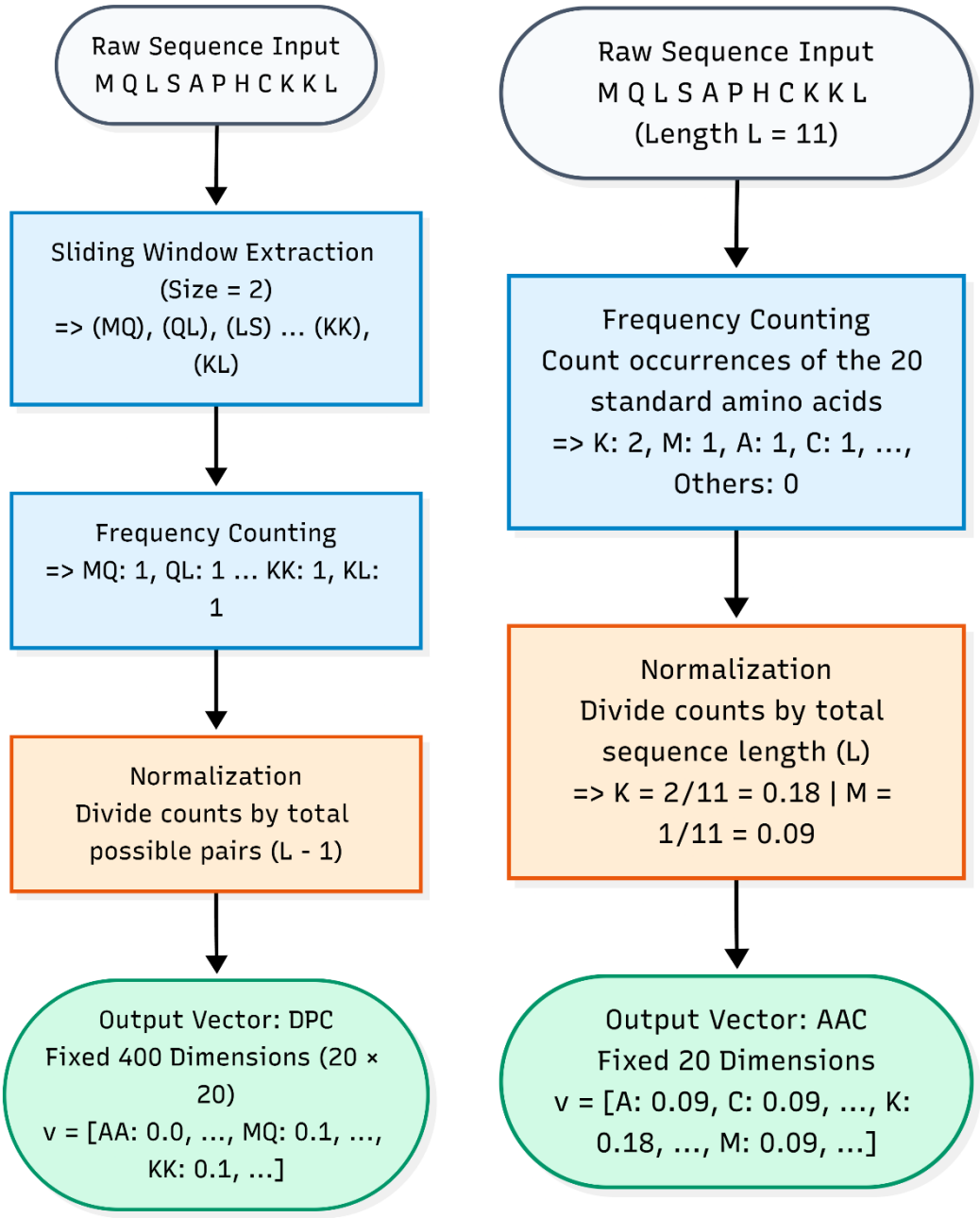


Figure 2.1: Workflow of the Dipeptide Composition (DPC) and Amino Acid Composition (AAC) feature extraction.

2.2.2. Evolutionary Information and PSSMs

Beyond simple composition, traditional pipelines attempted to capture the evolutionary context of peptides using Position-Specific Scoring Matrices (PSSMs). Generated via alignment tools like PSI-BLAST, a PSSM provides a matrix of scoring values that indicate how frequently specific amino acids are substituted at each position across a known protein family. [4]

Although PSSMs contain rich evolutionary data, utilizing them in traditional machine learning poses a structural problem. To conform to the strict fixed-input constraints of models like SVMs and RFs, these 2D evolutionary profiles are often flattened or summed together into 1D arrays (such as a fixed 20×20 matrix). This dimensionality reduction inevitably results in a severe loss of positional and sequential order information. [4]

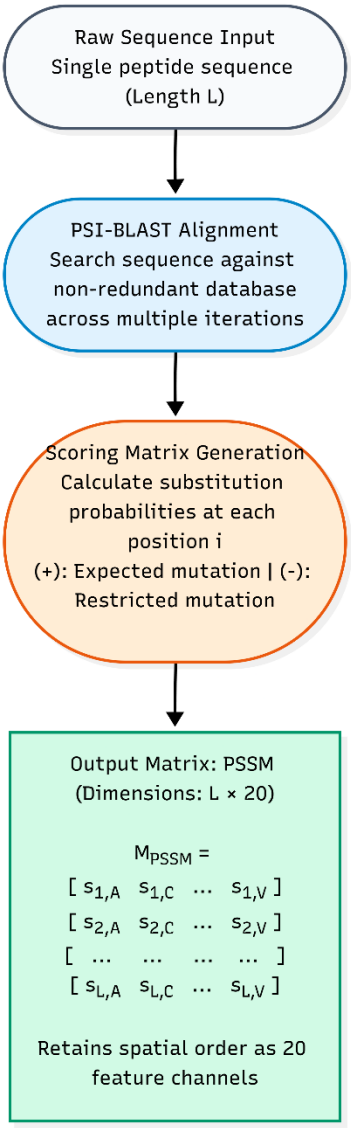


Figure 2.2: Workflow of the Position-Specific Scoring Matrix (PSSM) generation process, extracting evolutionary information and substitution probabilities while preserving the sequence's spatial order.

2.2.3. Limitations of the Traditional Pipeline

The primary bottleneck of traditional machine learning in peptide prediction is its heavy reliance on this manual feature engineering phase. Generating robust feature sets requires extensive time and expert biological knowledge. Furthermore, no matter how complex the hand-crafted features are, transforming sequences into flattened arrays inherently sacrifices the rich, multi-dimensional structural context of the peptide. These limitations directly motivated the shift toward deep learning architectures capable of automated representation learning. [1,4]

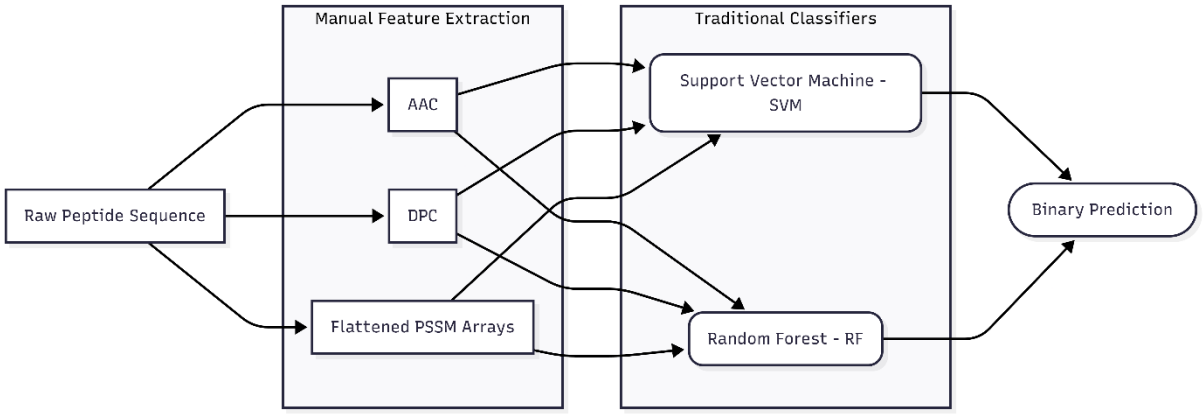


Figure 2.3: The traditional machine learning pipeline for peptide classification, highlighting the reliance on manual feature extraction to generate fixed-length inputs for standard statistical classifiers.

2.3. Representation Learning and Protein Language Models (PLMs)

The paradigm of biological sequence analysis has experienced a fundamental shift from traditional manual feature engineering to automated representation learning. In this modern approach, deep learning architectures automatically discover and extract the most relevant mathematical representations necessary for classification directly from raw amino acid sequences, bypassing the need for domain-specific, hand-crafted physicochemical features. [1,7]

2.3.1. Discrete Sequence Representations: One-Hot Encoding

Before complex contextual models can process peptide sequences, the alphabetical amino acid characters must be translated into a machine-readable numerical format. One-hot encoding has been widely utilized as a foundational discrete representation method. In this technique, each amino acid within a peptide is mapped to a unique binary vector. This transformation results in a two-dimensional matrix where the columns correspond to the 20 standard amino acids and the rows represent the sequential positions within the peptide chain. [6]

By assigning a value of 1 to the specific amino acid present at a given position and 0 to all other elements, convolutional and fully connected layers can process the sequence as numerical data. Despite its simplicity and lossless preservation of the primary sequence, one-hot encoding has significant limitations. The generated binary matrices are highly sparse and orthogonal. Consequently, the mathematical representation assumes all amino acids are equally distant from one another, completely failing to capture underlying physicochemical similarities (e.g., charge, hydrophobicity) or evolutionary relationships between different residues. Furthermore, while effective for identifying local features, this sparse representation struggles to capture long-range sequential dependencies. [6]

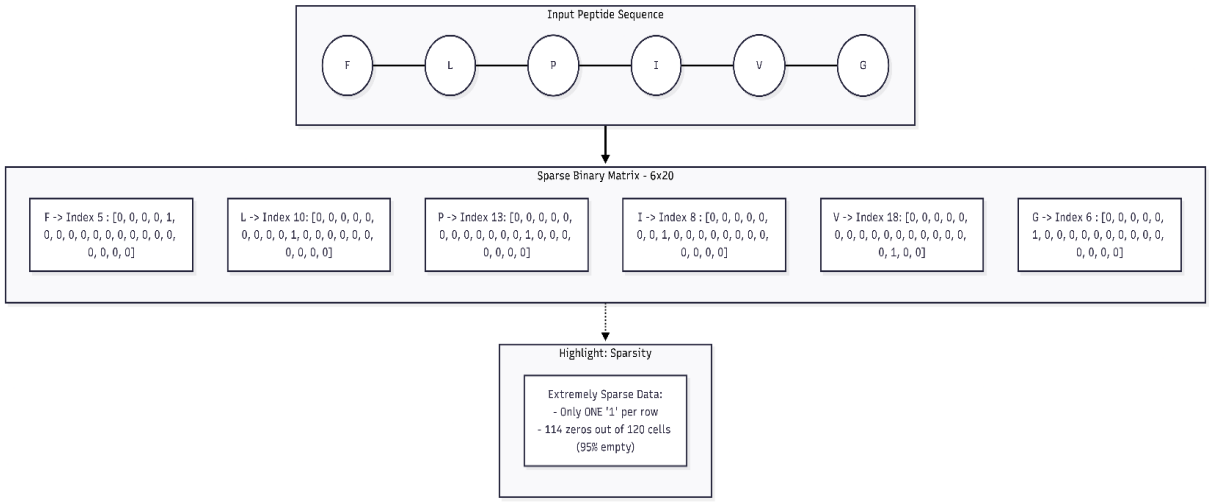


Figure 2.4: One-hot encoding representation of a peptide sequence, illustrating the mapping of individual amino acids into a sparse binary matrix.

2.3.2. Deep Contextual Embeddings and Protein Language Models (PLMs)

To transcend the limitations of orthogonal discrete representations, researchers initially applied Natural Language Processing (NLP) techniques, conceptualizing amino acids as "words" and entire sequences as "sentences". Early algorithms such as Word2Vec (Skip-Gram) and Bag of Words (BoW) were employed to capture basic semantic similarities between amino acids. However, these early NLP adaptations fundamentally failed to convey the complex contextual and structural information intricately encoded by the specific positional arrangement of amino acids within a sequence. [1]

This critical bottleneck was resolved with the introduction of Protein Language Models (PLMs) built upon the Transformer architecture. Unlike traditional models, PLMs are pre-trained on massive corpora of unlabeled protein databases—often encompassing millions or billions of sequences—utilizing self-supervised learning objectives such as masked amino acid prediction. For instance, models like Prot-BERT-BFD are trained on billions of protein fragments, learning to predict hidden amino acids based on surrounding context. Through this extensive pre-training, the self-attention mechanisms of the Transformer architecture learn to capture bi-directional context, generalizing the underlying "language of life". [1,7]

When applied to peptide sequences, PLMs like ESM-2 translate 1D character strings into dense, high-dimensional contextualized embeddings. These rich mathematical representations inherently encapsulate complex biochemical properties, local secondary structure propensities, and long-range evolutionary conservation without the need for manual feature calculation or explicit 3D structural simulations.

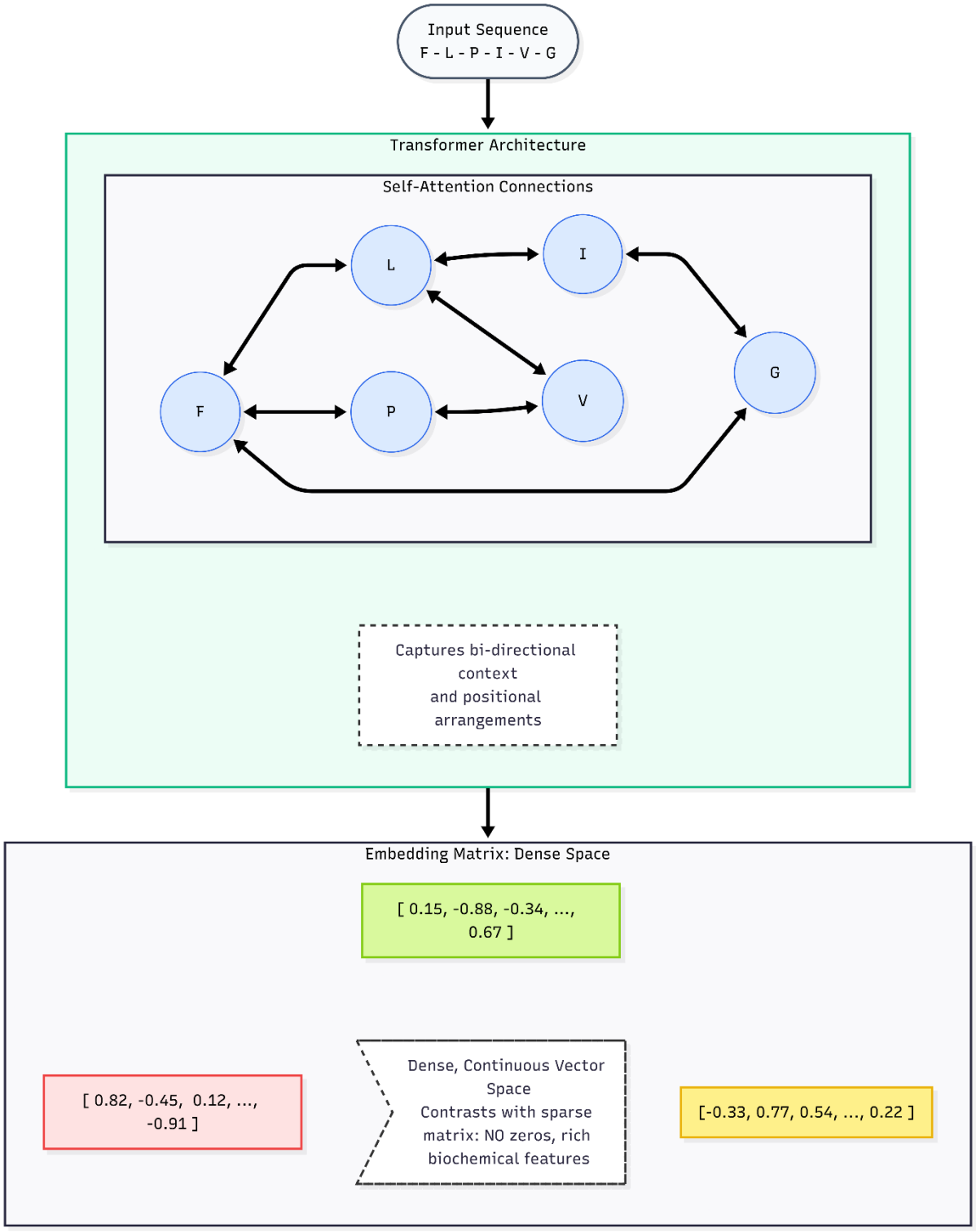


Figure 2.5: Generation of deep contextual embeddings via a Transformer-based Protein Language Model, highlighting the capture of bi-directional sequence dependencies

2.3.3. End-to-End Fine-Tuning vs. Selective Fine-Tuning

The integration of PLMs into classification pipelines typically follows one of two paradigms: utilizing the model as a static feature extractor or performing end-to-end fine-tuning.

In a full fine-tuning approach, the pre-trained language model is appended with a classification layer, and all internal parameters of the PLM are unfrozen and updated via stochastic gradient descent during training on a specific downstream task. While this allows the model to deeply adapt to specific sequence patterns, it introduces severe challenges in the domain of therapeutic peptides. State-of-the-art PLMs possess hundreds of millions of parameters, yet available datasets for tasks like hemolytic activity prediction are inherently small (often containing fewer than 3,000 data points). Fully fine-tuning such massive architectures on small datasets demands immense computational resources and significantly increases the risk of overfitting. [7]

To stabilize training and mitigate computational overhead, researchers have explored "selective fine-tuning". This methodology involves freezing the majority of the PLM's parameters—such as the complex self-attention heads and non-positional embeddings—while only unfreezing specific components like layer normalizations, positional embeddings, and the final classification layers. This strategy reduces the number of trainable parameters, effectively bridging the gap between the massive size of language models and the limited availability of biological data. [7]

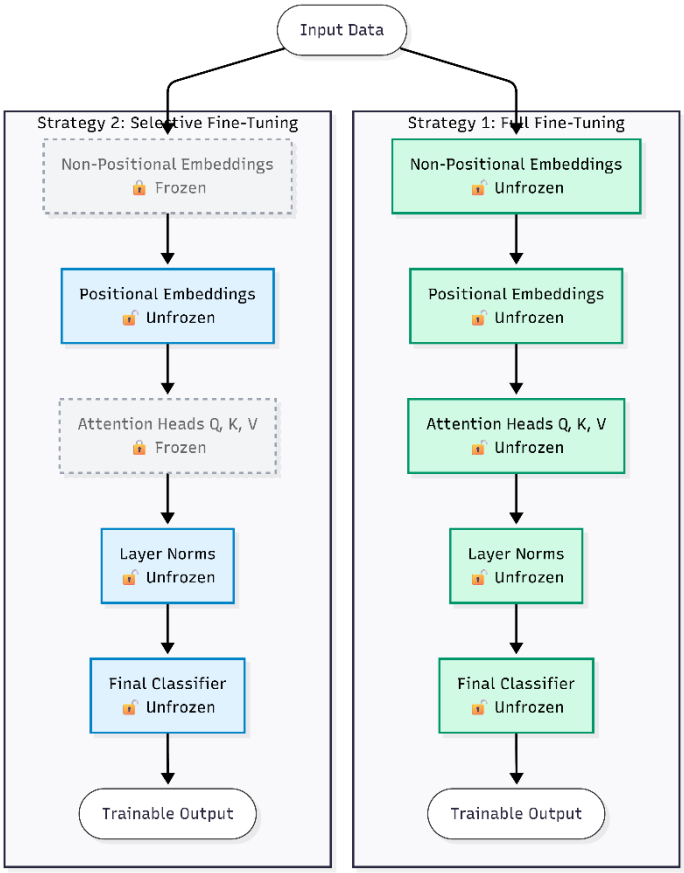


Figure 2.6: Architectural comparison of parameter freezing strategies during the fine-tuning of large-scale Protein Language Models on resource-constrained datasets.

2.4. Advanced Convolutional Architectures for Motif Scanning

While representation learning provides the necessary input format—whether as discrete binary matrices or dense continuous embeddings—the model architecture must be fully capable of interpreting it. Convolutional Neural Networks (CNNs) have proven highly successful in bioinformatics, acting as automated feature extractors that can capture the spatial and temporal dependencies between amino acids without requiring manual intervention. [1]

2.4.1. Standard 1D Convolutional Neural Networks

A standard 1D CNN processes an amino acid sequence by sliding a fixed-size mathematical filter (or kernel) across the data to detect local biochemical patterns. This localized scanning allows the network to identify sequence-order features that traditional compositional metrics (like AAC) completely miss. [1]

However, biological motifs—such as binding sites, structural helices, or toxicity signatures—do not possess a uniform length. A standard CNN utilizing a single, fixed kernel size encounters a significant biological limitation: it may miss wider structural contexts if the scanning window is too small, or fail to capture fine-grained local details if the window is too large. [4]

2.4.2. Multi-scale Convolutional Neural Networks (mCNNs)

To overcome the limitations of a fixed receptive field, the Multi-scale Convolutional Neural Network (mCNN) architecture was developed. Instead of relying on a single filter size, an mCNN deploys multiple convolutional layers with different window lengths operating entirely in parallel. By simultaneously scanning the input sequence with varying kernel sizes (for example, windows of 16, 24, and 32 amino acids), the network can efficiently detect multi-scale spatial dependencies and capture variable-length motifs in a single pass. [1,4]

Following this parallel convolution step, advanced architectures typically implement a pooling mechanism, such as a 1-max pooling layer. This operation aggressively filters the output of each convolutional window, retaining only the single maximum value. Biologically, this ensures that the most prominent and significant motif detected by each filter is preserved as a core feature for the final prediction, regardless of where that motif was physically located within the peptide sequence. [4]

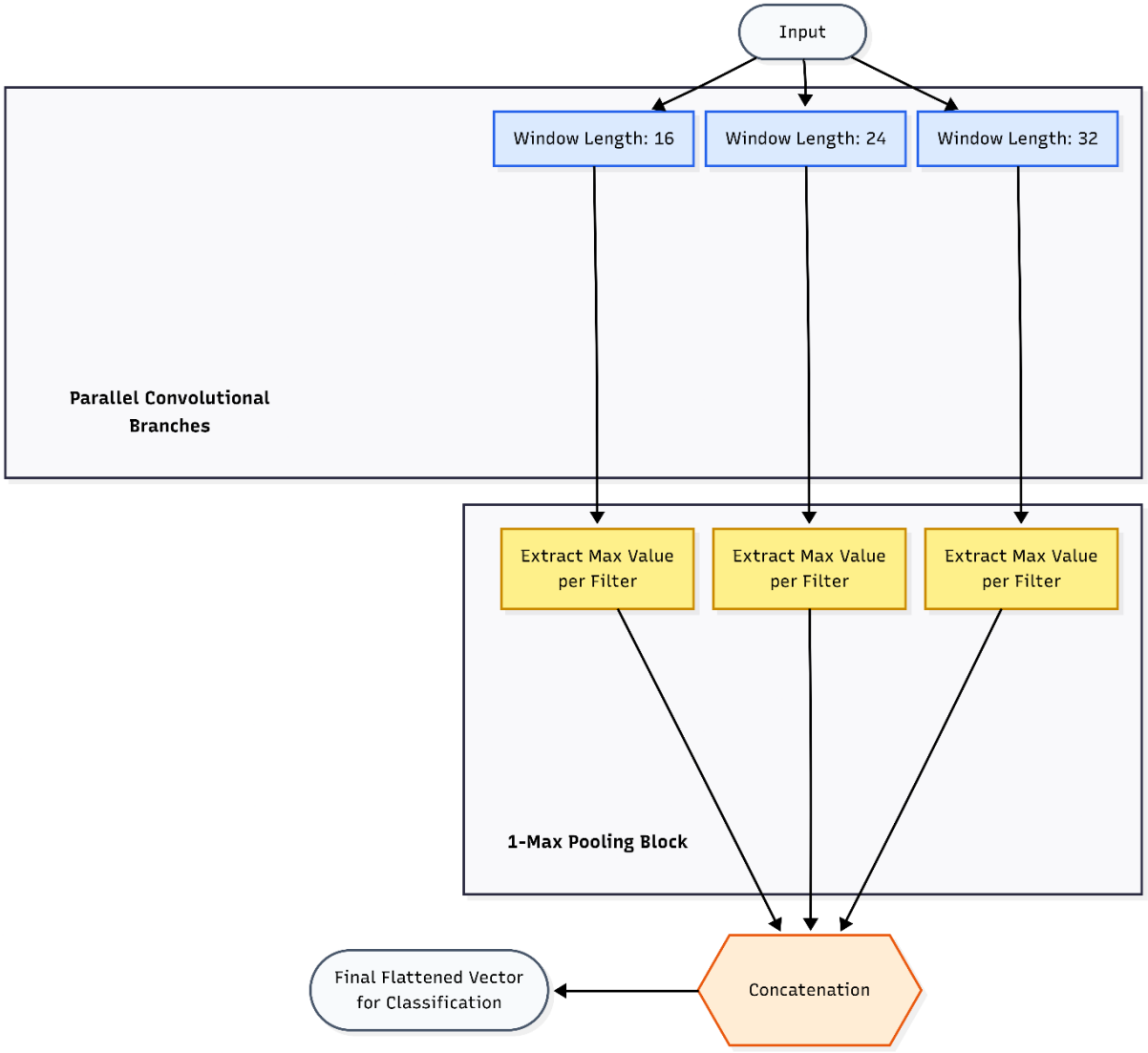


Figure 2.7: Architecture of a Multi-scale Convolutional Neural Network (mCNN) utilizing parallel variable-length windows and 1-max pooling for comprehensive motif extraction.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1. Overview of the Hybrid Cascade Architecture

To construct an efficient, resource-optimized, and highly interpretable *in silico* screening tool, this research proposes a "Simplified Waterfall Cascade" architecture. The system is designed to sequentially process raw amino acid sequences (up to 128 residues in length) through multiple deep learning and algorithmic checkpoints. By structuring the pipeline as a waterfall, the system strictly enforces conditional execution; subsequent computationally expensive assessments are only triggered if the sequence passes the preceding criteria, thereby conserving computational resources.

The proposed Hybrid Cascade Framework consists of three interconnected stages:

- **Stage 1: The Antimicrobial Gatekeeper (Primary Inference)** The initial stage processes the raw peptide sequence using a pre-trained ESM-2 35M model integrated with a Multi-scale Convolutional Neural Network (mCNN) classification head. This model evaluates the sequence's biophysical properties to determine its antimicrobial viability. If the model predicts the sequence as "NON-AMP" (probability ≤ 0.5), the sequence is immediately classified as inactive, and the pipeline is forcefully terminated under a "Resource Optimization Mode". Sequences that successfully pass this stage (probability > 0.5) are promoted to the next phase.
- **Stage 2: The Hemolytic Assessor and Motif Scanner** Sequences verified as AMPs undergo a stringent safety evaluation. This stage employs a second, independently trained ESM2-35M-mCNN model optimized specifically for toxicity prediction. Concurrently, the sequence is cross-referenced against a compiled database of 55 scientifically validated hemolytic and cytotoxic motifs. This dual-verification approach ensures that both the global contextual structure (analyzed by the PLM) and local toxic signatures (identified by regex pattern matching) are accounted for.
- **Stage 3: Explainable AI (XAI) and Wet-Lab Synthesis Recommendation** To provide transparency in the neural network's decision-making process, the system employs an Explainable AI (XAI) algorithm. This algorithm iteratively masks individual amino acids within the sequence and calculates the subsequent drop in prediction probability, thereby quantifying the importance of each residue. Based on these hotspot scores and the outputs of Stages 1 and 2, the system generates a final actionable recommendation: prioritizing safe AMPs for in vitro Solid-Phase Peptide Synthesis (SPPS) while advising against or cautioning the use of non-AMPs or highly hemolytic candidates.

- **High-Throughput Batch Screening Module** Alongside single-sequence Explainable AI analysis, the architecture features a high-throughput batch screening module designed for large-scale genomic mining. This module natively parses standard bioinformatics formats (FASTA) and tabular datasets (CSV). To maximize hardware efficiency and GPU utilization, sequences are tokenized and processed in parallel batches (e.g., 128 sequences per tensor). The cascade logic is strictly preserved during batch execution: the primary model filters the entire batch for AMPs, and only the positive candidates are subsequently routed to the Toxicity Assessor. The pipeline ultimately compiles the results into a structured format, isolating "Ideal Candidates" (sequences that are strictly AMP-positive and toxicity-negative). Furthermore, the system computes the global Amino Acid Composition (AAC) profile of the validated AMP hits, providing valuable statistical insights for downstream analysis and wet-lab synthesis.

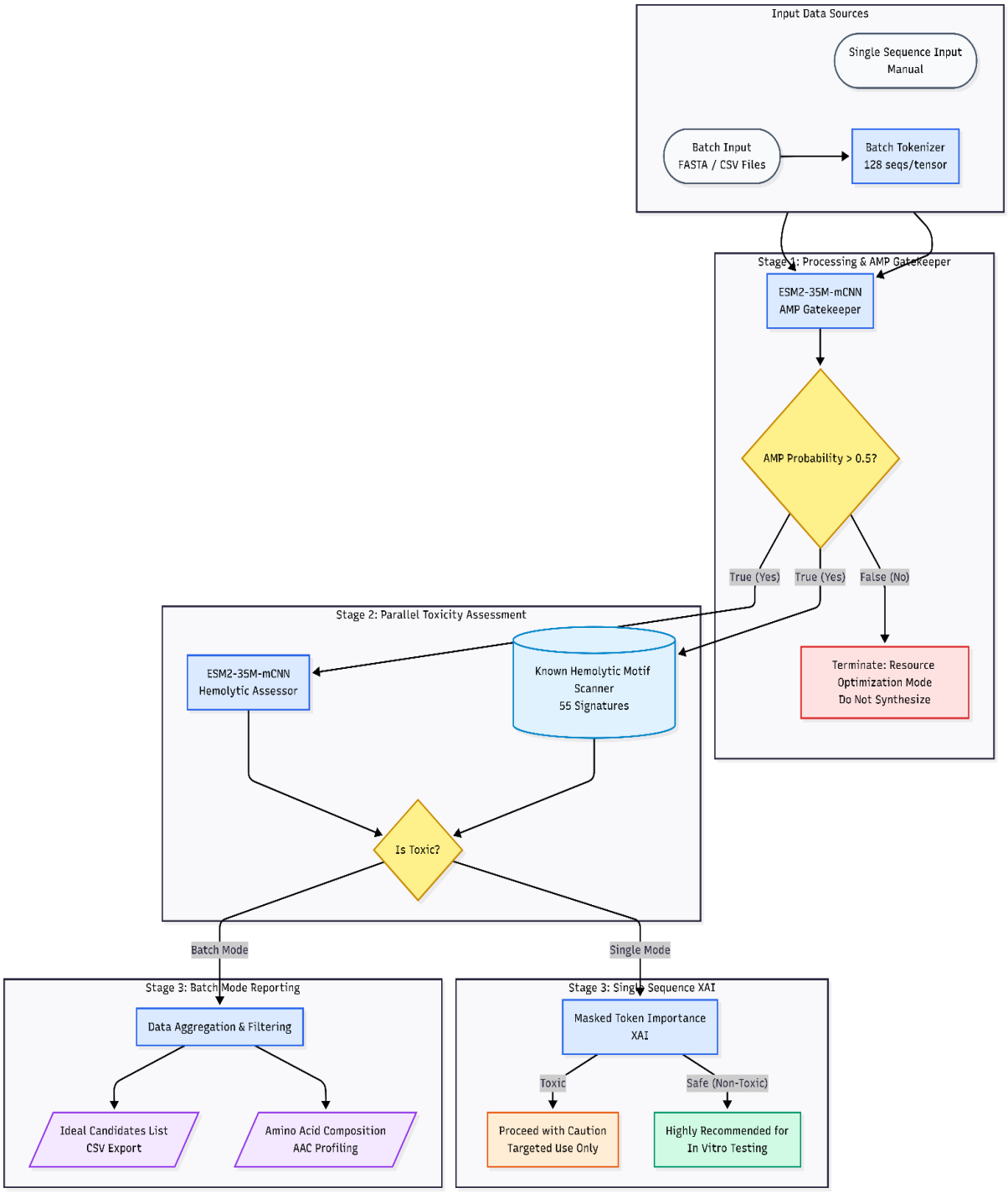


Figure 3.1: The Simplified Waterfall Cascade architecture of the proposed peptide screening pipeline, demonstrating the conditional execution flow from AMP Gatekeeping to Hemolytic Assessment and XAI extraction.

3.2. Dataset Curation and Preprocessing

The foundation of any robust deep learning framework lies in the quality, objectivity, and precise curation of its training data. To train the Hybrid Cascade Architecture effectively, data from multiple disparate sources must be aggregated, strictly filtered, and separated to fulfill the distinct binary classification objectives of Stage 1 (AMP Gatekeeper) and Stage 2 (Toxicity Assessor).

3.2.1. Full Dataset Collection and Homology Reduction

The raw sequence data for this research was aggregated from six primary sources. Five of these are established benchmark datasets commonly used in therapeutic peptide research: HemoPI-1[8], HemoPI-2[8], HemoPI-3[8], RNN-Hem[9], and Hlppredfuse[10]. The sixth source is the large-scale "Viri" dataset (ps29/amp_dataset_viri)[11], hosted on Hugging Face as Augmentation Subset, which provides extensive plant-based peptide sequences.

Because the cascade system executes two fundamentally different biological assessments, the data was rigorously partitioned using specific labeling logic:

I. Formulation of the AMP Dataset (Stage 1):

The objective of the first stage is to distinguish true Antimicrobial Peptides from non-functional sequences. To construct this dataset, positive labels (is_amp = 1) were assigned to all verified AMPs. Crucially, this includes both the highly toxic and the safe/weak AMPs found in HemoPI-2 and HemoPI-3, as well as the positive samples from HemoPI-1, RNN-Hem, Hlppredfuse, and Viri. Conversely, negative labels (is_amp = 0) were assigned to the background noise sequences (e.g., Swiss-Prot or randomly generated sequences) found in HemoPI-1, RNN-Hem, Hlppredfuse, and Viri.

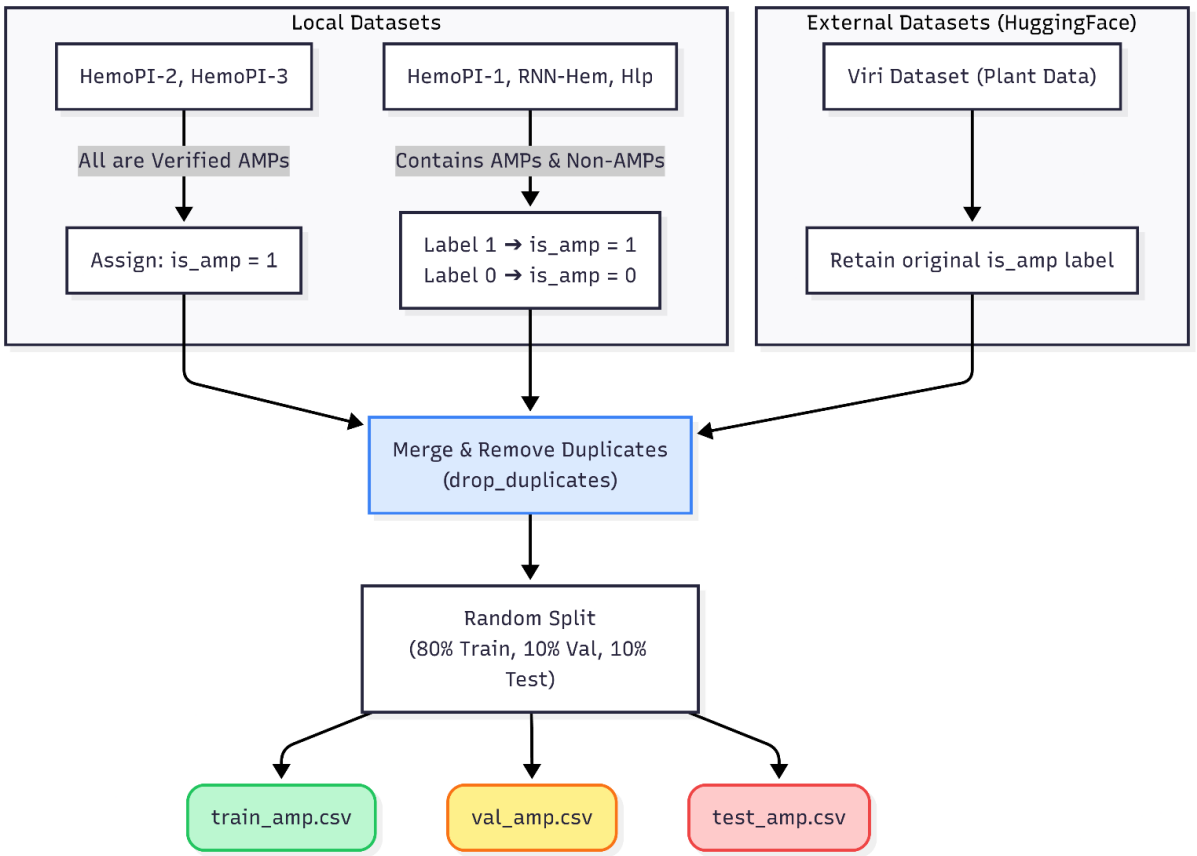


Figure 3.2: Data aggregation and splitting workflow for the AMP Gatekeeper (Stage 1).

II. Formulation of the Toxicity Dataset (Stage 2 - Strict Mode):

The objective of the second stage is to distinguish *Toxic AMPs* from *Safe AMPs*. A critical data engineering challenge in previous literature is the inclusion of non-AMPs in toxicity training sets, which causes neural networks to learn spurious correlations (i.e., learning to differentiate AMPs from non-AMPs rather than recognizing actual toxicity signatures). To prevent this, a "Strict Mode" filtering protocol was implemented.

- **Positive Class (Toxic):** All sequences labeled as '1' (hemolytic) across the five local datasets were retained as positive toxic samples.
- **Negative Class (Safe):** Sequences labeled as '0' were strictly filtered. Only the negative sequences from HemoPI-2 and HemoPI-3 were retained, as these exclusively represent verified AMPs with weak or no hemolytic activity.
- **Discarded Data:** All negative sequences from HemoPI-1, RNN-Hem, and Hlppredfuse were completely discarded, as they are non-AMP "imposters". Furthermore, the entire Viri dataset was excluded from this stage because the clinical toxicity of its sequences is unverified.

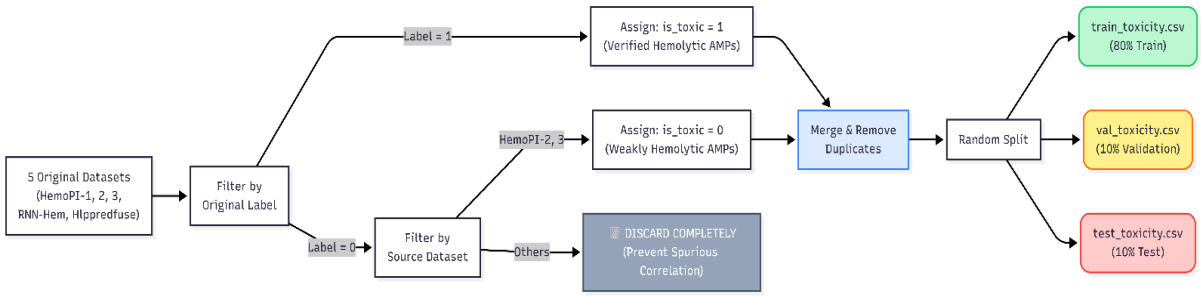


Figure 3.3: Strict-mode data filtering for the Toxicity Assessor (Stage 2), explicitly discarding non-AMP imposters.

Data Cleaning and Splitting:

Following the logical separation, sequences underwent a programmatic cleaning phase where specific string artifacts (e.g., b' prefixes, '$ suffixes, and whitespace) were stripped. To prevent data leakage and artificially inflated performance metrics, a strict homology reduction was applied using the drop_duplicates function, eliminating any 100% exact sequence matches across the merged datasets. Finally, the cleaned datasets were randomly split into 80% Training, 10% Validation, and 10% Testing subsets, and exported in both CSV and FASTA formats to facilitate downstream training and batch inference.

The held-out test sets were isolated before the data-centric intervention and were kept unchanged during length-matched negative sampling. They were used only for final model evaluation.

3.3. Scientific Justification for Proxy Task

3.3.1. The "Proxy Subset" Methodology

Training massive Protein Language Models (PLMs) such as ESM-2 across multiple architectural configurations requires extraordinary computational resources and time. To efficiently conduct this Architecture Search without incurring the prohibitive hardware costs of full-scale system training, a computationally economical "Proxy Subset" was deployed.

Rather than utilizing the full sequence corpus, this subset was generated by randomly extracting exactly 5,000 sequences from the pre-processed AMP training dataset (train_amp.csv), alongside exactly 1,000 sequences sampled from the validation dataset (val_amp.csv). To ensure scientific reproducibility across all ablation runs, this sampling process was mathematically controlled using a fixed algorithmic seed (random_state=42). Consequently, the training cycles for all configurations in this phase were capped at exactly 3 epochs.

3.3.2. Scientific Justification (Multi-Fidelity Architecture Search)

While 3 epochs on a 5,000-sequence subset may seem brief, this micro-benchmarking strategy—formally known in Deep Learning literature as a "Proxy Task" or "Multi-Fidelity Optimization"—is a mathematically rigorous approach for Neural Architecture Search. The objective of Phase 1 is not to achieve absolute peak global accuracy, but rather to conduct a rapid screening of various ESM-2 parameter sizes and mCNN window combinations.

By evaluating the early convergence trajectory of the proxy task, we can accurately measure the *relative* Return on Investment (ROI) of each architecture and effectively isolate the optimal configuration. Architectures that suffer from fundamental mathematical mismatches will exhibit performance collapse almost immediately, allowing us to identify bottlenecks without exhaustively processing the entire dataset.

3.4. Representation Learning and Motif Scanning Architectures

This section defines the core technological components evaluated in the experimental setup. The study is designed as a modular pipeline consisting of two main stages: **Feature Extraction** (generating representations from sequences) and **Downstream Classification** (scanning for motifs). By systematically pairing various extraction methods with different classifier heads, the study creates a comprehensive combinatorial framework to identify the optimal architecture.

3.4.1. Static Feature Extraction (Baselines)

To establish a fundamental performance benchmark, the first module of the pipeline utilizes traditional, sequence-based feature extraction methods. These include Amino Acid Composition (AAC), Dipeptide Composition (DPC), Position-Specific Scoring Matrix (PSSM), and One-hot encoding.

These static encodings are evaluated by pairing them with a diverse spectrum of classifiers. They are fed into classic machine learning algorithms—specifically Support Vector Machine (SVM) and Random Forest (RF)—as well as the advanced convolutional architectures defined in Section 3.4.3. This pairing strategy quantifies the absolute limits of traditional descriptors before introducing deep contextual models.

3.4.2. Contextual Representation Learning (ESM-2)

As a sophisticated alternative to static features, the pipeline integrates the Evolutionary Scale Modeling 2 (ESM-2) model family. To map out the bottleneck of information saturation, these Protein Language Models (PLMs) are implemented across a varying parameter spectrum: 8M, 35M, 150M, and 650M.

The ESM-2 models are tested under two distinct operational paradigms:

1. **Frozen Feature Extractor:** The pre-trained weights of the ESM-2 models are locked. The rich, high-dimensional embeddings they generate are then routed directly into the universal downstream convolutional architectures (detailed in Section 3.3.3) for a direct, controlled comparison against the baseline encodings.
2. **Fine-tuning End-to-End:** As an independent baseline for training cost, the entire ESM-2 architecture is unfrozen and trained seamlessly with its native linear classification head.

3.4.3. Universal Downstream Motif Scanners (1D CNN & mCNN) and Hyperparameter Rationales

These convolutional architectures serve as the universal downstream classification backend, consistently coupled with the features extracted from both static baseline descriptors (Section 3.4.1) and the frozen ESM-2 contextual embeddings (Section 3.4.2). The topological design and hyperparameter bounds of these networks are not arbitrarily chosen but are deeply rooted in mathematical principles, biological constraints, and rigorous empirical validation:

- **Single-branch 1D CNN:** Designed to evaluate localized sequence motif extraction using a single receptive field. The experimental setup sweeps through a continuous progression of strictly odd-numbered window sizes—specifically **K3, K5, K7, K9, and K11**.
    - *Rationale for Odd Kernels (Symmetric Padding):* The exclusive use of odd-numbered kernel sizes ensures the presence of a distinct central element within the convolutional filter. This mathematical property allows for symmetric zero-padding, which perfectly preserves the exact spatial alignment of amino acid residues without inducing translational shifts or spatial distortions during feature map generation [12], [13].

- **Multi-branch 1D CNN (mCNN):** Formulated to assess the representational power of parallel, multi-threaded motif scanning. The architecture consistently utilizes a tri-branch design strategically grouped into distinct field-of-view levels: **Micro (1, 3, 5), Standard (3, 5, 7), Macro (5, 7, 9), Mega (7, 9, 11), and Giga (9, 11, 13)**.
    - *Rationale for Tri-branch Architecture:* Heavily inspired by foundational multi-scale sequence processing models [14], [15], deploying exactly three parallel branches effectively captures a hierarchy of local features—spanning short, medium, and long-range structural motifs—while intrinsically preventing parameter bloat. Excessive branching drastically increases computational overhead and exacerbates the risk of overfitting on constrained biological datasets. The superiority of this multi-branch integration is empirically validated in our ablation studies; for instance, the AAC mCNN Micro configuration achieved an F1-Score of 0.7661, significantly outperforming the isolated AAC Single K3 configuration (F1-Score: 0.7058).
    - *Rationale for Receptive Field Upper Bounds:* The upper bound of the convolutional receptive field is strictly capped at a kernel size of 13 (the Giga configuration). Biologically, Antimicrobial Peptides (AMPs) are inherently short sequences, typically comprising only 10 to 50 amino acids [3], [16]. A kernel size exceeding 13 would either span the entire length of shorter peptides—failing to extract localized motifs—or encompass an excessive amount of zero-padding tokens, thereby introducing severe informational noise. Our ablation studies empirically corroborate this theoretical limitation. For example, deploying the ESM2_650M core reveals that increasing the single kernel size beyond K5 (Sensitivity: 0.9779) to K9 or K11 results in a performance drop and subsequent plateau (Sensitivity: 0.9716). Similarly, the ESM2_150M model peaks at an ROC-AUC of 0.9919 with K7 but regresses to 0.9912 when scaled up to K11.

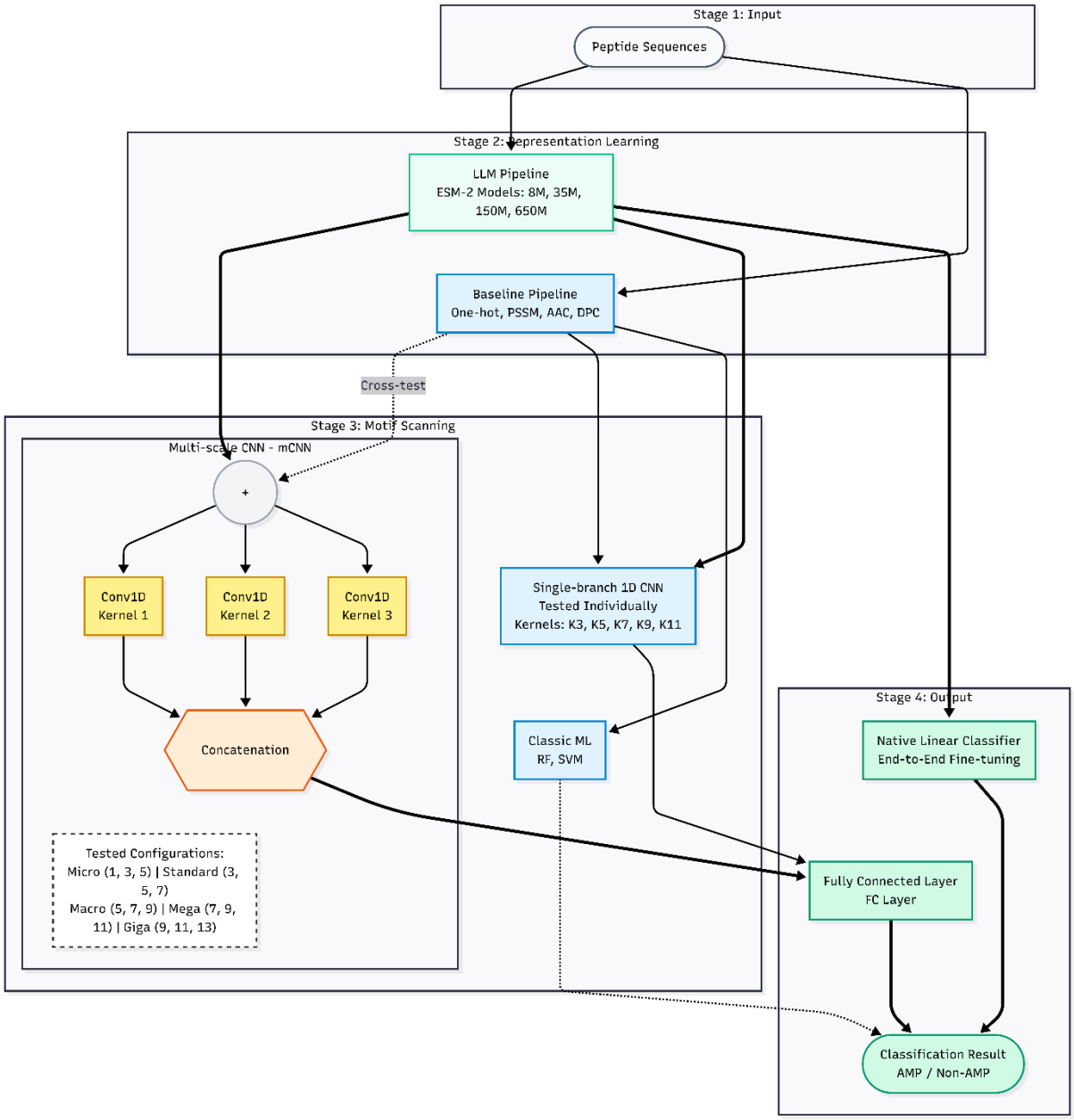


Figure 3.4: Modular pipeline architecture and combinatorial testing framework for peptide classification.

3.5. Progressive Architecture Evolution and Ablation Strategy

Rather than deploying a pre-determined complex architecture, this study adopts an iterative, step-wise approach to model evolution. The ablation strategy is designed to progressively increase computational complexity, evaluating the performance bottlenecks at each stage. This ensures that the final selected architectures are systematically evaluated rather than arbitrarily chosen. The evolution of the experimental setup follows a clear trajectory:

3.5.1. Stage 1: Shallow Baselines and Static Descriptors

The initial phase evaluates fundamental machine learning paradigms. Traditional sequence encoding techniques, including Amino Acid Composition (AAC), Dipeptide Composition (DPC), Position-Specific Scoring Matrix (PSSM), and One-hot encoding, are paired with standard classifiers such as Support Vector Machines (SVM) and Random Forests (RF). This stage aims to determine the absolute performance ceiling of shallow models relying solely on static, context-free sequence features.

3.5.2. Stage 2: Spatial Feature Extraction via Deep Learning

To overcome the limitations observed in the shallow baselines, the study introduces deep learning algorithms capable of capturing spatial relationships. The static descriptors are sequentially evaluated using a Single-branch 1D Convolutional Neural Network (1D CNN), and subsequently, a Multi-branch 1D CNN (mCNN). This transition tests the hypothesis that capturing localized sequence motifs at varying receptive fields significantly improves classification accuracy compared to traditional machine learning.

3.5.3. Stage 3: Contextual Paradigms and Hybrid Architectures

Driven by the need to capture deep evolutionary grammar, Protein Language Models (ESM-2) are introduced. The ESM-2 model is first evaluated as an isolated, end-to-end architecture. Upon analyzing its convergence behavior, the strategy shifts toward a hybrid paradigm. The generated ESM-2 embeddings are systematically paired with the previous classifiers (SVM, RF) and ultimately with the deep learning modules (Single 1D CNN and mCNN). This final hybridization step isolates the optimal pairing between state-of-the-art contextual embeddings and multi-scale motif scanning architectures.

3.6. Final Model Training and Dual-Task Deployment

Following the rigorous ablation study in Section 3.4, the most optimal architectures are selected for deployment. The objective of this final phase is to train robust, standalone predictive models tailored for two independent biological tasks: AMP identification and Toxicity prediction.

3.6.1. AMP Classification Task

The winning architectures (e.g., the top two performing ESM-2 + mCNN configurations) are trained from scratch on the complete, full-scale AMP dataset. The models in this phase are optimized to distinguish highly effective antimicrobial peptides from non-AMP sequences, forming the primary predictive engines of the study.

3.6.2. Toxicity Prediction Task and Transfer Justification

The architectures optimized during the AMP ablation phase are subsequently applied to the Toxicity prediction task. Since the physical mechanisms of membrane disruption in bacterial cells (AMPs) and mammalian red blood cells (Toxicity) share fundamental structural similarities, transferring the motif-scanning architecture (mCNN) from the AMP task to the Toxicity task is highly justified. Furthermore, due to the limited size and imbalanced nature of the Toxicity dataset, Focal Loss is explicitly integrated during training to heavily penalize misclassified minority classes, thereby preventing model overfitting and ensuring robust toxicity assessment.

3.7. Model Interpretability and Explainable AI (XAI)

A fundamental challenge in the application of Deep Learning within the biological sciences is the inherent opacity of deep neural networks. While architectures such as the ESM2-mCNN demonstrate exceptional predictive capabilities, their "black-box" nature obscures the specific biological mechanisms driving their classifications. To establish computational transparency and clinical trust, this research actively integrates an Explainable AI (XAI) mapping mechanism directly into the inference pipeline.

Technique and Mechanism:

To interpret the model's decision-making process, the system employs **Occlusion-based Sensitivity Analysis** (often referred to as Leave-One-Out Masking). Rather than relying on the raw, multi-headed attention matrices extracted from the Transformer layers—which can be mathematically dense and biologically ambiguous—this algorithm evaluates feature importance through systematic perturbation.

For a given peptide sequence of length L, the algorithm iteratively isolates each spatial position. During each iteration, a single amino acid token is replaced with a designated [MASK] (or [PAD]) token. The perturbed sequence is then passed through the full forward-propagation topology of the frozen ESM2-mCNN architecture.

Mathematical Scoring and Normalization:

The core of this interpretability framework relies on calculating the **"Logit Drop."** For each masked position i, the algorithm computes the differential between the original prediction logit and the structurally degraded prediction logit:

$$\Delta Logit_i = Logit_{original} - Logit_{masked}^{(i)}$$

A significant positive $\Delta Logit_i$ indicates that the targeted amino acid was highly crucial for the network's predictive confidence; its absence actively damages the model's ability to classify the sequence. Conversely, a negligible drop implies the residue acts as structural filler with minimal impact on the target biological property.

To transition these raw mathematical variances into an interpretable visual format, the array of logit drops across the sequence is subjected to Min-Max normalization, scaling the values into a continuous 0 to 1 range. This process dynamically generates a high-resolution 1D attention heatmap, effectively projecting the neural network's internal, high-dimensional algorithmic decisions back into the linear biological sequence space.

CHAPTER 4: EXPERIMENTAL RESULTS AND DISCUSSION

4.1. Data Collection and Corpus Composition

The foundation of any robust deep learning model relies on the quality and diversity of its training corpus. To ensure broad generalization, raw peptide sequences were aggregated from multiple well-established biological databases, including HemoPI, DBAASP, Swiss-Prot, and specialized plant data repositories. The data processing pipeline was strictly divided to cater to the dual-stage nature of this study.

4.1.1. Origin and Aggregation of the AMP Dataset

The objective of the first stage (AMP Gatekeeper) is to distinguish true Antimicrobial Peptides from non-AMP sequences. To achieve this, a massive aggregation strategy was employed, combining experimentally verified AMPs with a large corpus of background noise (Non-AMPs) to expose the model to maximum sequence diversity. The detailed composition of the raw AMP corpus is presented in Table 4.1.

Dataset Name	Source / Classification	AMP (is_amp = 1)	Non-AMP (is_amp = 0)
HemoPI-1	Experimental (Hemolytik / Swiss-Prot)	552	552
HemoPI-2	Categorized by Toxicity Level	1,014 *(552 toxic + 462 safe)*	0
HemoPI-3	Categorized by Toxicity Level (DBAASP)	1,623 *(885 toxic + 738 safe)*	0
RNN-Hem	DBAASP / Swiss-Prot	1,359	1,198
Hlppredfuse	Hemolytik / PEPred-SUITE	1,066	2,422
Viri Dataset (ps29)	Plant data (Toxicity unverified)	21,550	50,348

Table 4.1: Sources and distribution of raw data aggregated for the AMP classification task.

4.1.2. Filtering Protocol for the Toxicity Dataset

While the AMP dataset relies on massive aggregation, the Toxicity prediction task requires extreme clinical precision. The fundamental logic applied in this study is that *toxicity should only be assessed on peptides that are already verified as true AMPs*. Consequently, a strict filtering protocol was enforced to discard irrelevant background noise and unverified sequences. As detailed in Table 4.2, all non-AMP sequences (imposters) and plant-derived data (Viri dataset) were completely excluded, as their toxicity profiles on mammalian cells are unverified.

Dataset Name	Source	Toxic AMP (is_toxic = 1)	Safe AMP (is_toxic = 0)	Processing Notes (Based on Pipeline)
HemoPI-1	Hemolytik / Swiss-Prot	552	0	Completely discarded 552 label-0 sequences (Non-AMPs/Imposters).
HemoPI-2	Categorized by toxicity level	552	462	Retained all, as they are verified true AMPs.
HemoPI-3	Categorized by toxicity level	885	738	Retained all, as they are verified true AMPs.
RNN-Hem	DBAASP / Swiss-Prot	1,359	0	Completely discarded 1,198 label-0 sequences (Non-AMPs).
Hlppredfuse	Hemolytik / PEPred-SUITE	1,066	0	Completely discarded 2,422 label-0 sequences (Non-AMPs).
Viri Dataset	Plant data	0	0	Toxicity is unverified.

Table 4.2: Filtering protocol and source tracking for the Toxicity assessment task.

4.2. Dataset Statistics and Data Splitting

4.2.1. Global Distribution and Sequence Length Constraints

Following the aggregation and filtering protocols described above, the global class distribution and sequence length density were thoroughly analyzed. As depicted in **Figure 4.1**, the empirical distribution of the curated datasets reveals that the sequences naturally terminate at a maximum length of 100 amino acids. This observation aligns flawlessly with the biological consensus that AMPs are characteristically short peptides.

However, when configuring the neural network architectures, the maximum input dimension (for tokenization and zero-padding) was strategically standardized to **128 amino acids**. Rather than capping the network strictly at the current dataset's maximum, this 128-residue threshold acts as a crucial **structural buffer**. This design choice strikes an optimal balance for real-world deployment: it is sufficiently expansive to accommodate unseen, slightly extended peptides during future inference without triggering sequence truncation, yet compact enough to prevent excessive zero-padding. Excessive padding must be avoided, as it would mathematically dilute the localized motif features during CNN extraction and unnecessarily escalate computational overhead.

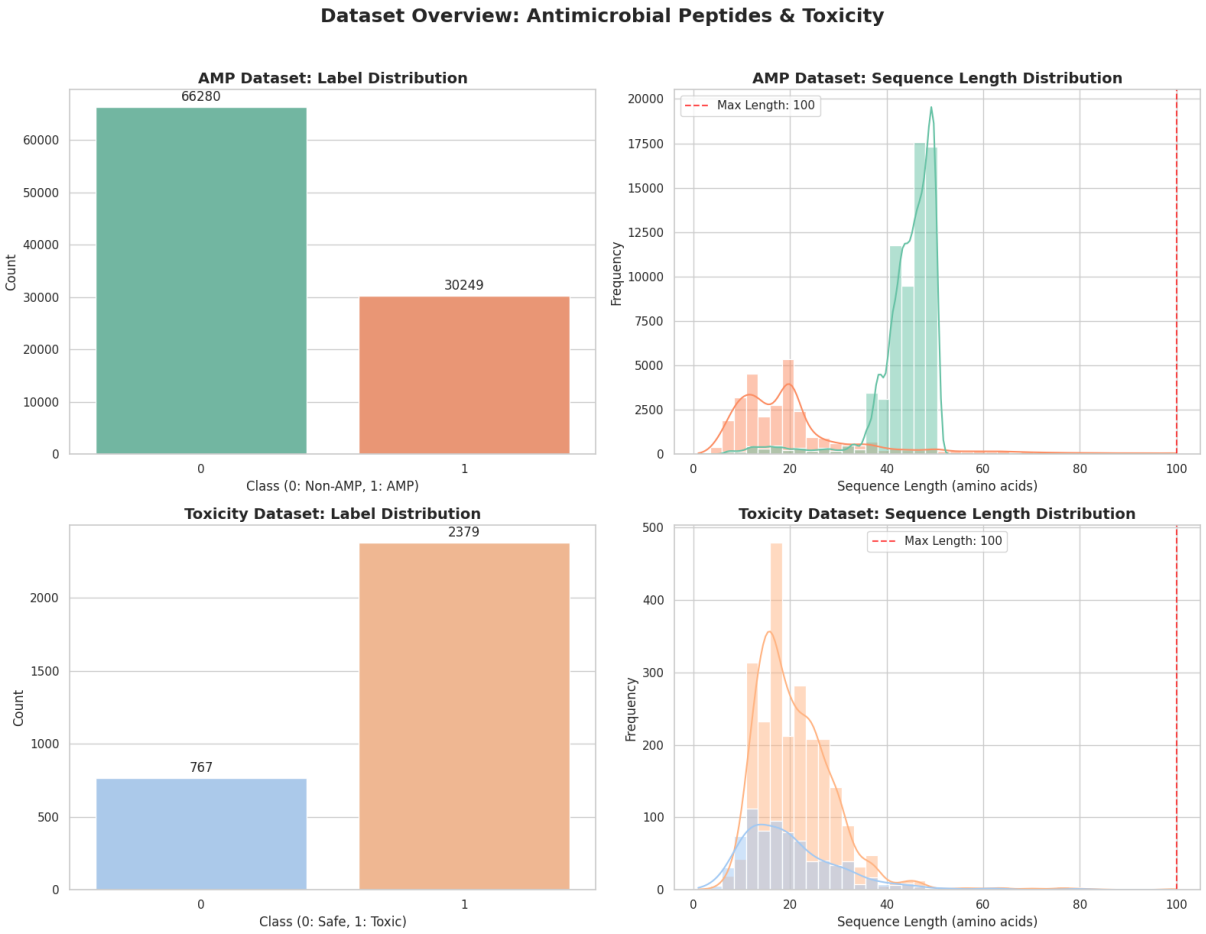


Figure 4.1: Global class distribution and sequence length density of the AMP and Toxicity datasets.

The pre-split class distribution reveals significant characteristics:

- **AMP Dataset:** The corpus contains 96,529 sequences, reflecting a moderate class imbalance with 66,280 Non-AMPs (Class 0) and 30,249 AMPs (Class 1).
- **Toxicity Dataset:** The refined corpus is exceptionally constrained, consisting of only 3,146 sequences (767 Safe vs. 2,379 Toxic). This severe scarcity and inverse class imbalance necessitate the deployment of architectural transfer and Focal Loss techniques in subsequent training phases.

Target Task	Total Sequences	Negative Class (0)	Positive Class (1)
AMP Classification	96,529	66,280 (Non-AMP)	30,249 (AMP)
Toxicity Prediction	3,146	767 (Safe)	2,379 (Toxic)

Table 4.3: Global Dataset Statistics prior to data splitting.

4.2.2. Post-Split Dataset Statistics

To rigorously evaluate model performance and prevent data leakage, both the AMP and Toxicity datasets were partitioned using an **80:10:10** ratio for Training, Validation, and Testing, respectively. Stratified sampling was applied to maintain the original class imbalance ratios across all subsets. The detailed statistical composition, including sequence length variations post-splitting, is presented in Table 4.4.

Dataset	Split Role	Ratio	Total Samples	Negative (0)	Positive (1)
AMP Dataset	Training	80%	77,223	53,053	24,170
	Validation	10%	9,653	6,659	2,994
	Testing	10%	9,653	6,568	3,085
Toxicity Dataset	Training	80%	2,516	604	1,912
	Validation	10%	315	86	229
	Testing	10%	315	77	238

Table 4.4: Detailed statistical breakdown of the Training, Validation, and Testing subsets.

4.3. Experimental Setup and Evaluation Metrics

4.3.1. Installation Environment and Hardware Configurations

The computational experiments, including the ablation studies and the final system deployment, were conducted across two distinct environments to balance massive training requirements with real-world application testing.

1. Cloud Training Environment (Ablation & Model Training) Due to the immense parameter scale of the ESM-2 model family (ranging up to 650M parameters), the core training and ablation screening were executed on the **Kaggle cloud computing platform**.

- **Hardware:** Dual NVIDIA Tesla T4 GPUs. While considered older-generation accelerators, the dual T4 setup provides a combined 32GB of VRAM (16GB each), which is an absolute necessity for accommodating the deep contextual embeddings and large batch sizes without encountering Out-Of-Memory (OOM) errors.
- **Software Stack:** Python 3.12 [17], PyTorch [18], HuggingFace transformers [19], datasets [20], and scikit-learn [21].

2. Local Deployment Environment (System Integration & Inference)

To validate the real-world feasibility of the final selected architectures and to construct the end-to-end classification system, the models were deployed on a local workstation. The system integration leveraged the highly efficient Python ecosystem, utilizing **FastAPI** [22] for the backend inference engine and **Streamlit** [23] for the frontend user interface.

- **OS (Operating System):** Windows 11 Pro 64-bit (10.0, Build 26200)
- **Processor (CPU):** Intel Core i5-14600KF (20 CPUs), ~3.5GHz.
- **Memory (RAM):** 32.0 GB (32768 MB).
- **Graphics Card (GPU):** NVIDIA GeForce RTX 3070 with 8.0 GB VRAM.

This hardware division ensures that the models are not only theoretically powerful when backed by cloud computing but also practically deployable on standard consumer-grade hardware.

4.3.2. Justification of Evaluation Metrics

To quantitatively assess the performance of the classification models, a standard confusion matrix is utilized, comprising True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The primary evaluation metrics are calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Sensitivity\ (Recall) = \frac{TP}{TP + FN}$$

$$F1\text{-}Score = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity}$$

The Biological and Computational Rationale:

While **Accuracy** is a common metric in standard machine learning, it is fundamentally flawed and heavily misleading when applied to imbalanced bioinformatics datasets (such as the Toxicity corpus, which contains a 3:1 ratio of toxic to safe peptides). Relying on Accuracy could result in a model that simply predicts the majority class while failing its primary objective. Therefore, this study prioritizes the following metrics:

- **Sensitivity (Recall) - The Ultimate Safety Standard:** In the context of predicting Antimicrobial Peptides and, more critically, Hemolytic Toxicity, a False Negative (classifying a toxic peptide as safe) has catastrophic clinical consequences. Sensitivity directly measures the model's ability to identify *all* actual positive cases without missing any. Maximizing Sensitivity ensures that no dangerous peptide slips through the screening pipeline.
- **F1-Score and ROC-AUC:** To prevent the model from blindly predicting all sequences as positive (which would yield 100% Sensitivity but terrible Precision), the **F1-Score** is employed as the harmonic mean, ensuring a strict balance. Furthermore, the **ROC-AUC** (Receiver Operating Characteristic - Area Under the Curve) serves as the supreme threshold-independent metric, evaluating the model's absolute discriminative capability between the positive and negative classes.
- **Training Time (Computational ROI):** Beyond predictive accuracy, the time required to train the model is introduced as an empirical metric. In Phase 1 (Ablation Study), Training Time is continuously plotted against Sensitivity and ROC-AUC to analyze the **Return on Investment (ROI)**. This allows for the selection of an architecture that achieves state-of-the-art performance while remaining computationally economical.

Note on Mathematical Notation: Throughout this thesis, the notation $\mathbb{R}^d$ is utilized to denote a d-dimensional real coordinate space. This is the standard mathematical convention in deep learning literature to explicitly define the shape and the continuous real-valued nature of extracted feature vectors and embedding matrices.

4.4. Phase 1 Results: Architecture Screening and Ablation Study

Experimental Setup Reminder: As established in the methodology, all configurations in this ablation study were evaluated on a controlled proxy subset (5,000 training and 1,000 validation sequences) over exactly 3 epochs. This constraint is strictly maintained to isolate architectural bottlenecks without exhaustive computational overhead.

4.4.1. Stage 1: Shallow Baselines and Static Descriptors

This initial stage evaluates the performance ceiling of static, context-free sequence representations Amino Acid Composition (AAC), Dipeptide Composition (DPC), and One-hot encoding when paired with standard Machine Learning classifiers: Support Vector Machines (SVM) and Random Forests (RF).

Feature	Classifier	Training Time (s)	Sensitivity	F1-Score	ROC-AUC
AAC ($\mathbb{R}^{20}$)	Random Forest	0.506	0.8899	**0.9188**	0.9843
AAC ($\mathbb{R}^{20}$)	SVM	2.119	0.8396	0.8725	0.9582
DPC ($\mathbb{R}^{400}$)	Random Forest	0.763	0.8773	0.9043	0.9862
DPC ($\mathbb{R}^{400}$)	SVM	10.811	0.8773	0.8942	0.9502
One-Hot ($\mathbb{R}^{2560}$)	Random Forest	1.579	0.8490	0.8794	0.9754
One-Hot ($\mathbb{R}^{2560}$)	SVM	121.278	0.8805	0.8888	0.9697

Table 4.5: Performance evaluation of static descriptors paired with Classic ML algorithms.

Technical Analysis & Dimensionality Bottleneck:

The empirical logs reveal strict computational and representational limits when processing peptide sequences using shallow models:

1. **Efficiency on Low-Dimensional Flat Vectors:** The Random Forest algorithm demonstrates high efficiency when handling global frequency distributions. The AAC + Random Forest configuration establishes a strong early baseline, achieving an F1-Score of 0.9188 with an extremely lightweight training time of 0.506 seconds.
2. **The Curse of Dimensionality for SVM:** As the feature space expands linearly and quadratically, the computational cost of the SVM classifier inflates exponentially. Moving from the compact AAC vector ($\mathbb{R}^{20}$) to DPC ($\mathbb{R}^{400}$), the SVM training time increases by a factor of 5 (from 2.11s to 10.81s).
3. **Scalability Failure on Spatial Matrices:** When processing the spatial One-Hot encoding—which requires flattening a 128×20 matrix into a high-dimensional vector of $\mathbb{R}^{2560}$—the SVM completely bottlenecks. The training time severely spikes to 121.278 seconds, yet the predictive accuracy stagnates (F1-Score: 0.8888), underperforming the simple AAC + Random Forest baseline.

Conclusion of Stage 1:

While Classic ML models can provide rapid classifications on heavily compressed, flat sequence descriptors (AAC/DPC), they lack the computational scalability to efficiently process dense, spatial sequence representations (One-Hot). To capture localized amino acid motifs without triggering the curse of dimensionality, the pipeline must transition to Convolutional Deep Learning architectures in Stage 2.

4.4.2. Stage 2: Spatial Feature Extraction via Deep Learning

Following the evaluation of shallow classifiers, Stage 2 transitions to Deep Learning architectures designed to capture localized amino acid motifs. This stage compares standard Single-branch 1D Convolutional Neural Networks (Single 1D CNN) against a Multi-branch 1D CNN (mCNN).

To systematically evaluate the impact of different receptive fields on peptide sequences, the dynamic mCNN architecture was parameterized into five distinct configurations based on their parallel kernel window sizes (K):

- **Micro:** $K \in \{1,3,5\}$ — Designed to capture ultra-short, adjacent sequence motifs.
- **Standard:** $K \in \{3,5,7\}$ — The default configuration for typical secondary structures (e.g., α-helices).
- **Macro:** $K \in \{5,7,9\}$ — Expanded receptive field for longer functional domains.
- **Mega:** $K \in \{7,9,11\}$ — Aggressive pooling for extended spatial relationships.
- **Giga:** $K \in \{9,11,13\}$ — Maximum theoretical window for global sequence scanning.

1. The Spatial Mismatch: Applying CNNs to Flat Descriptors

Convolutional filters mathematically rely on the spatial and topological arrangement of the input matrix. To empirically demonstrate the necessity of spatial order, both Single CNNs and all mCNN configurations were deliberately applied to static global frequency features: Amino Acid Composition (AAC) and Dipeptide Composition (DPC).

Feature	Model Architecture	Training Time (s)	Sensitivity	F1-Score	ROC-AUC
AAC	Single_K3	4.061	0.5849	0.7058	0.8715
	Single_K5	2.586	0.5691	0.6895	0.8624
	Single_K7	2.611	0.5786	0.6982	0.8632
	Single_K9	2.683	0.5628	0.6884	0.8514
	Single_K11	2.778	0.5534	0.6769	0.8542
	mCNN_Micro	5.024	0.6698	**0.7661**	0.8988

Feature	Model Architecture	Training Time (s)	Sensitivity	F1-Score	ROC-AUC
DPC	mCNN_Standard	3.326	0.6509	0.7527	0.8979
	mCNN_Macro	3.334	0.6226	0.7319	0.8880
	mCNN_Mega	3.286	0.6352	0.7399	0.8879
	mCNN_Giga	3.297	0.6383	0.7408	0.8827
	Single_K3	3.035	0.6132	0.7249	0.8371
	Single_K5	3.061	0.6194	0.7309	0.8386
	Single_K7	3.007	0.6163	0.7286	0.8381
	Single_K9	3.025	0.6226	0.7333	0.8403
	Single_K11	3.032	0.6194	0.7309	0.8370
	mCNN_Micro	3.479	0.6540	0.7495	0.8619
	mCNN_Standard	3.452	0.6540	0.7495	0.8605
	mCNN_Macro	3.427	0.6540	0.7495	0.8564
	mCNN_Mega	3.388	0.6572	**0.7531**	0.8568
	mCNN_Giga	3.404	0.6509	0.7472	0.8554

Table 4.6: Performance collapse when applying Convolutional Networks to non-spatial AAC and DPC features.

Technical Analysis:

The empirical logs highlight a catastrophic algorithmic mismatch across the board. Even when utilizing the highly parameterized mCNN architectures on both AAC and DPC, the F1-Score collapses to a maximum of ~0.76 (AAC_Micro). Because AAC and DPC are global frequency counts, the original sequential order of the amino acids is entirely destroyed. Scanning sliding convolutional windows across an unordered frequency vector extracts topological noise rather than biological grammar, proving that Deep Learning requires intact spatial configurations to function. This performance is a severe downgrade compared to the Random Forest baseline operating on the exact same AAC feature (F1-Score: 0.9188).

2. The Sparsity Bottleneck of One-Hot Encoding

To satisfy the spatial prerequisite of CNNs, One-Hot Encoding was evaluated. This technique preserves the sequence length, transforming each peptide into a sparse binary matrix ($\mathbb{R}^{128\times20}$). We benchmarked this representation across Classic ML (established in Stage 1) and the full spectrum of Convolutional architectures.

Feature	Model Architecture	Training Time (s)	Sensitivity	F1-Score	ROC-AUC
OneHot	Random Forest	1.579	0.8490	0.8794	0.9754
	SVM	121.278	0.8805	**0.8888**	0.9697
	Single_K3	34.326	0.7798	0.8507	0.9347
	Single_K5	32.656	0.7830	0.8498	0.9421
	Single_K7	32.727	0.7798	0.8464	0.9400
	Single_K9	33.392	0.7704	0.8404	0.9455
	Single_K11	33.002	0.7924	0.8527	0.9479
	mCNN_Micro	33.410	0.8238	**0.8718**	0.9458
	mCNN_Standard	33.394	0.8113	0.8672	0.9479
	mCNN_Macro	33.252	0.8144	0.8691	0.9486
	mCNN_Mega	33.681	0.8207	0.8714	0.9488
	mCNN_Giga	33.417	0.8238	**0.8718**	0.9516

Table 4.7: The performance ceiling of One-Hot Encoding across Shallow and Deep Learning architectures.

Technical Analysis:

Restoring the spatial sequence via One-Hot encoding significantly rehabilitates the neural network's performance, pushing the F1-Score from ~0.76 (with AAC/DPC) up to 0.8718 (with mCNN_Micro and mCNN_Giga). Furthermore, the multi-branch logic of the mCNN proves its superiority over Single CNNs; by combining multiple receptive fields, the mCNN effectively captures a richer hierarchy of motifs, consistently outperforming the Single-kernel models.

However, a strict performance ceiling is observed. The fundamental bottleneck here is *matrix sparsity*. Because the One-Hot matrix consists almost entirely of zeros with only a single active unit per row, it physically starves the parallel convolutional filters of contextual density. This sparsity prevents the mCNN from surpassing the raw predictive power of the SVM (F1-Score: 0.8888), despite the SVM requiring nearly 4 times longer to train.

Conclusion of Stage 2:

While the dynamic mCNN architecture proves mathematically superior to Single 1D CNNs in resolving spatial hierarchies, its potential is severely bottlenecked by the sparsity and lack of evolutionary context in static representations like One-Hot encoding. To fully unlock the parallel processing power of the mCNN without hitting this sparsity ceiling, the pipeline must discard static encoding and integrate high-density, context-aware PLM embeddings.

4.4.3. Stage 3: Contextual Paradigms and Protein Language Models

Having established the limitations of static descriptors and the sparsity bottlenecks of One-Hot encoding in CNNs, Stage 3 transitions to the core paradigm of this research: **Deep Evolutionary Context via Protein Language Models (ESM-2)**. Unlike static features that merely count frequencies or map spatial positions, ESM-2 processes peptide sequences as a biological language, extracting high-dimensional contextual grammar learned from millions of evolutionary protein structures.

1. Native End-to-End Fine-Tuning: The Unrestricted Baseline

To establish the raw predictive capability of the ESM-2 architecture, we first evaluated the models in a strictly **Native End-to-End Fine-Tuning** configuration. In this setup, the entire Transformer architecture is unfrozen, and gradients are computed and updated across all internal layers (from 8M up to 650M parameters) alongside a native linear classification head.

Run Configuration	Model Architecture	Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_8M	Native End-to-End	57.74	0.9559	0.9310	0.9904
ESM2_35M	Native End-to-End	103.18	**0.9654**	0.9388	0.9903
ESM2_150M	Native End-to-End	287.88	**0.9654**	0.9359	0.9917
ESM2_650M	Native End-to-End	805.60	0.9402	**0.9462**	**0.9936**

Table 4.8: Performance trajectory of Native End-to-End Fine-Tuning across scaling ESM-2 parameter sizes.

Technical Analysis: The Power and The Penalty

The empirical logs from the end-to-end evaluation reveal profound insights into the behavior of massive Transformer models when adapted for Antimicrobial Peptide classification:

- **The Paradigm Shift in Raw Power:** Even the smallest language model (**ESM2_8M**) operating natively achieves an F1-Score of 0.9310. This significantly exceeds the performance ceiling of all previous methodologies evaluated in Stage 1 and Stage 2 (where the highest recorded F1 was 0.9188 using AAC + Random Forest, and 0.8888 using One-Hot + SVM). This confirms that pre-trained evolutionary grammar is mathematically superior to context-free feature engineering.
- **The 150M Instability and Overfitting Risk:** A critical anomaly is observed at the 150-million parameter scale. While the model size increases exponentially from 35M to 150M, the predictive performance actually regresses (F1-Score dropping from 0.9388 to 0.9359). When fully unfreezing a massive Transformer and fine-tuning it on a constrained dataset (5,000 sequences), the model is highly susceptible to *catastrophic forgetting*—unlearning its robust pre-trained biological representations and beginning to overfit the local training distribution.
- **The Computational Explosion:** While the **ESM2_650M** model ultimately forces its way to the highest F1-Score (0.9462), it incurs a staggering computational penalty. The training time sharply spikes to 805.60 seconds—a **nearly 800% increase** in compute time compared to the 35M model (103.18 seconds), in exchange for a marginal +0.0074 gain in F1-Score.

Interim Conclusion for End-to-End Tuning:

Native End-to-End fine-tuning proves the strong representation capabilities of ESM-2's contextual embeddings. However, unfreezing hundreds of millions of parameters for a specialized downstream task like AMP classification is mathematically inefficient and computationally prohibitive. This severe Return on Investment (ROI) bottleneck necessitates a Hybrid Paradigm: extracting frozen ESM-2 representations and projecting them into dynamic, lightweight multi-scale Convolutional Networks (mCNN).

2. Frozen ESM-2 with Classic ML: The Mean Pooling Bottleneck

Given the severe computational penalty and overfitting risks associated with Native End-to-End fine-tuning, the study evaluated a computationally economical alternative: utilizing the pre-trained ESM-2 architectures strictly as frozen feature extractors. The continuous high-dimensional output representations were subsequently passed to standard Machine Learning classifiers (Random Forest and SVM).

Because Classic ML algorithms require fixed-size 1D arrays, the variable-length spatial matrix of the transformer output ($\mathbb{R}^{L \times D}$, where L is sequence length and D is the embedding dimension) had to be compressed using a **Mean Pooling** operation to generate a global representation vector ($\mathbb{R}^{1 \times D}$) for each peptide.

Run Configuration	Embedding Dimension	ML Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_8M_RandomForest	$\mathbb{R}^{320}$	4.611070	**0.940252**	**0.928571**	0.984099
ESM2_8M_SVM	$\mathbb{R}^{320}$	5.216835	0.930818	0.920684	0.982907
ESM2_35M_RandomForest	$\mathbb{R}^{480}$	5.108846	0.933962	0.928125	0.984689
ESM2_35M_SVM	$\mathbb{R}^{480}$	7.394060	0.927673	0.916149	0.981187
ESM2_150M_RandomForest	$\mathbb{R}^{640}$	6.952642	0.930818	0.922118	0.984337
ESM2_150M_SVM	$\mathbb{R}^{640}$	8.533945	**0.940252**	0.924266	0.983723
ESM2_650M_RandomForest	$\mathbb{R}^{1280}$	8.572562	0.921384	0.919937	**0.987500**

Run Configuration	Embedding Dimension	ML Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_650M_ SVM	$\mathbb{R}^{1280}$	21.744862	0.927673	0.921875	0.982755

Table 4.9: Performance evaluation of Mean-Pooled ESM-2 embeddings paired with Classic Machine Learning algorithms.

Technical Analysis: Spatial Destruction vs. Computational Speed

1. **Extreme Computational Efficiency:** The immediate advantage of this frozen paradigm is speed. Extracting features and training a Random Forest on the 8M model takes a mere 4.61 seconds—a drastic reduction compared to the 57.74 seconds required for Native End-to-End tuning on the same model architecture.
2. **The "Mean Pooling" Performance Ceiling:** Despite the high quality of the ESM-2 representations, the predictive performance across all models and classifiers stagnates around an F1-Score of ~ 0.928 (peaking at the 8M_RandomForest configuration). This hard mathematical ceiling is a direct consequence of the Mean Pooling operation. By computing the average across the entire sequence axis, the algorithm inherently destroys the strict topological arrangement of the amino acids. Localized structural motifs critical for AMP identification are "blurred" into a single global summary vector.
3. **Dimensionality Curse on Massive PLMs:** A counter-intuitive phenomenon occurs when scaling up the PLM sizes. The 650M model generates massively rich, 1280-dimensional vectors, yet its downstream Random Forest F1-Score drops to 0.9199. Concurrently, the SVM training time spikes by over 400% (from 5.21s for the 8M model up to 21.74s for the 650M model). This proves that feeding hyper-dense, pooled representations into shallow classifiers introduces excessive noise and re-triggers the curse of dimensionality without providing additional predictive boundaries.

Interim Conclusion for Classic ML Coupling:

While coupling frozen ESM-2 embeddings with Classic ML bypasses the computational bloat of End-to-End training, it fails to fully exploit the spatial grammar of the protein sequences due to the destructive nature of Mean Pooling. To preserve the sequential arrangement ($\mathbb{R}^{L \times D}$) while avoiding catastrophic forgetting, the embeddings must be routed through architectures explicitly engineered for spatial scanning.

3. Spatial Preservation: The Geometric Instability of Single-Scale Convolutions

To overcome the spatial destruction caused by Mean Pooling, the raw, unpooled sequence embeddings ($\mathbb{R}^{L\times D}$) from the frozen ESM-2 models were fed directly into Convolutional Neural Networks. We initiated a rigorous Bottleneck Test using a Single-branch 1D CNN, systematically sweeping the receptive field from an ultra-short motif window ($K = 3$) to an extended domain window ($K = 11$).

Run Config	Kernel Size	Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_8M_K_3	K_3	13.026643	0.952830	0.928025	0.989955
ESM2_8M_K_5	K_5	11.107634	0.952830	0.929448	0.990677
ESM2_8M_K_7	K_7	10.972988	0.949686	0.932099	0.990425
ESM2_8M_K_9	K_9	11.103652	0.959119	0.934150	0.990453
ESM2_8M_K_11	K_11	11.252143	0.952830	0.932308	0.990476
ESM2_35M_K_3	K_3	26.303194	0.959119	0.935583	0.990204
ESM2_35M_K_5	K_5	26.829637	0.955975	0.939722	0.989563
ESM2_35M_K_7	K_7	26.422578	0.959119	0.938462	0.989681
ESM2_35M_K_9	K_9	26.423699	0.959119	**0.941358**	0.989967
ESM2_35M_K_11	K_11	26.645523	0.943396	0.930233	0.989275
ESM2_150M_K_3	K_3	80.249531	0.959119	0.935583	0.991516
ESM2_150M_K_5	K_5	80.209242	0.965409	**0.943164**	0.991876
ESM2_150M_K_7	K_7	79.941612	0.965409	**0.943164**	0.991901
ESM2_150M_K_9	K_9	80.218890	0.962264	0.935780	0.991029
ESM2_150M_K_11	K_11	80.156801	0.959119	0.931298	0.991242
ESM2_650M_K_3	K_3	248.990804	0.974843	0.949464	0.991601

Run Config	Kernel Size	Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_650M_K_5	K_5	248.748305	0.977987	**0.952527**	0.991424
ESM2_650M_K_7	K_7	248.127637	0.968553	0.946237	0.991504
ESM2_650M_K_9	K_9	248.812184	0.971698	0.944954	0.990688
ESM2_650M_K_11	K_11	248.799963	0.971698	0.944954	0.991537

Table 4.10: Single Kernel Performance Comparison across scaling ESM-2 models.

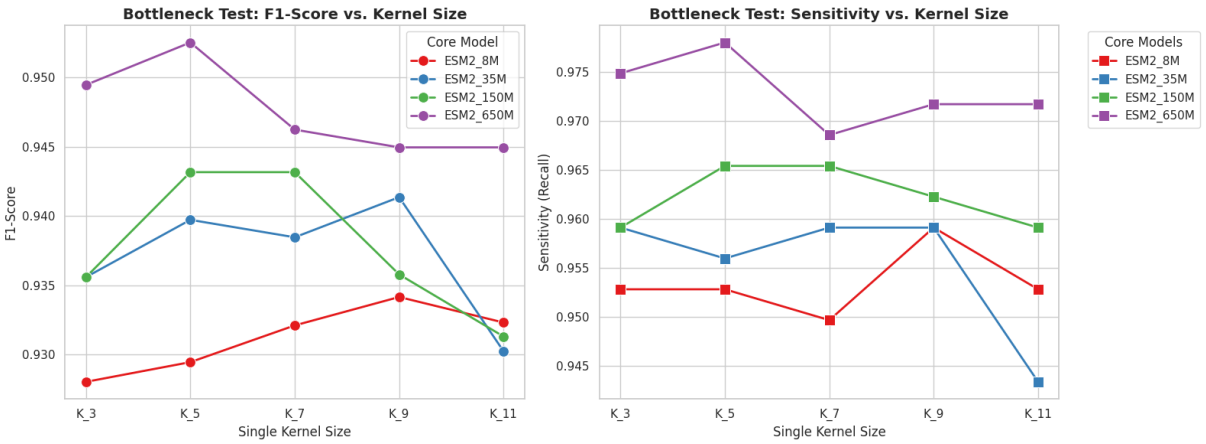


Figure 4.2: Bottleneck Test: F1-Score vs. Kernel Size

Technical Analysis: The "Nosedive" Phenomenon and Feature Mismatch

The empirical data exposes a critical architectural vulnerability when applying single-scale convolutions to complex biological representations:

1. **Immediate Performance Recovery:** Compared to the flat ML baseline, simply preserving the spatial order allows the models to perform significantly better. For example, processing the 35M embeddings through a $K = 9$ convolution yields a peak F1-Score of 0.941358, fundamentally outperforming the Random Forest's 0.9281.
2. **Geometric Instability (The Nosedive):** As clearly depicted in the plotted trajectories, scanning the sequence with a single, rigid kernel size forces the network to hunt for motifs of an exact, fixed length. This creates severe instability. The **ESM2_35M** model reaches its zenith at $K = 9$, but when the window is expanded slightly to $K = 11$, the F1-Score experiences a sharp nosedive down to 0.930233.

3. **Variable Motif Dependencies:** This instability demonstrates that Antimicrobial Peptides do not rely on a single structural dimension. Some crucial biological signals (like tight α-helical turns) require narrow receptive fields ($K = 3,5$), while broader β-sheet regions require extended scanning ($K = 9,11$). A single kernel network is structurally incapable of decoding both simultaneously.

Interim Conclusion for Single CNN:

While Single 1D CNNs preserve spatial integrity and prevent catastrophic forgetting, their rigid architecture cannot capture the multi-scale hierarchy of peptide grammar. This geometrical limitation directly mandates the final architectural evolution: deploying a multi-branch, dynamic CNN (mCNN).

4.4.4. Resolving Geometric Instability: The Multi-Scale CNN (mCNN) Hybrid

To counter the geometric instability of single-scale convolutions and the spatial destruction of Mean Pooling, the final architectural evolution deploys the dynamic Multi-branch CNN (mCNN). By processing the frozen, unpooled ESM-2 sequence embeddings ($\mathbb{R}^{L\times D}$) through parallel convolutional branches (e.g., K=3, 5, 7 simultaneously), the network can theoretically capture both ultra-short α-helical turns and extended β-sheet regions without being forced into a rigid receptive field.

To rigorously evaluate the Return on Investment (ROI) and identify the architectural "sweet spot," the mCNN configurations (Micro, Standard, Macro) were applied to the 8M, 35M, and 150M models. To verify the absolute limits of extended spatial scanning, the massive 650M model was subjected to the full spectrum of configurations, including the extreme Mega and Giga windows.

Run Combination	mCNN Window	Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_8M_Micro	Micro	16.307420	0.949686	0.934985	0.989413
ESM2_8M_Standard	Standard	14.106914	0.949686	0.929231	0.990202
ESM2_8M_Macro	Macro	14.466062	0.952830	0.936631	0.990202
ESM2_35M_Micro	Micro	37.924598	0.955975	0.941176	0.990017
ESM2_35M_Standard	Standard	40.178414	**0.962264**	**0.941538**	0.990096
ESM2_35M_Macro	Macro	42.448711	0.959119	0.939908	0.989441
ESM2_150M_Micro	Micro	145.563229	0.962264	0.935780	0.991253
ESM2_150M_Standard	Standard	146.627788	0.959119	0.938462	0.991825
ESM2_150M_Macro	Macro	147.757813	0.952830	0.930876	0.991345

Run Combination	mCNN Window	Training Time (s)	Sensitivity (Recall)	F1-Score	ROC-AUC
ESM2_650M_Micro	Micro	256.847239	0.968553	0.949153	0.992346
ESM2_650M_Standard	Standard	261.616190	0.962264	0.942989	0.991693
ESM2_650M_Macro	Macro	262.178564	**0.981132**	**0.951220**	0.991816
ESM2_650M_Mega	Mega	254.653601	0.974843	0.949464	0.992014
ESM2_650M_Giga	Giga	257.389429	0.959119	0.942813	0.991931

Table 4.11: Performance Evaluation of the Dynamic mCNN Hybrid Architecture across scaling ESM-2 models.

1. The Ultimate Sweet Spot (Efficiency vs. Performance)

The **ESM2_35M_Standard** configuration emerges as the absolute optimal mathematical and computational balance. It achieves a highly robust F1-Score of 0.9415 in just ~40.17 seconds of training time. The parallel branches ($K \in \{3,5,7\}$) successfully capture both short α-helical turns and regional motifs without triggering the curse of dimensionality or computational bloat.

2. Spatial Overfitting & The "Early Stopping" Heuristic

A critical methodological decision in this phase was the intentional omission of the Mega ($K \in \{7,9,11\}$) and Giga ($K \in \{9,11,13\}$) configurations for the 8M, 35M, and 150M models.

As the empirical logs demonstrate, scaling the base model to 150M parameters actively degrades performance compared to the 35M variant (F1 dropping to 0.9384).

Furthermore, expanding the window to the Macro setting on the 150M model drops the F1-Score further down to 0.9308.

Mathematical deduction: If the spatial resolution of a lower-dimensional embedding ($\mathbb{R}^{320}$ to $\mathbb{R}^{640}$) is already breaking down and extracting noise at a maximum kernel of $K = 9$ (Macro), forcing it through $K = 11$ (Mega) or $K = 13$ (Giga) will only exacerbate spatial overfitting. Therefore, a strictly controlled **Early Stopping** heuristic was applied to prevent computational waste.

3. Absolute Peak Performance vs. The Giga Collapse

To verify the absolute limits of extended spatial scanning, only the massive 650M model (producing a hyper-dense $\mathbb{R}^{1280}$ embedding matrix) was subjected to the extreme Mega and Giga windows.

While the **ESM2_650M_Macro** configuration achieves the absolute peak F1-Score of the entire study (0.9512), this comes at an exorbitant computational cost (~262.17 seconds). More importantly, when pushed into the Mega and Giga configurations, the network suffers from severe diminishing returns, with the F1-Score collapsing to 0.9428. This geometric failure proves that across *all* PLM scales, excessively wide parallel kernels blur localized antimicrobial signals into indecipherable topological noise.

4.5. Return on Investment (ROI) Analysis and Final Architecture Selection

To conclude the ablation study and select the candidate architectures for full-scale dataset training, a visual **Return on Investment (ROI)** analysis was conducted. In deep learning architecture search, the ROI is evaluated by plotting the computational cost (Training Time on the X-axis) against the predictive yield (Sensitivity and ROC-AUC on the Y-axis). The ideal configuration must reside in the upper-left quadrant—maximizing predictive power while minimizing computational overhead.

(Note: The extreme Mega and Giga configurations are excluded from these plots due to their previously established geometric bottleneck and severe performance degradation).

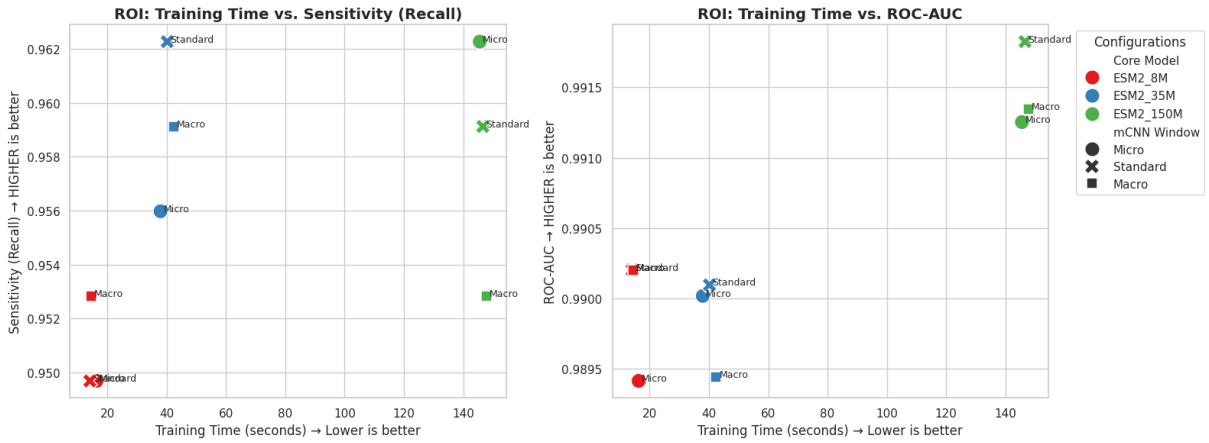


Figure 4.3: ROI Scatter Plots for Compact and Mid-tier PLMs (8M, 35M, 150M) coupled with mCNN configurations.



Figure 4.4: ROI Scatter Plots for the Massive 650M PLM coupled with mCNN configurations.

1. Visualizing the Computational Wall (Diminishing Returns)

The scatter plots vividly illustrate the phenomenon of diminishing returns when blindly scaling Protein Language Models. Looking at the X-axis (Training Time), there is a massive computational gulf between the models. While the **8M** (Red) and **35M** (Blue) models tightly cluster on the left side of the cost spectrum (under 45 seconds), the **150M** (Green) model violently shifts to the right (~146 seconds). The **650M** model (Figure 4.4) pushes this extreme even further, consuming over 260 seconds per run. Crucially, looking at the Y-axis, this exponential increase in training time does *not* translate into proportional vertical gains. The 150M cluster is visually trapped at the same, or even lower, performance horizontal plane as the 35M cluster.

2. Isolating the "Sweet Spot"

By analyzing the cross-section of both time and performance, the **ESM2_35M (Blue)** cluster represents the optimal topological boundary. Specifically, the **Standard** configuration (represented by the blue 'X' marker) hits a distinct peak in Sensitivity (>0.962) and maintains a near-perfect ROC-AUC (>0.990) while remaining computationally economical (~40 seconds). While the 650M model theoretically achieves a slightly higher absolute Sensitivity (0.981 for Macro, as seen in Figure 4.4), its position on the extreme far-right of the X-axis yields an unviable ROI. Demanding over a 600% increase in computational cost for a marginal ~0.019 gain in Sensitivity is unjustifiable for large-scale, high-throughput biological screening.

3. The Final Verdict: The Two-Tiered Winning Configurations

Concluding the exhaustive multiphase ablation study which screened classic descriptors, evaluated single-kernel geometric constraints, and mapped the PLM computational wall this research officially declares two distinct optimal architectures tailored for different deployment environments.

- **The ROI Champion (Standard-Scale Tier):** The hybrid architecture comprising **Frozen ESM2_35M embeddings coupled with a Standard mCNN ($K \in \{3,5,7\}$)** is identified as the optimal practical model. This specific configuration provides the best Return on Investment, seamlessly balancing the deep evolutionary grammar of Transformers with the multi-scale spatial resolution of Convolutional Networks, all without succumbing to the curse of dimensionality or computational bloat.
- **The Absolute Apex (The High-Capacity Tier):** For server-grade environments where computational cost is not a limiting factor and maximum predictive performance is the sole objective, the **Frozen ESM2_650M embeddings coupled with a Macro mCNN ($K \in \{5,7,9\}$)** is identified as the high-capacity comparison model. Despite its exorbitant training time, it establishes the absolute performance ceiling on large-scale datasets.

Having isolated these two optimal structures, Phase 2 of this research (Section 4.7) will proceed to train both the Standard-Scale (35M) and High-Capacity (650M) architectures on the full, unconstrained datasets to establish the final global benchmarks.

4.6. Data-Centric Intervention: Resolving Spurious Length Correlations

Before scaling the optimized ESM2-mCNN architecture to the full dataset for final production training, the architectures selected from Phase 1 were initially evaluated under the original, full-scale negative-sampling distribution. As detailed in Table 4.12 and Figure 4.5, these preliminary experiments produced exceptionally high benchmark performance with the ESM2-650M configuration achieving the strongest AMP performance under the original data distribution.

Model	Task	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	ROC AUC
ESM2-35M (AMP)	AMP	95.08	88.46	98.19	91.99	0.9914
ESM2-650M (AMP)	**AMP**	**96.85**	**95.36**	**97.55**	**95.09**	**0.9944**

Table 4.12: Initial Benchmark Results (Before Data Intervention)

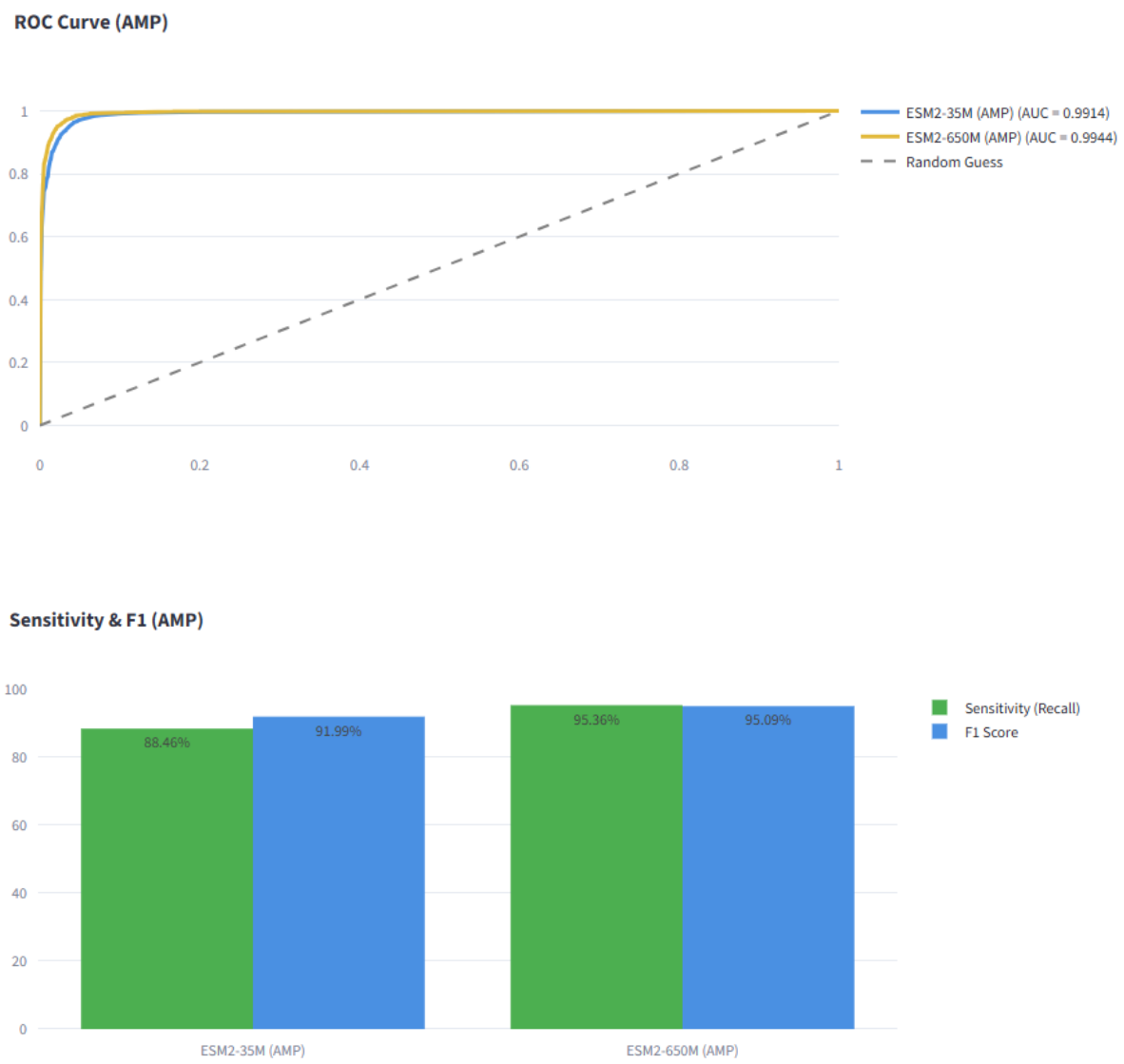


Figure 4.5: Initial AMP ROC Curve and Performance Metrics (Before length-matching intervention), showcasing the strong initial performance of the 650M model before the length-matching intervention.

4.6.1. Discovery of Spurious Correlation at Scale

However, this spectacular performance of the 650M model was an artifact of the data distribution. A rigorous exploratory data analysis conducted on the global corpus revealed a critical structural flaw inherent in heavily aggregated public biological databases.

A blatant "Length Bias" was discovered: the mean sequence length of positive AMPs was approximately 22 Amino Acids (AAs), whereas negative non-AMPs (which often consist of fragmented or full-length proteins) averaged approximately 44 AAs. Crucially, for sequence lengths strictly under 40 AAs, the positive-to-negative ratio was heavily and unnaturally skewed at 3:1.

In the paradigm of Deep Learning, neural networks operate as "lazy learners"; they will mathematically exploit the easiest path to minimize the loss function. This phenomenon is known as "Shortcut Learning" or "Spurious Correlation." If the ESM2-mCNN were evaluated on this raw, unbalanced distribution, it would likely bypass learning complex biochemical grammar. Instead, the model would devolve into a naive "length estimator," automatically assigning high AMP probabilities to any short sequence and low probabilities to longer ones. While this would yield deceptively high benchmark accuracy, the model would fail significantly in real-world, in-vitro clinical screening.

4.6.2. Engineering the Solution: Length-Matched Negative Sampling

To ensure the final production model possessed true biological generalization, a robust data-centric intervention was engineered prior to the final full-scale training phase. **Crucially, this intervention was applied strictly to the AMP dataset.** The Toxicity dataset inherently bypassed this requirement, as both its positive (toxic) and negative (safe) classes are comprised entirely of verified AMPs, naturally sharing an identical short-length distribution.

For the AMP task, rather than relying on standard undersampling which would destroy massive amounts of valuable data a **Length-Matched Negative Sampling** algorithm was implemented. The methodology operates as follows: instead of feeding the network full-length negative proteins, the algorithm programmatically traverses the non-AMP negative sequence pool and randomly crops continuous subsequences. These crops are algorithmically constrained to strictly match the target length distribution of the true positive AMP dataset (predominantly 15 to 40 AAs).

Conclusion of Data-Centric Intervention:

By matching the length distributions of the positive and negative training instances, the intervention reduces the opportunity for the classifier to use sequence length as a dominant shortcut. The resulting models are therefore expected to rely more strongly on sequence-level features beyond length alone. However, the cropped subsequences are computationally constructed negative instances derived from annotated non-AMP source sequences, and their individual antimicrobial activity was not experimentally verified.

4.6.3. Validation of Ablation Integrity on Sub-Sampled Datasets

A critical methodological question arises: *Does this underlying data bias invalidate the Phase 1 architectural ablation study?*

As detailed in Section 4.4, the multiphase ablation studies (comparing classic descriptors, Single CNNs, and multi-scale CNNs across varying ESM-2 parameter sizes) were executed on a proxy subset (5,000 Training / 1,000 Validation sequences). This sub-sampling strategy was strictly implemented to optimize GPU computational resources and enable rapid architectural prototyping.

Because all Phase 1 architectures were evaluated under the same proxy sampling procedure, the relative comparison remains useful for architecture screening. These results therefore provided sufficient evidence for selecting candidate architectures for subsequent full-scale evaluation, while not guaranteeing that the same ranking would be preserved after changes to the training distribution.

Consequently, Phase 1 serves functionally as an architectural screening step, while the subsequent Phase 2 (Section 4.7) acts as the definitive final evaluation. By re-evaluating these isolated candidate architectures on the cleansed, length-matched dataset, Phase 2 establishes the true, unbiased global benchmarks.

4.7. Phase 2: Full-Scale System Training and Global Benchmarking

Having successfully isolated the optimal architectures during the Phase 1 Ablation Study, the pipeline advances to full-scale dataset training. To bridge the gap between theoretical Deep Learning accuracy and practical software deployment, a **Resource-Aware Multi-Tiered Strategy** was implemented for both the Antimicrobial (AMP) and Toxicity prediction tasks.

4.7.1. Deployment-Driven Multi-Tiered Strategy

Processing thousands of peptide sequences simultaneously imposes severe hardware constraints. While massive Protein Language Models (e.g., the 650M parameter model) are theoretically capable of capturing ultra-dense representations, their high computational overhead makes them susceptible to Out-Of-Memory (OOM) failures on standard machines. To accommodate these constraints and systematically evaluate the models, the deployment framework was structured into two distinct operational tiers:

- **The Standard-Scale Tier (ESM2_35M + Standard mCNN):** The Standard-Scale Tier uses the ESM2-35M + Standard mCNN configuration as the preferred operational configuration based on the observed performance-efficiency trade-off.
- **The High-Capacity Tier (ESM2_650M + Macro mCNN):** The High-Capacity Tier is retained as a comparison configuration for evaluating the effect of increased parameter scale.

4.7.2. Architecture Flowchart and Algorithmic Configurations

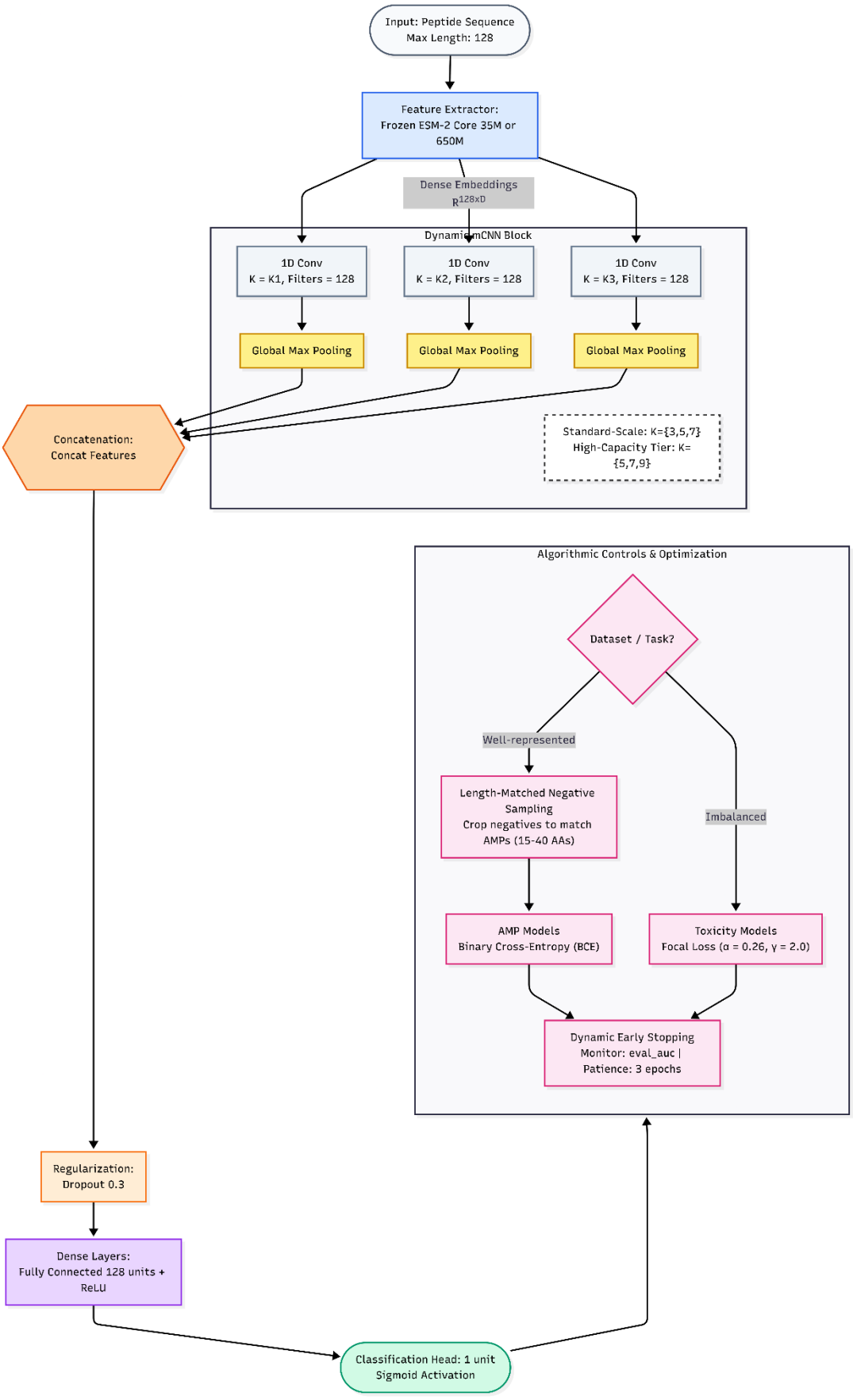


Figure 4.6: The Generalized Hybrid Deep Learning Architecture for AMP and Toxicity Prediction

As illustrated in Figure 4.6, to ensure systemic consistency, all four production models share an identical, unified forward-propagation topology. The architecture acts as a generalized template, dynamically adjusting its internal tensor dimensions (D) and convolutional receptive fields (K) based on the assigned deployment tier:

- **Feature Extraction:** Raw peptide sequences are padded and truncated to a strict maximum length of 128 tokens. These tokens are passed through the frozen ESM-2 cores, generating dense spatial embeddings. The embedding dimensionality dynamically scales depending on the tier: $\mathbb{R}^{128\times480}$ for the 35M Standard-Scale models, and a hyper-dense $\mathbb{R}^{128\times1280}$ for the 650M High-Capacity models.
- **Dynamic Multi-Branch CNN (mCNN):** The embedding matrices are permuted and scanned by three parallel 1D Convolutional branches (128 filters each). The Standard-Scale tier utilizes a *Standard* motif-scanning window ($K_1 = 3, K_2 = 5, K_3 = 7$), whereas the High-Capacity tier utilizes an extended *Macro* window ($K_1 = 5, K_2 = 7, K_3 = 9$) to capture broader structural domains.
- **Classification Head:** The spatially extracted features from the parallel branches undergo Global Max Pooling, are concatenated into a fixed $\mathbb{R}^{384}$ representation, and passed through a 0.3 Dropout layer for robust regularization. Finally, the features are compressed via a 128-unit Fully Connected layer before reaching the Sigmoid classification output.

Dataset Scaling and Imbalance Handling: To execute the full-scale training, the AMP systems processed a massive, well-represented corpus of 77,223 training sequences and 9,653 validation sequences. In stark contrast, the Toxicity systems operated on a highly constrained and imbalanced dataset comprising only 2,516 training and 315 validation sequences.

To optimize these divergent datasets, three critical algorithmic controls were mandated:

1. **Dynamic Early Stopping:** Applied universally across all four models, the training loop monitored the eval_auc metric with a strict patience of 3 epochs. If the network failed to improve its classification boundaries within this window, training was automatically halted to prevent global overfitting, restoring the optimal checkpoint.
2. **Focal Loss Mathematical Intervention:** While the AMP models utilized standard Binary Cross-Entropy (BCE) loss, the Toxicity models required a strict mathematical intervention to counteract severe dataset imbalance. A **Focal Loss** function was explicitly implemented ($\alpha = 0.26,\ \gamma = 2.0$) for the Toxicity models. This mechanism down-weights the gradient contribution of easily classified non-toxic samples, forcing the network to focus its learning capacity entirely on the rare, hard-to-classify toxic minority instances.

3. **Length-Matched Negative Sampling:** To prevent the neural network from exploiting spurious structural correlations specifically the natural length disparity where true AMPs are significantly shorter than negative background proteins a strict data-centric intervention was applied prior to training. The negative sequence pool was algorithmically cropped to match the exact length distribution of the positive AMP class (predominantly 15 to 40 amino acids). This intervention neutralized sequence length as a discriminative mathematical shortcut, encouraging the model to rely on sequence-level features beyond length alone.

4.7.3. Global Independent Benchmarking

To evaluate the real-world predictive generalization of the fully trained systems, a final independent benchmarking phase was conducted. The four production models were evaluated against strict, unseen hold-out test datasets (**test_amp.csv** and **test_toxicity.csv**). This separation reduces the risk of evaluation leakage and preserves the independence of the final assessment and ensures that the reported metrics reflect the models' true performance on novel, uncharacterized peptide sequences. The held-out test sets were kept unchanged during the length-matching intervention.

The overall comparison summary across both tasks and architectural tiers is detailed in Table 4.13.

Task	Core Architecture	Accuracy	Sensitivity (Recall)	Specificity	F1-Score	ROC-AUC
AMP	**ESM2-35M + Standard mCNN**	**91.88%**	**81.91%**	96.56%	**86.57%**	0.9739
	ESM2-650M + Macro mCNN	91.69%	79.45%	**97.44%**	85.94%	**0.9801**
TOX	**ESM2-35M + Standard mCNN + Focal**	**77.78%**	**75.21%**	**85.71%**	**83.64%**	**0.8950**
	ESM2-650M + Macro mCNN + Focal	74.92%	71.43%	**85.71%**	81.15%	0.8870

Table 4.13: Final Independent Benchmark Results on Hold-out Test Datasets.

A. Performance Analysis: Antimicrobial Peptide (AMP) Task

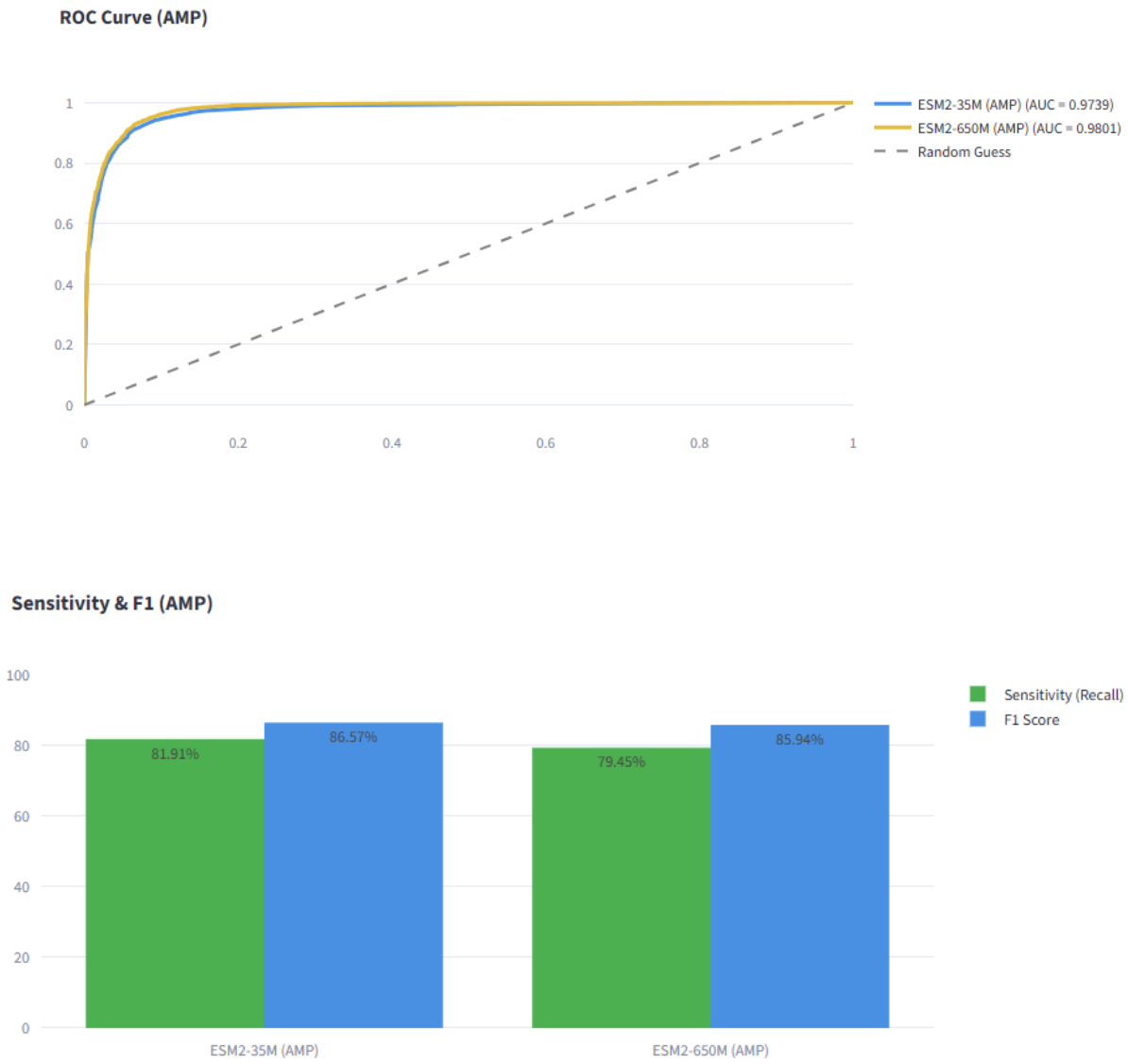


Figure 4.7: ROC Curve and Performance Metrics for the AMP Task

The visualization of the AMP benchmark (Figure 4.7) yields an extraordinary, counter-intuitive revelation following the strict Length-Matched Negative Sampling intervention. Within the evaluated configurations, the ESM2-35M model achieved higher Sensitivity and F1-score than the ESM2-650M model on the AMP task, while the 650M model retained a slightly higher ROC-AUC.

By avoiding the spatial noise and over-parameterization inherent in hyper-dense embedding spaces when processing short sequences, the 35M model effectively pushes Sensitivity (Recall) to **81.91%**, compared to the 650M model's 79.45%.

Furthermore, it establishes a higher F1-Score of **86.57%**. This robust retention of Sensitivity ensures that significantly fewer genuine antimicrobial sequences are missed during critical clinical screening, validating the compact 35M architecture as the highly practical and preferred operational standard for AMP discovery.

B. Performance Analysis: Toxicity Task

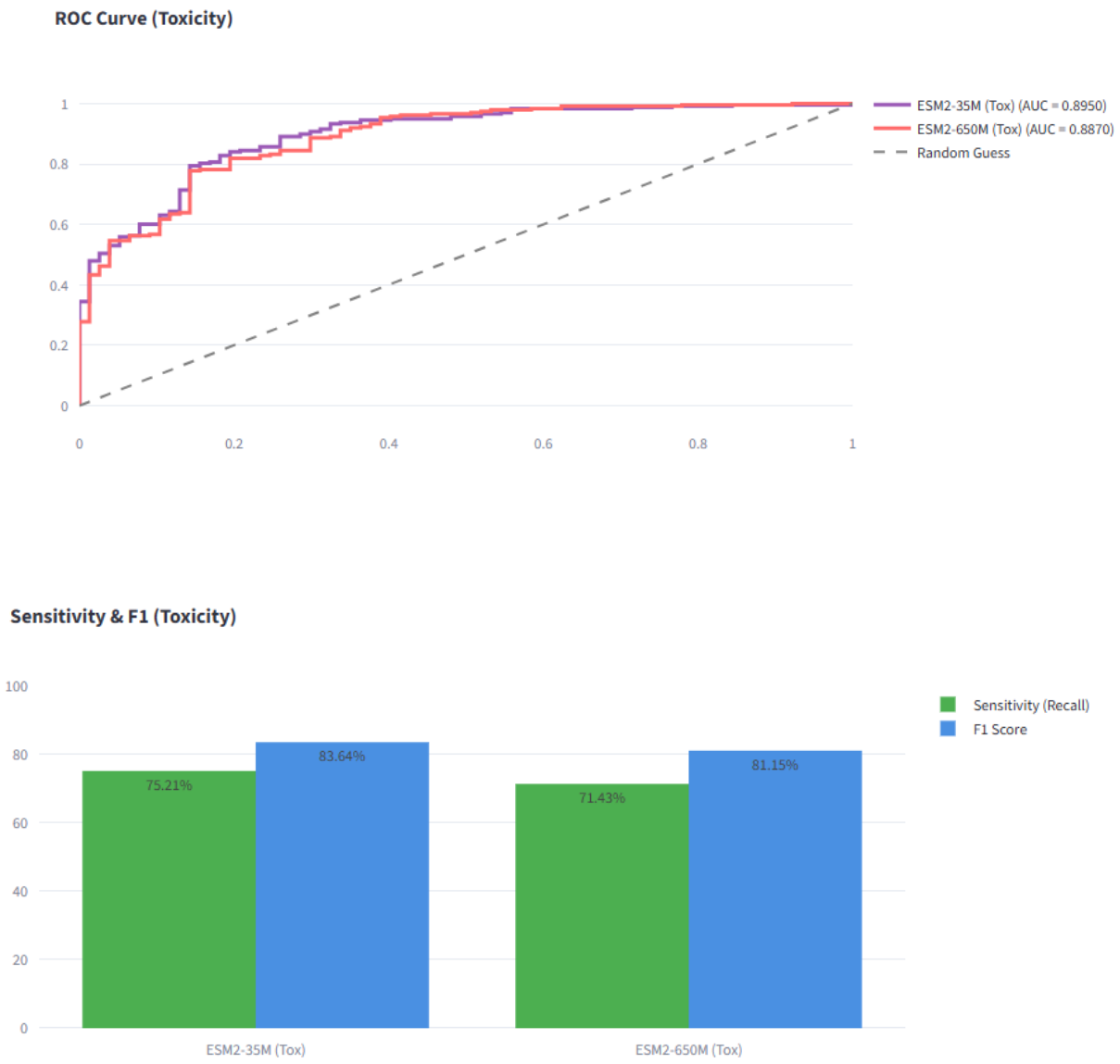


Figure 4.8: ROC Curve and Performance Metrics for the Toxicity Task

The visualizations for the Toxicity benchmark (Figure 4.8) illustrate the performance of both models on the secondary task. Despite being structurally identical in the forward-propagation methodology, the ESM2-35M configuration achieved higher F1-score, Sensitivity, and ROC-AUC than the 650M configuration on the evaluated toxicity test set. As depicted in the bar charts, the 35M model achieved an F1-Score of 83.64% and a Sensitivity of 75.21%, along with a higher ROC-AUC (0.8950) compared to the 650M architecture (0.8870). Both models equalized on Specificity (85.71%), indicating that the Focal Loss formulation successfully aided both networks in classifying the negative samples.

This observation across both the length-matched AMP dataset and the highly constrained Toxicity dataset suggests that increasing parameter count did not provide a performance advantage under the evaluated toxicity setting. Rather than confirming a fundamental architectural flaw, it highlights the empirical challenges of scaling deep learning models on highly constrained biological data, further contextualizing the "650M Paradox" discussed in the subsequent section.

4.7.4. The Universal "650M Paradox" and Architectural Limitations

The empirical benchmarks reveal a profound structural paradox that redefines the scaling laws of Protein Language Models in the context of short peptides. The massive 650M High-Capacity model underperformed the lightweight 35M Standard-Scale model across **both** the Toxicity and AMP tasks in crucial evaluation metrics.

Initially, this regression was hypothesized to be a symptom of extreme over-parameterization exclusively triggered by the small, imbalanced Toxicity dataset. However, observing the 650M model's performance on the massive AMP dataset post-length-matching shifts this perspective. In this study, the term "650M Paradox" refers to the observed performance reversal in which the evaluated ESM2-650M configuration did not outperform the ESM2-35M configuration after the length-matching intervention. This result suggests that scaling parameter count does not necessarily yield improved performance under the present short-peptide data and architectural conditions.

Architectural Hypothesis:

When the structural "shortcut" of sequence length was removed from the dataset, the models were forced to rely purely on complex biochemical grammar. The 650 million parameter model maps sequences into a hyper-dense R^{1280} embedding space, whereas the 35M model maps into a highly concise R^{480} embedding space. One possible explanation is that the higher-dimensional representation may introduce additional complexity that is not fully supported by the available short-peptide data, increasing the difficulty of downstream motif extraction. However, this interpretation should be considered a hypothesis rather than a directly established causal mechanism.

4.8. Interpretability and Sequence Analysis via XAI

The true utility of a bioinformatics framework lies not merely in its benchmark accuracy, but in its capacity to generate actionable biological insights. The integration of the Occlusion-based Sensitivity Analysis elevates the system beyond standard classification, allowing for profound sequence-level interpretation.

Bridging Artificial Intelligence with Empirical Biology:

The generated XAI heatmaps provide a residue-level map of prediction sensitivity. To achieve this, the predictive pipeline integrates the XAI outputs with empirical **Biological Motif Heuristics**—a rule-based engine designed to match AI-identified hotspots with known physicochemical properties.

Antimicrobial Structural Discoveries:

When analyzing sequences classified as Antimicrobial Peptides (AMPs), the XAI heatmaps consistently illuminate biologically critical structural domains. The algorithm demonstrates a strong, autonomous bias toward alpha-helix formations and localized **Cationic patches**. Specifically, the logit drop algorithm frequently isolates regions rich in Lysine (K) and Arginine (R) residues. This computational behavior perfectly aligns with established microbiological paradigms: these positively charged residues are the primary mechanical agents required to electrostatically bind to and disrupt negatively charged bacterial phospholipid membranes. The highlighted regions were consistent with sequence properties previously associated with antimicrobial activity, such as cationic and hydrophobic residues.

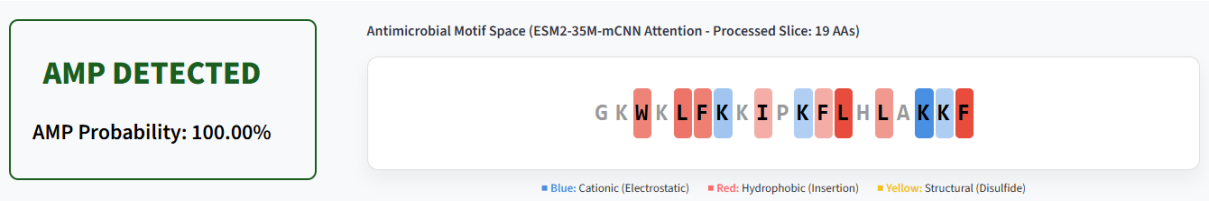


Figure 4.9: XAI heatmap of Stage 1 (AMP Gatekeeper) predicting a verified Antimicrobial Peptide, highlighting critical cationic and hydrophobic motifs.

Hemolytic and Toxicity Discoveries:

Conversely, when evaluating non-AMP or explicitly Toxic sequences, the XAI mapping reveals distinctly different structural dependencies. The system actively pinpoints concentrated **Hydrophobic clusters** regions dominated by Leucine (L), Valine (V), Isoleucine (I), Phenylalanine (F), and Tryptophan (W) as well as hyper-cationic segments. By feeding these mapped coordinates into the integrated heuristic engine, the system automatically translates algorithmic hotspots into human-readable biological warnings. The UI provides explicit rationale, such as flagging a "Hemolytic Hydrophobic Cluster" or a "Non-specific Binding Motif." This explains the *why* behind a toxicity prediction, highlighting the specific structural domains likely to cause non-selective rupturing of human red blood cells.

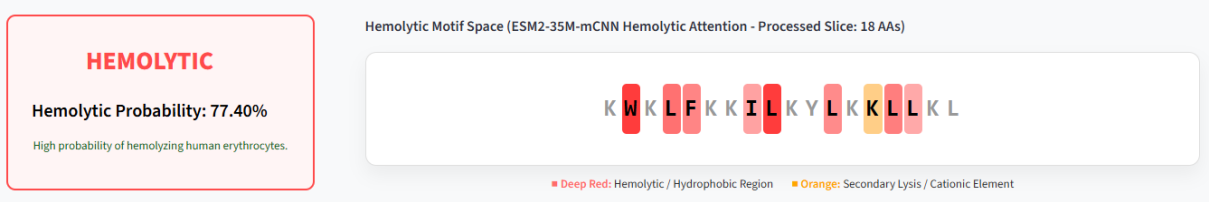


Figure 4.10: XAI heatmap of Stage 2 (Toxicity Assessor) flagging a hemolytic sequence, emphasizing concentrated hydrophobic clusters associated with membrane interaction and hemolytic activity.

Conclusion of Interpretability:

The successful synthesis of Occlusion-based XAI with empirical structural heuristics represents a major methodological advancement in this research. It formally transforms the ESM2-mCNN pipeline from an opaque, "black-box predictor" into an **interpretable biological analysis tool**. By proving that the model's mathematical optimizations actively mirror real-world biomolecular physics, the system provides an interpretable view of the sequence regions that contribute most strongly to the model's predictions.

4.9. Chapter Conclusion and key findings

The exhaustive empirical evaluation conducted throughout Chapter 4 provides critical insights into the scaling behavior of Protein Language Models for short-peptide classification.

The experiments indicate that the ESM2-35M + Standard mCNN configuration provided the most favorable performance-efficiency trade-off among the evaluated architectures. The results also showed that increasing parameter scale did not consistently improve performance after the length-matching intervention.

Ultimately, this compact architecture ensures rapid, offline training and minimal hardware overhead, validating it as a highly practical and robust solution for real-world software deployment in bioinformatics.

CHAPTER 5: REAL-WORLD DEPLOYMENT AND CONCLUSION

5.1. System Architecture and Software Integration

Transitioning deep learning models from isolated experimental environments into functional, real-world bioinformatics applications requires a robust and scalable software engineering framework. To achieve this, the predictive system was built upon a strictly decoupled Client-Server architecture. This modular design explicitly separates the deep learning inference logic from the presentation layer, ensuring high maintainability, structural scalability, and seamless performance.

- **The Backend Server (FastAPI & Uvicorn):** Functioning as the computational engine of the application, the backend is driven by the high-performance FastAPI framework and served via Uvicorn. Its primary responsibility is the heavy lifting: pre-loading the optimized PyTorch ESM-2 mCNN model weights into memory (RAM or GPU VRAM), actively listening for incoming biological sequences via HTTP POST requests, and executing the tensor-based AI inference. Upon completion, it returns structured JSON payloads containing the precise AMP and Toxicity classification probabilities, alongside Explainable AI (XAI) attention matrices.
- **The Frontend Client (Streamlit):** Operating as the interactive user interface (UI), the frontend is developed entirely in Streamlit. It acts as a lightweight presentation layer that communicates with the backend asynchronously through REST API calls. Upon receiving the JSON payload, the frontend dynamically renders the biological metrics, sequence visualizations, and complex XAI heatmap overlays for intuitive user analysis.

- **Integration Rationale:** The architectural decision to decouple the frontend from the backend solves a fundamental bottleneck in bioinformatics tool development. By isolating the computational environment, the heavy GPU/CPU tensor operations do not block the main thread, preventing the UI from freezing during intense inference tasks. Furthermore, this separation of concerns allows the underlying AI models (such as swapping inference parameters or updating core checkpoints) to be hot-swapped without requiring a single modification to the Streamlit frontend codebase, adhering to strict software engineering best practices.

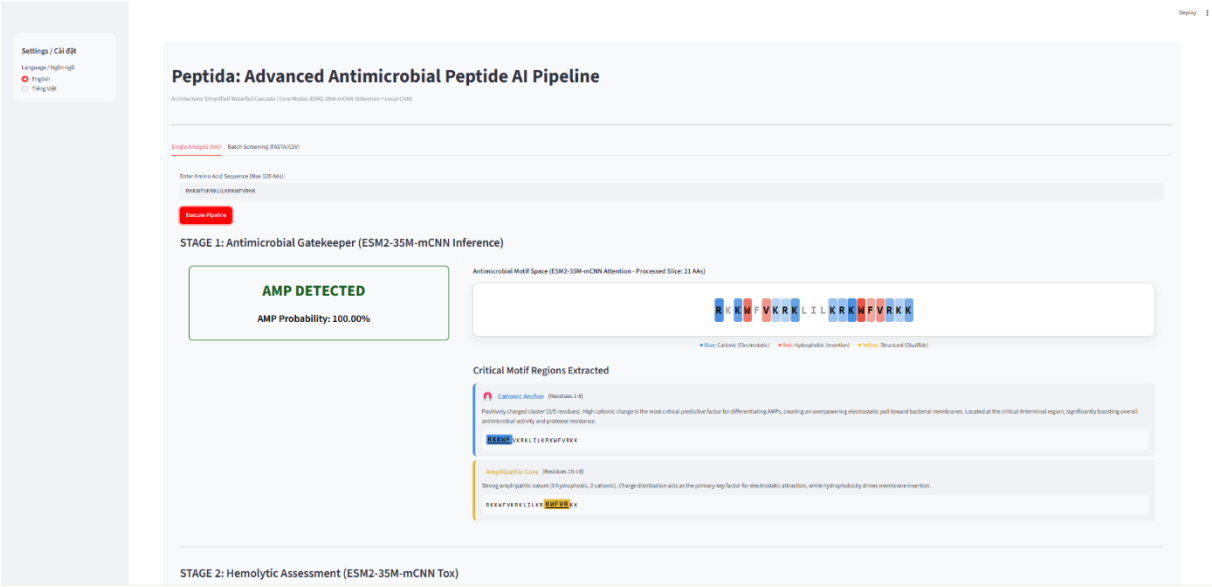


Figure 5.1: The decoupled Streamlit-based graphical user interface, demonstrating real-time peptide sequence entry, automated cascade screening, and interactive XAI heatmap visualization.

5.2. System Deployment Strategies

A persistent challenge in deploying advanced predictive models is resolving the inherent trade-off between maximizing software accessibility for general users and maintaining strict, high-performance computational reproducibility for academic researchers. To elegantly bridge this gap, a Dual-Tier Deployment Strategy was engineered around the optimized **ESM2-35M mCNN architecture** (the optimized evaluation model), ensuring uniform predictive power across both local and containerized environments.

5.2.1. Tier 1: Lightweight Portable Release (General Accessibility)

The first tier is engineered to maximize accessibility, prioritizing a "zero-configuration" philosophy. It utilizes the highly compact **ESM2-35M mCNN architecture** (selected ESM2-35M + Standard mCNN configuration) for both the Antimicrobial Peptide (AMP) and Toxicity prediction tasks.

- **Target Audience:** This release is specifically targeted at students, educators, and casual researchers who require a rapid, offline screening tool without the overhead of configuring complex computational environments or virtualization software.
- **Architecture & Mechanism:** The deployment utilizes a Single-node, Multi-process Orchestration pattern. The system is packaged as a standalone directory equipped with an automated batch script (1_Click_Run.bat). Upon execution, this script dynamically initializes an isolated Python Virtual Environment (venv), seamlessly installs all operational dependencies, and spawns both the FastAPI backend and Streamlit frontend processes concurrently on the local host machine.
- **System & Environment Requirements:**
    - **OS:** Windows 10 / 11
    - **Hardware:** Minimum 4GB RAM. Optimized for CPU-only inference (No dedicated GPU required).
    - **Environment:** Python 3.10+ (Installed globally on the host machine).

5.2.2. Tier 2: Pro Academic Docker Release (Research & Server)

The second tier is designed to function as a high-throughput diagnostic tool for institutional environments, integrating the optimized **ESM2-35M mCNN pipeline** within a standardized microservices architecture to guarantee rapid, concurrent screening without hardware bottlenecks.

- **Target Audience:** This tier is strictly intended for computational biologists, academic laboratories, and institutional servers where operational stability, high-throughput batch parsing, and absolute environmental reproducibility are foundational scientific standards.
- **Architecture & Mechanism:** To satisfy these rigorous demands, the deployment mechanism utilizes a Containerized Microservices architecture orchestrated via docker-compose. The API (backend) and Web (frontend) services are isolated into separate Linux containers. Crucially, the backend container integrates optional hardware-accelerated GPU (CUDA) execution via the NVIDIA Container Toolkit while retaining full CPU compatibility. The frontend container securely communicates with the backend via Docker's internal virtual bridge network using a globally injected environment variable (BACKEND_URL=http://api:8000).
- **System & Environment Requirements:**
    - **OS:** Linux (Ubuntu/CentOS) or Windows (with WSL2 enabled).
    - **Hardware:** Minimum 8GB RAM. (Optional: NVIDIA GPU with CUDA support for accelerated batch tensor processing).
    - **Environment:** Docker Engine and Docker Compose installed.

- **Architectural Advantage:** This containerized approach provides unprecedented fault tolerance and scalability. Most importantly, it eliminates the persistent "it works on my machine" discrepancy, ensuring consistent computational reproducibility across any operating system a critical validation standard in modern scientific computing.

5.3. Limitations and Future Work

Despite achieving strong benchmark performance within the evaluated experimental scope, this research acknowledges fundamental limitations dictated by current data availability, time constraints, computational resources, and natural biophysical boundaries.

1. **Dataset Constraints and the Over-parameterization Paradox:** The most prominent limitation encountered was the highly constrained and imbalanced nature of the Toxicity dataset. Containing merely a few thousand samples, it actively prevented the viable implementation of massive architectures. The attempt to evaluate the ESM2-650M model exposed a critical flaw in deep learning scaling laws (the "650M Paradox"): scaling up parameter complexity without a proportional increase in high-quality data volume leads to immediate memorization of noise, spatial overfitting, and a catastrophic loss of predictive generalization.
2. **Scope of Explored Architectures:** Due to strict academic timeframes and constraints in available computational processing power, it was not feasible to exhaustively test every available Protein Language Model (such as ProtT5, ESM-3, or Ankh) or to benchmark infinitely complex spatial scanning mechanisms. The ablation study was purposefully narrowed to the ESM-2 family and 1D-CNN topologies to ensure the research remained focused and achievable within the current scope.

3. **The Ultra-Short Peptide Blind Spot (Architectural & Biophysical Limits):** A notable limitation of the Hybrid Cascade Framework is its diminished predictive accuracy when processing ultra-short peptide sequences (fewer than 10 amino acids). Architecturally, the ESM-2 Transformer requires sufficient sequence length to establish meaningful self-attention matrices. Furthermore, the mCNN utilizes large receptive fields (e.g., $K \in \{3,5,7\}$) which encompass an entire ultra-short sequence in a single stride, causing premature spatial collapse during Max-Pooling and preventing the extraction of localized hierarchical features. However, this algorithmic blind spot serendipitously aligns with empirical biophysical constraints: to physically span and disrupt a bacterial lipid bilayer (30–40 Ångströms thick), an alpha-helical peptide requires a minimum chain length of 15 to 20 amino acids. Naturally occurring AMPs under 10 amino acids are exceedingly rare and operate through highly atypical intracellular mechanisms. The network's tendency to reject these ultra-short fragments demonstrates that it has accurately internalized the structural boundaries of its training distribution, learning that insufficient physical length overwhelmingly correlates with a lack of standard membrane-disruptive capacity.
4. **A limitation of the length-matched negative sampling strategy** is that cropped subsequences inherit the negative label from their source sequences, but their individual antimicrobial activity was not experimentally verified.

Future Work and Scalability:

This research does not represent a culmination, but rather a robust, scalable blueprint. The rigorous ablation methodology and the decoupled deployment pipeline established herein are primed for future expansion. Future research trajectories will focus on:

- **Dataset Expansion:** Collaborating with wet-lab facilities or utilizing generative data augmentation (e.g., GANs, diffusion models) to aggregate massive, experimentally validated toxicity datasets.
- **Next-Generation Integration:** Once sufficient data volume is acquired, the system is perfectly positioned to integrate and unlock the true evolutionary representation capabilities of billion-parameter PLMs without the risk of overfitting.
- **Advanced Scanning Mechanisms:** Exploring multi-dimensional attention heads and graph-based neural networks to replace or augment the mCNN block.
- **Specialized Micro-Peptide Models:** Developing distinct, lightweight statistical models specifically tailored to evaluate ultra-short peptides (< 10 amino acids) that operate via non-membrane-disruptive mechanisms, bypassing the constraints of large-kernel spatial convolutions.

The foundational architecture has been deployed. As biological data scales and computational paradigms evolve, this system is structurally designed to seamlessly assimilate new technologies, ensuring its continued relevance at the forefront of computational peptide discovery.

5.4. Conclusion

This research aimed to establish a definitive, highly optimized deep learning architecture for the identification of Antimicrobial Peptides (AMPs) and the prediction of their associated mammalian toxicity. By systematically transitioning from classic Machine Learning baselines (Random Forest, SVM) using hand-crafted descriptors (AAC, DPC) to cutting-edge Protein Language Models (ESM-2), this study rigorously mapped the topological boundaries of peptide sequence classification.

Through an exhaustive ablation study, it was empirically proven that rigid, single-scale convolutions suffer from geometric instability, failing to capture the variable structural motifs of short peptides. This necessitated the engineering of the Multi-branch CNN (mCNN), which dynamically scans sequences across multiple receptive fields to resolve feature hierarchies.

Within the architectures evaluated in this thesis, the ESM2-35M + Standard mCNN configuration provided the most favorable balance between predictive performance and computational cost. The experiments further highlighted the importance of addressing sequence-length confounding when training AMP classifiers on heterogeneous negative datasets.

Furthermore, this thesis successfully bridged the gap between theoretical AI and practical software engineering. By engineering a decoupled, Multi-Tiered Client-Server architecture (FastAPI and Streamlit) alongside strict XAI interpretability mapping, the predictive models were transformed into accessible and reproducible bioinformatics tools. The proposed cascade therefore provides a practical computational framework for AMP and hemolytic toxicity screening, while further experimental validation and broader benchmarking remain necessary.

REFERENCES

[1] W. Dee, "LMPred: predicting antimicrobial peptides using pre-trained language models and deep learning," *Bioinformatics Advances*, vol. 2, no. 1, p. vbac021, Mar. 2022, doi: 10.1093/bioadv/vbac021.

[2] P. Bhadra, J. Yan, J. Li, S. Fong, and S. W. I. Siu, "AmPEP: Sequence-based prediction of antimicrobial peptides using distribution patterns of amino acid properties and random forest," *Scientific Reports*, vol. 8, p. 1697, Jan. 2018, doi: 10.1038/s41598-018-19752-w.

[3] J. Yan et al., "Deep-AmPEP30: Improve Short Antimicrobial Peptides Prediction with Deep Learning," Molecular Therapy: Nucleic Acids, vol. 20, pp. 882–894, Jun. 2020, doi: 10.1016/j.omtn.2020.05.006.

[4] Q.-T. Ho, N. Q. K. Le, and Y.-Y. Ou, "mCNN-ETC: identifying electron transporters and their functional families by using multiple windows scanning techniques in convolutional neural networks with evolutionary information of protein sequences," *Briefings in Bioinformatics*, vol. 23, no. 1, p. bbab352, Jan. 2022, doi: 10.1093/bib/bbab352.

[5] K. Castillo-Mendieta *et al.*, "Peptide hemolytic activity analysis using visual data mining of similarity-based complex networks," *npj Systems Biology and Applications*, vol. 10, p. 115, Oct. 2024, doi: 10.1038/s41540-024-00429-2.

[6] I. Abdelbaky, M. Elhakeem, H. Tayara, E. Badr, and M. A. Salam, "Enhanced prediction of hemolytic activity in antimicrobial peptides using deep learning-based sequence analysis," *BMC Bioinformatics*, vol. 25, p. 368, Nov. 2024, doi: 10.1186/s12859-024-05983-4.

[7] M. Salem, A. K. Arshadi, and J. S. Yuan, "AMPDeep: hemolytic activity prediction of antimicrobial peptides using transfer learning," *BMC Bioinformatics*, vol. 23, p. 389, Sep. 2022, doi: 10.1186/s12859-022-04952-z.

[8] K. Chaudhary *et al.*, "A web server and mobile app for computing hemolytic potency of peptides," *Sci Rep.*, vol. 6, no. 1, p. 1, 2016. https://doi.org/10.1038/srep22843.

[9] M. M. Hasan *et al.*, "HLPpred-Fuse: improved and robust prediction of hemolytic peptide and its activity by fusing multiple feature representation," *Bioinformatics*, vol. 36, no. 11, pp. 3350-3356, 2020. https://doi.org/10.1093/bioinformatics/btaa160.

[10] A. Capecchi, X. Cai, H. Personne, T. Köhler, C. van Delden, and J.-L. Reymond, "Machine learning designs non-hemolytic antimicrobial peptides," *Chem Sci.*, vol. 12, no. 26, pp. 9221-9232, 2021. https://doi.org/10.1039/D1SC01713F .

[11] ps29, "amp_dataset_viri," Hugging Face Dataset, DOI: 10.57967/hf/2249.

[12] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.

[13] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[14] Y. Kim, "Convolutional Neural Networks for Sentence Classification," in *Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP)*, 2014, pp. 1746–1751.

[15] C. Szegedy *et al.*, "Going Deeper with Convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 1-9.

[16] D. Veltri, U. Kamath, and A. Shehu, "Deep-AmPEP: a deep learning approach for predicting antimicrobial peptides," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 15, no. 6, pp. 2069-2078, 2018.

[17] G. Van Rossum and F. L. Drake, *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace, 2009.

[18] A. Paszke *et al.*, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in *Advances in Neural Information Processing Systems 32*, 2019, pp. 8024–8035.

[19] T. Wolf *et al.*, "Transformers: State-of-the-Art Natural Language Processing," in *Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP): System Demonstrations*, 2020, pp. 38–45.

[20] Q. Lhoest *et al.*, "Datasets: A Community Library for Natural Language Processing," in *Proc. Conf. Empir. Methods Nat. Lang. Process. (EMNLP): System Demonstrations*, 2021, pp. 175–184.

[21] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[22] S. Ramírez, "FastAPI: High performance, easy to learn, fast to code, ready for production," 2018. [Online]. Available: https://fastapi.tiangolo.com/

[23] Streamlit Inc., "Streamlit: The fastest way to build and share data apps," 2019. [Online]. Available: https://streamlit.io/

APPENDIX

APPENDIX A: OPEN-SOURCE REPOSITORY AND SOURCE CODE

In adherence to the principles of Open Science and to ensure strict computational reproducibility, the entire source code developed during this research is made publicly available. This project is released under an open-source license strictly for educational and academic research purposes, aiming to provide the bioinformatics community with accessible, transparent, and high-performance tools for Antimicrobial Peptide and Toxicity screening.

The repository contains the complete deep learning pipelines, architectural implementations, and software deployment orchestrations discussed throughout this thesis.

GitHub Repository: https://github.com/QuocBaoHo/AMP_Project

Researchers and developers are encouraged to clone, audit, and build upon this repository to further advance computational peptide discovery.

APPENDIX B: EVALUATION OF POSITION-SPECIFIC SCORING MATRIX (PSSM) AND ALIGNMENT-BASED LIMITATIONS

During the initial phases of the ablation study, the traditional Position-Specific Scoring Matrix (PSSM) was evaluated to establish a classical evolutionary baseline for Antimicrobial Peptide (AMP) prediction. The PSSM profiles were extracted using MMseqs2 against the massive Swiss-Prot database. However, the empirical results revealed a catastrophic algorithmic failure across all tested configurations.

When the PSSM features were propagated through the engineered Deep Learning architectures (ranging from single-kernel 1D-CNNs to the massive Giga mCNN configurations), the models uniformly returned a Sensitivity (Recall) of 0.0, an F1-Score of 0.0, and an ROC-AUC hovering around 0.5 (equivalent to random guessing). Classic Machine Learning classifiers (Random Forest, SVM) exhibited similarly negligible predictive power.

This total failure is not an implementation anomaly, but rather a fundamental biological and mathematical limitation of alignment-based profiling for short sequences.

B.1. The Alignment Blind Spot of Short Peptides

Antimicrobial Peptides are inherently short, predominantly consisting of fewer than 50 amino acids, with many active motifs spanning less than 20 residues. Traditional PSSM generation relies on local alignment algorithms (e.g., PSI-BLAST, MMseqs2) to scan sequence databases and identify evolutionary homologs. However, for ultra-short sequences, these algorithms mathematically fail to find statistically significant matches, as the Expected Value (E-value) of short hits exceeds standard significance thresholds. Consequently, the search algorithm hits an "alignment blind spot," returning virtually no meaningful evolutionary relationships.

B.2. Sparse Matrices and Convolutional Model Collapse

Because no homologous sequences are retrieved, the resulting PSSMs fail to capture sequence conservation. They degenerate into highly sparse matrices, populated almost entirely by zeros or uniform background frequencies. When these barren profiles are structurally padded to fit the system's strict 128×20 input tensor geometry, the signal-to-noise ratio is diluted to near absolute zero.

When these sparse, noise-heavy matrices are fed into the Multi-branch CNN (mCNN) or SVM models, the mathematical filters are unable to extract any discriminative spatial motifs. Confronted with a complete lack of meaningful gradient signals, the neural network optimizer rapidly descends into a severe local minimum. The network undergoes **Model Collapse**, resorting to continuously predicting the majority negative class (Non-AMP = 0) simply to minimize the loss function. This mechanism perfectly explains the strict 0.0 Sensitivity observed during the evaluation.

B.3. Justification for Alignment-Free Architectures

The documented failure of PSSM unequivocally demonstrates that traditional alignment-based feature extraction is structurally obsolete for short-sequence peptide discovery. This empirical evidence mathematically justifies the core methodological shift of this research: the adoption of **Protein Language Models (PLMs)**. Unlike PSSM, ESM-2 operates on an entirely "Alignment-Free" paradigm. Through its deep, multi-head Self-Attention mechanism, ESM-2 circumvents explicit database querying. It dynamically infers 3D structural topologies and hidden biochemical grammar directly from the standalone sequence context, successfully solving the short-peptide blind spot and achieving the state-of-the-art results presented in Chapter 4.

APPENDIX C: SOFTWARE USER INTERFACE (STREAMLIT DEMO)

This section provides a comprehensive visual overview of the graphical user interface (GUI) developed using the Streamlit framework. The application is designed to handle both interactive single-sequence analysis via Explainable AI (XAI) and high-throughput batch screening.

C.1. Single Sequence Analysis (Cascade Stages 2 & 3)

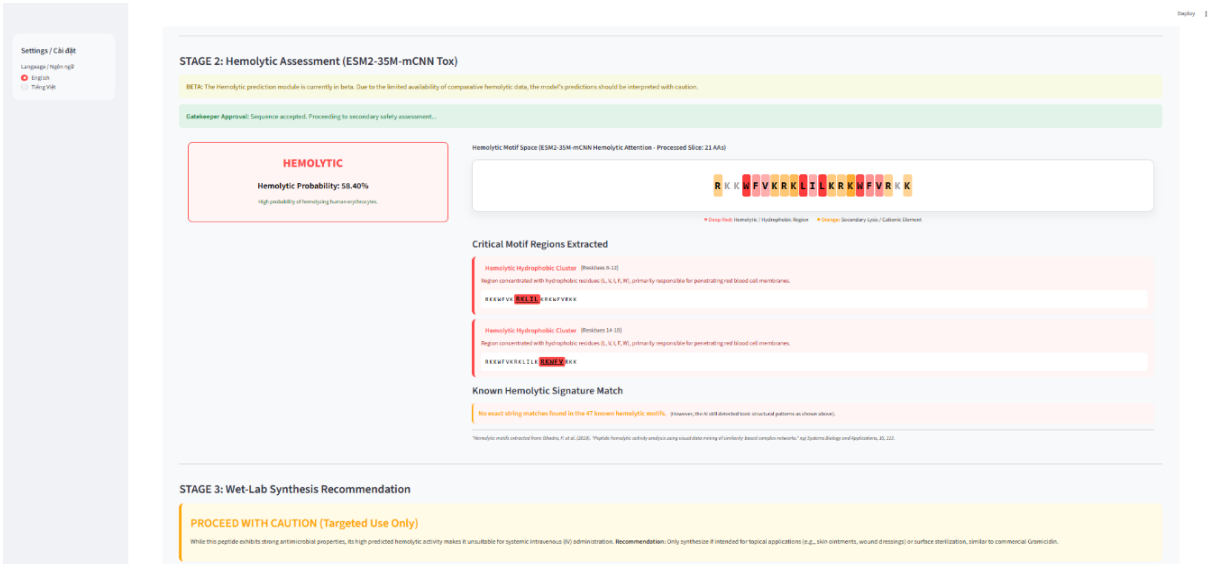


Figure C.1: Stage 2 (Hemolytic Assessment) interface. Following AMP detection, the sequence is evaluated for toxicity. The UI renders the XAI Hemolytic Attention heatmap and automatically extracts critical sub-sequences, specifically flagging Hemolytic Hydrophobic Clusters responsible for membrane penetration.

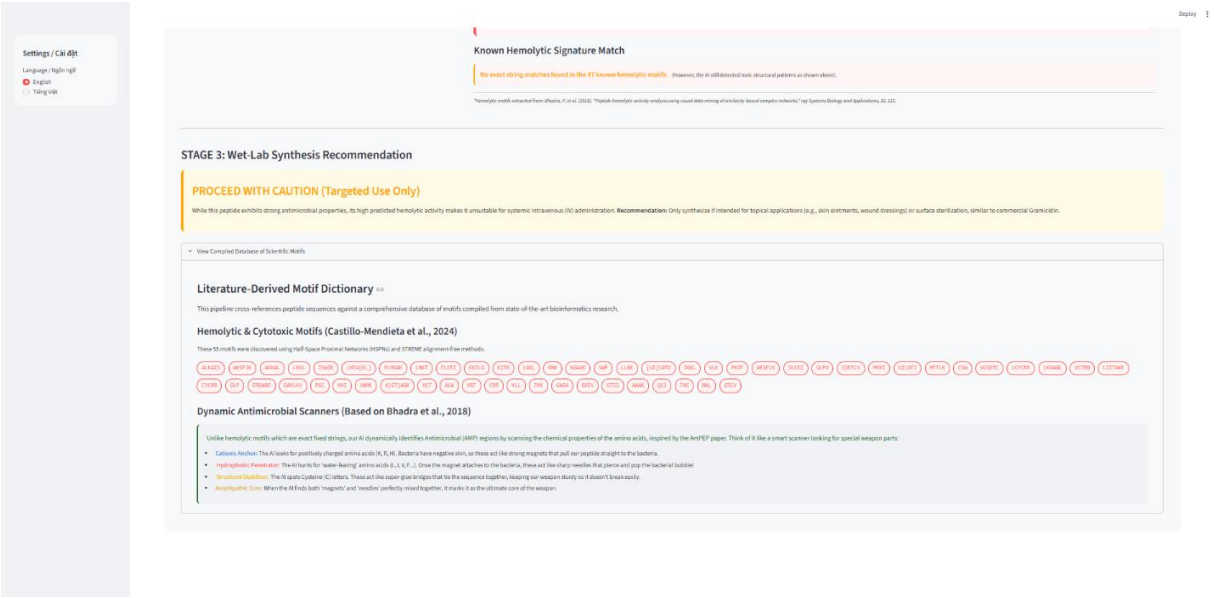


Figure C.2: Stage 3 (Wet-Lab Synthesis Recommendation). Based on the cascade results, the system outputs actionable synthesis advice. The UI also features a built-in Literature-Derived Motif Dictionary, allowing researchers to cross-reference AI findings with established wet-lab databases.

C.2. High-Throughput Batch Screening Module

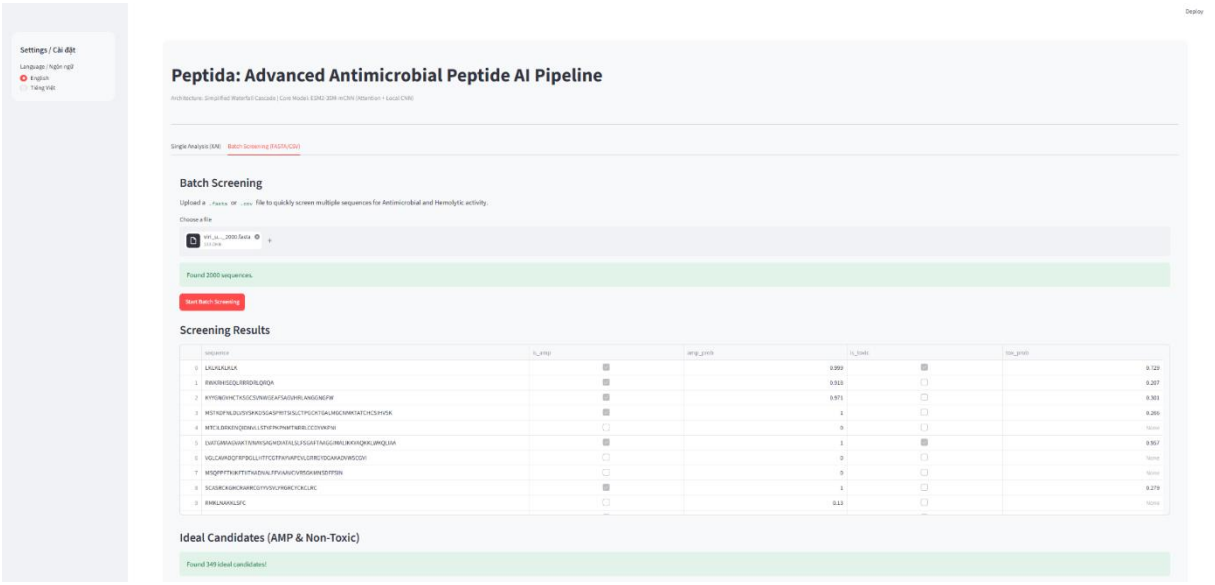


Figure C.3: The Batch Screening interface. Users can upload standard .fasta or .csv files. The system processes sequences in parallel utilizing the ESM2-35M core, outputting a comprehensive tabular results matrix tracking both AMP and Toxicity probabilities.

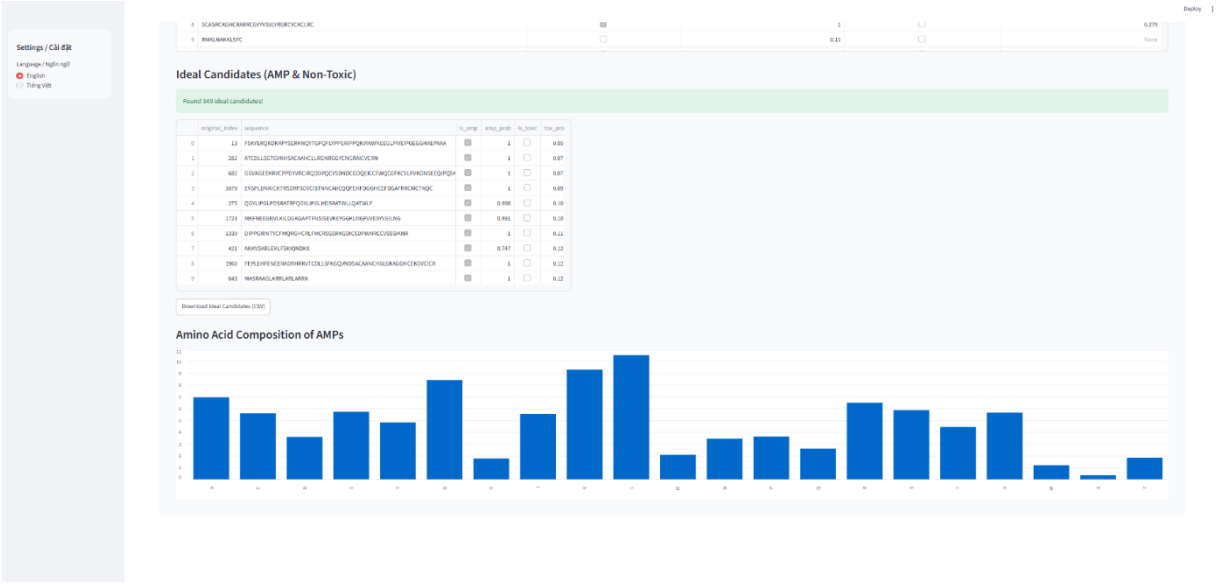


Figure C.4: Batch Analytics Dashboard. The system automatically filters the batch to isolate 'Ideal Candidates' (High AMP, Low Toxicity). Additionally, it generates a global Amino Acid Composition (AAC) profile of the successful candidates to aid in downstream synthesis planning.